

Basic Principles of Medicine II
Module: Nervous System and Behavioral
Sciences

Lecture No: NB 02

General Morphology of the Nervous
System

Neuroscience Faculty
St. George's University

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Note: The lecture slides that are available for download and print contain complementary slides that are not presented during the lecture.

These slides have a blue background and contain additional explanation of the material discussed during the lecture.

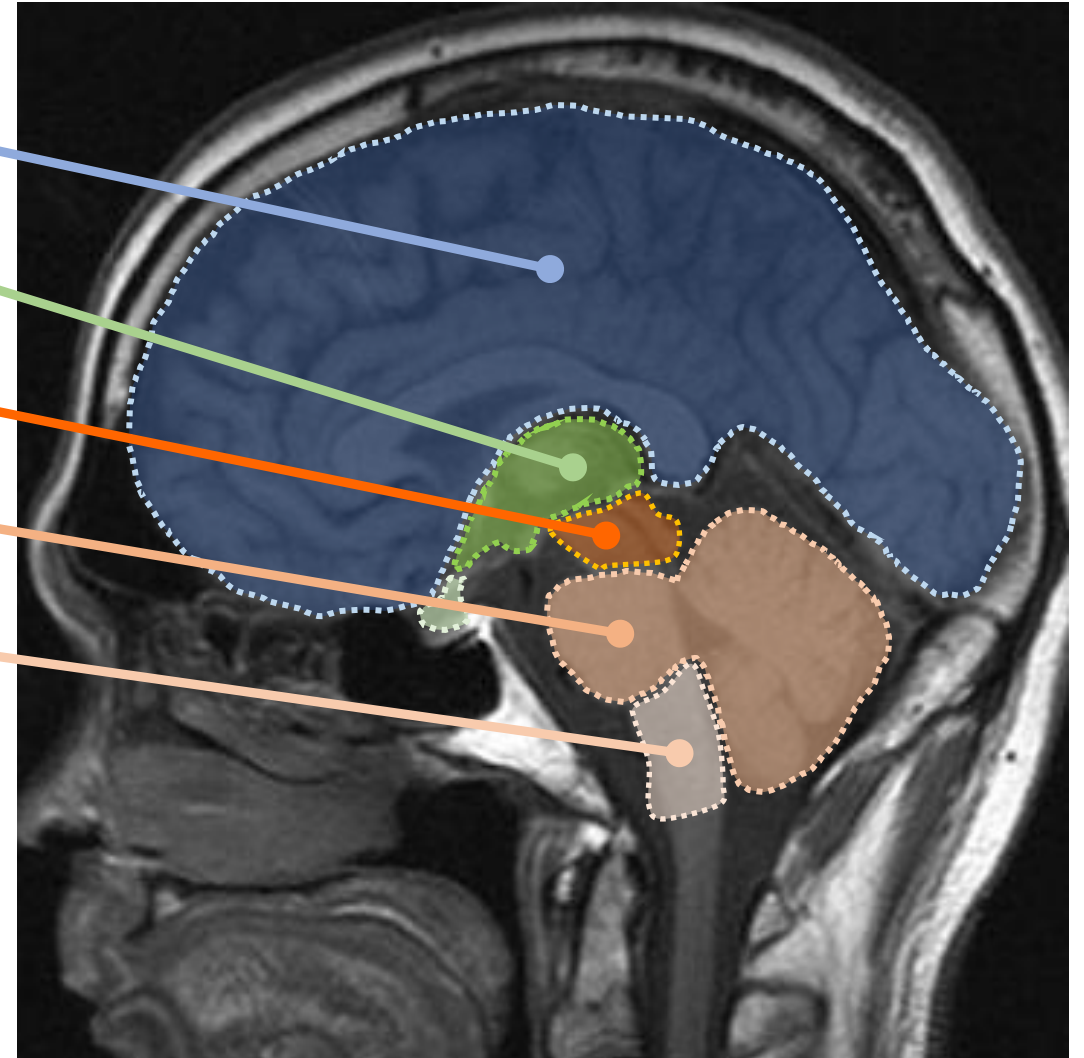
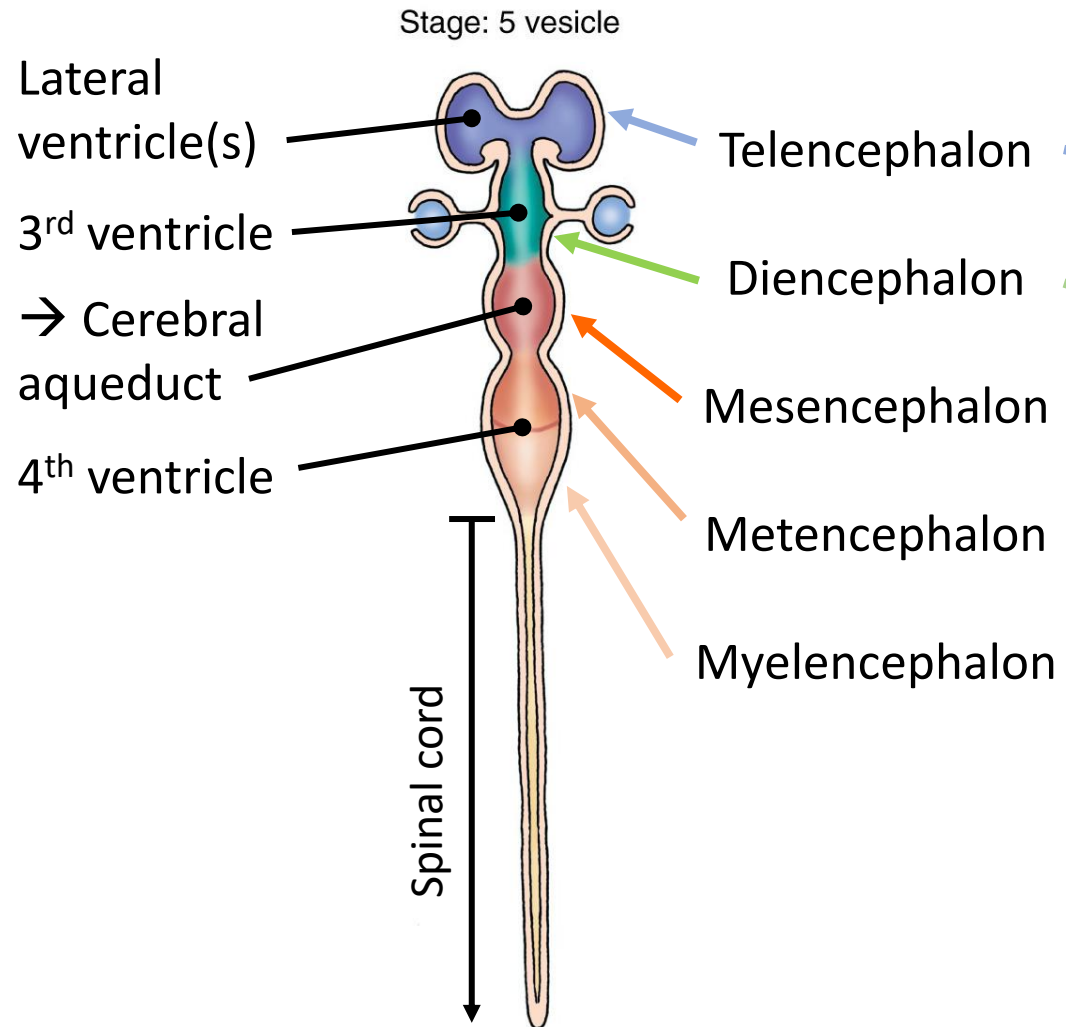
Learning Objectives

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0001** Describe the divisions of the nervous system
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0550** Describe the ventricular system of the central nervous system and the relationship of its elements to the embryological origin of the five parts of the brain
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0008** Apply the use of anatomical vectors in the representation of the nervous system
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0009** Apply the use of anatomical vectors as represented in neuroimages
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0021** Functionally characterize the cerebral lobes from surface views
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0023** Functionally characterize the major structures of the diencephalon
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0024** Functionally characterize the brainstem and cerebellum
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0551** Functionally characterize the spinal cord

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Five Parts of the Brain



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Central Nervous System (CNS) – Terminology Overview

- **Brain**

- **Telencephalon** (Endbrain, Cerebrum)

- **Diencephalon**

- Thalamus
 - Hypothalamus
 - Subthalamus
 - Epithalamus

- **Mesencephalon** (Midbrain)

- **Metencephalon** (Pons and Cerebellum)

- **Myelencephalon** (Medulla)

Prosencephalon (Forebrain)

Brainstem

- Midbrain
- Pons
- Medulla

Cerebellum

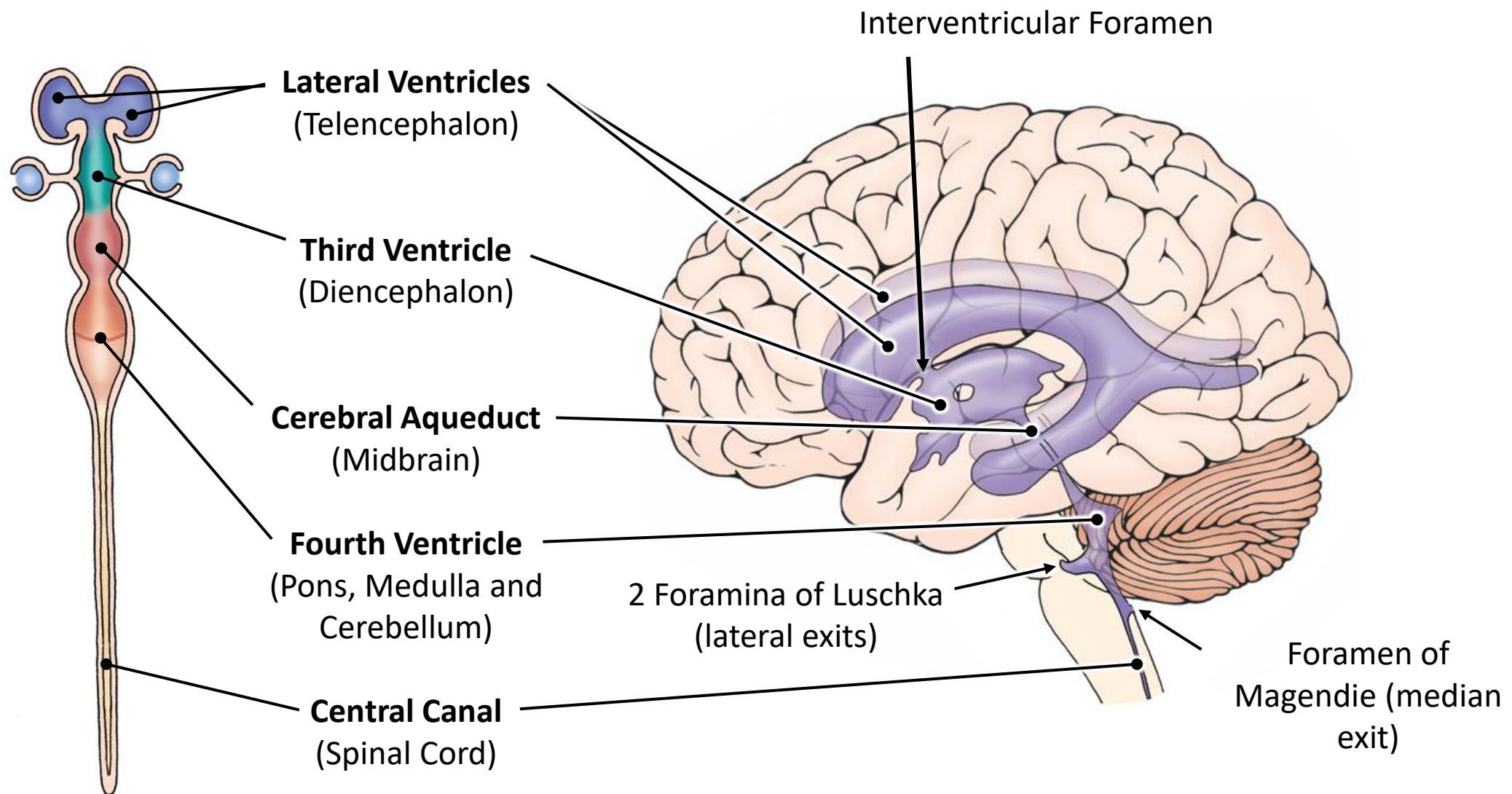
Rhombencephalon (Hindbrain)

- **Spinal cord**

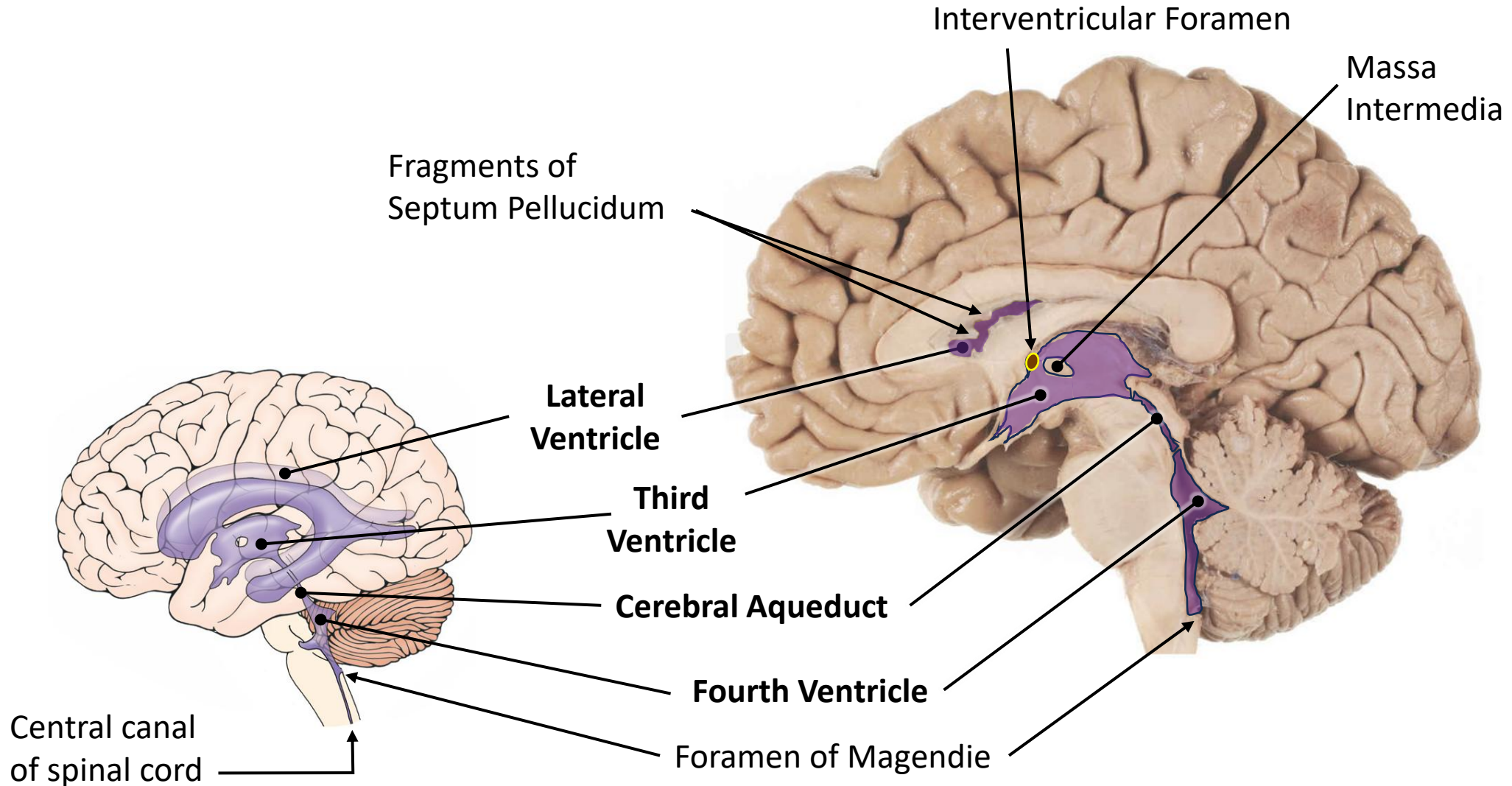
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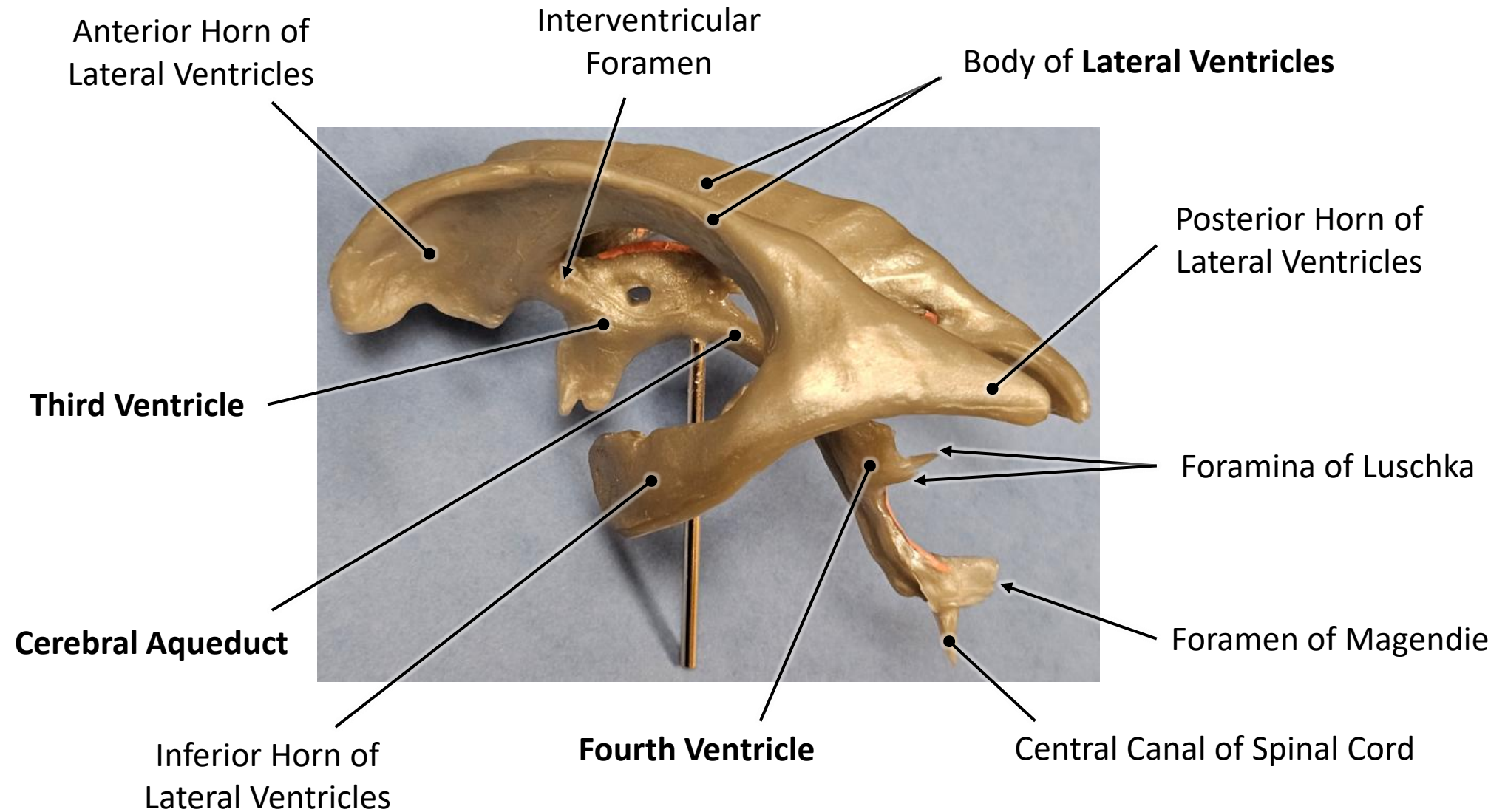
Embryological Origin / Ventricular System and Parts of the Brain



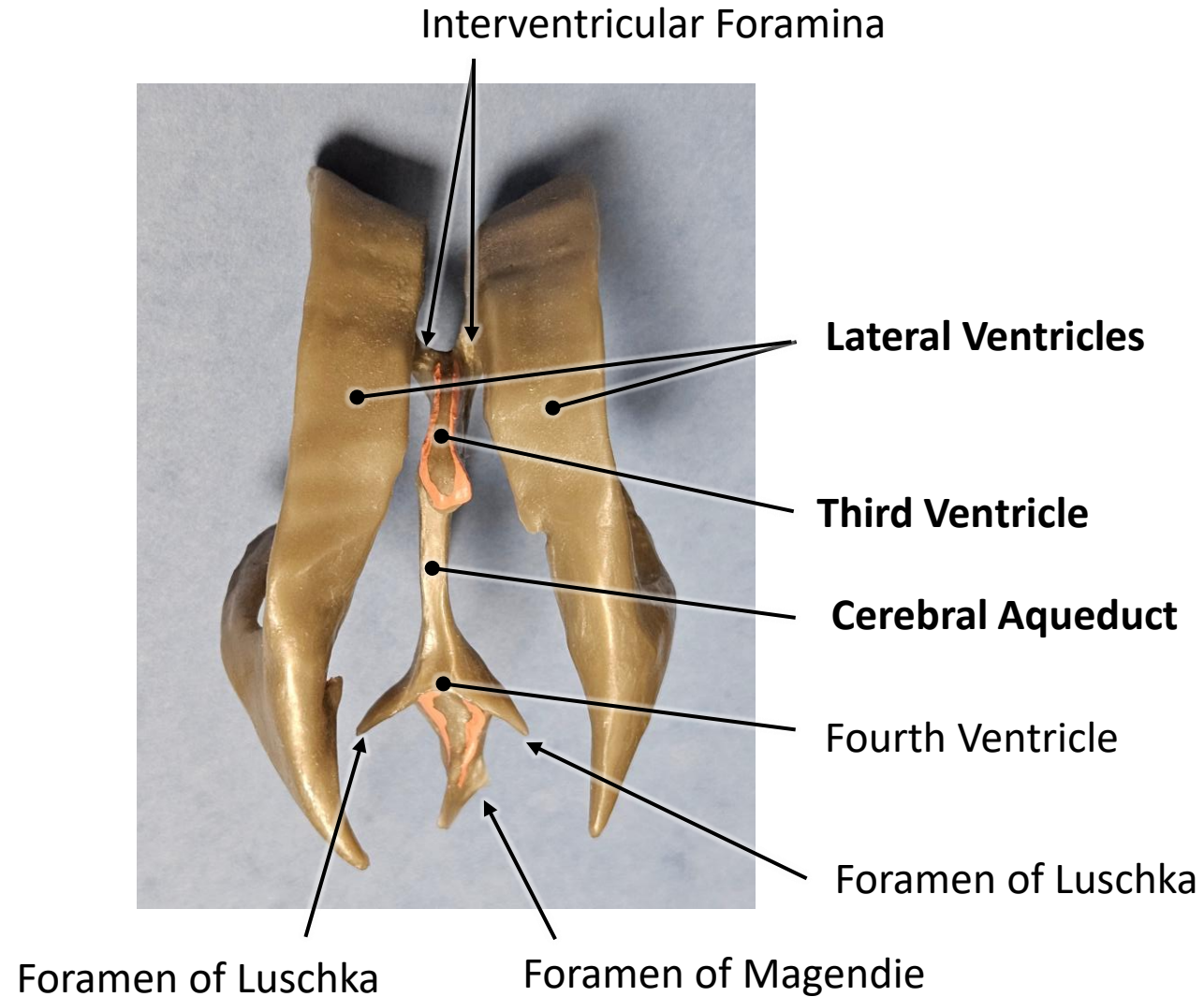
Ventricular System of the Brain – Medial (Midsagittal) View



Ventricular System – Lateral View



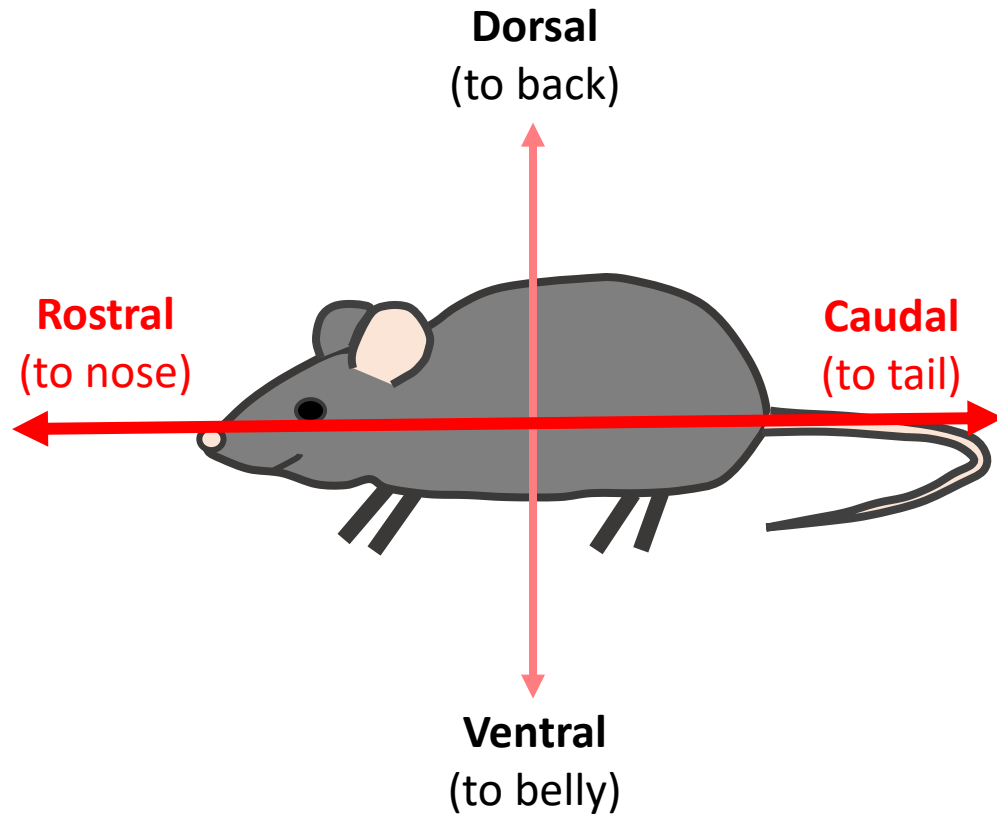
Ventricles – Superior View



Learning Objectives

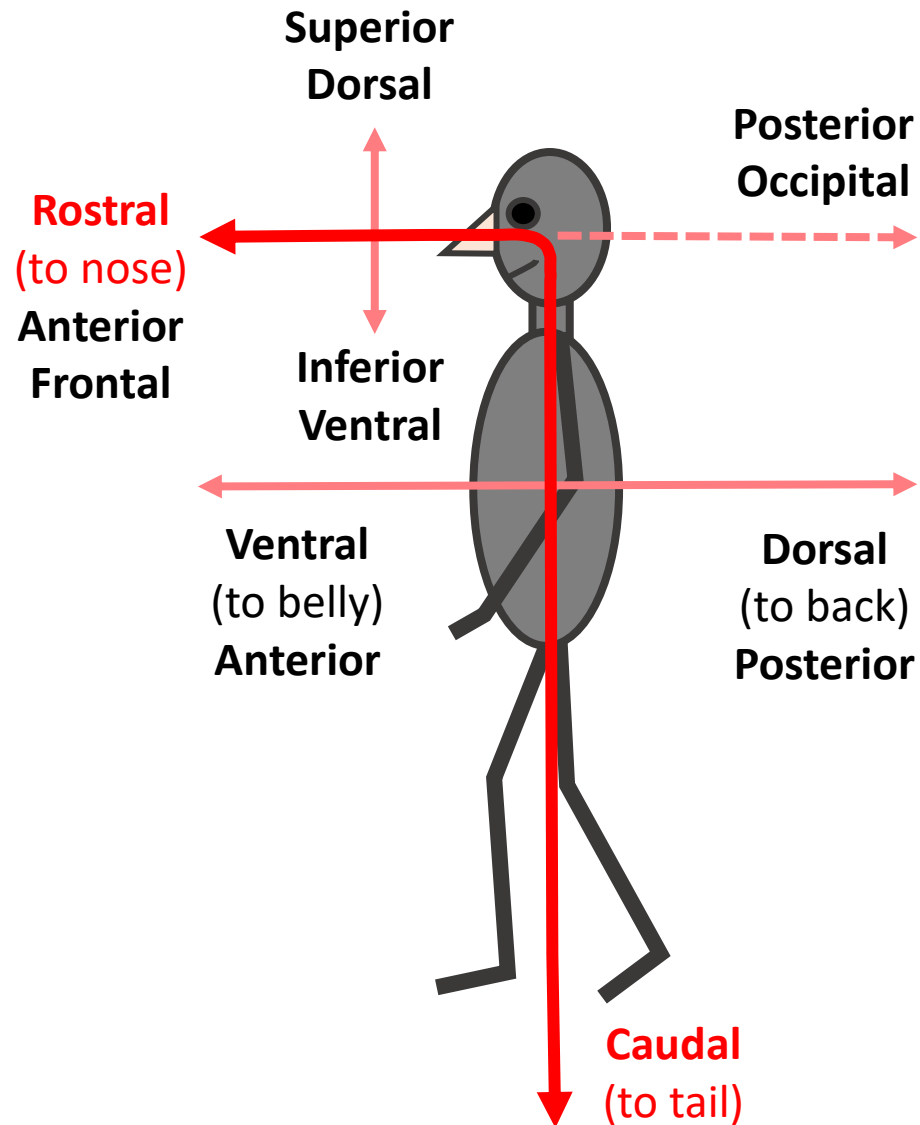
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Orientation – Lateral View



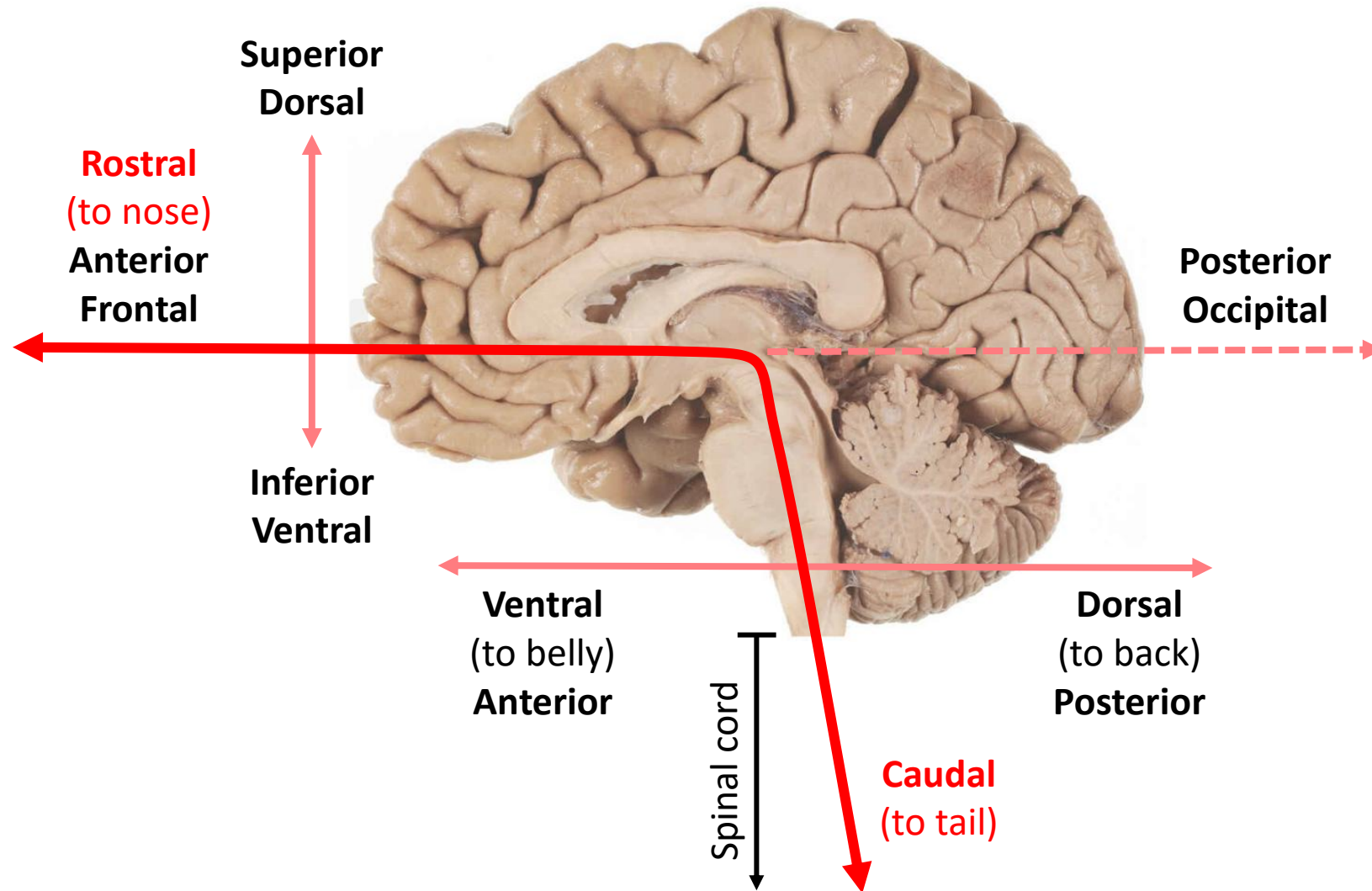
- Looking at a 4-legged animal, orientation within the central nervous system is rather simple
- The rostro-caudal axis is going straight from nose to the tail
- The dorso-ventral axis is perpendicular to the rostro-caudal axis from back to belly

Orientation – Lateral View

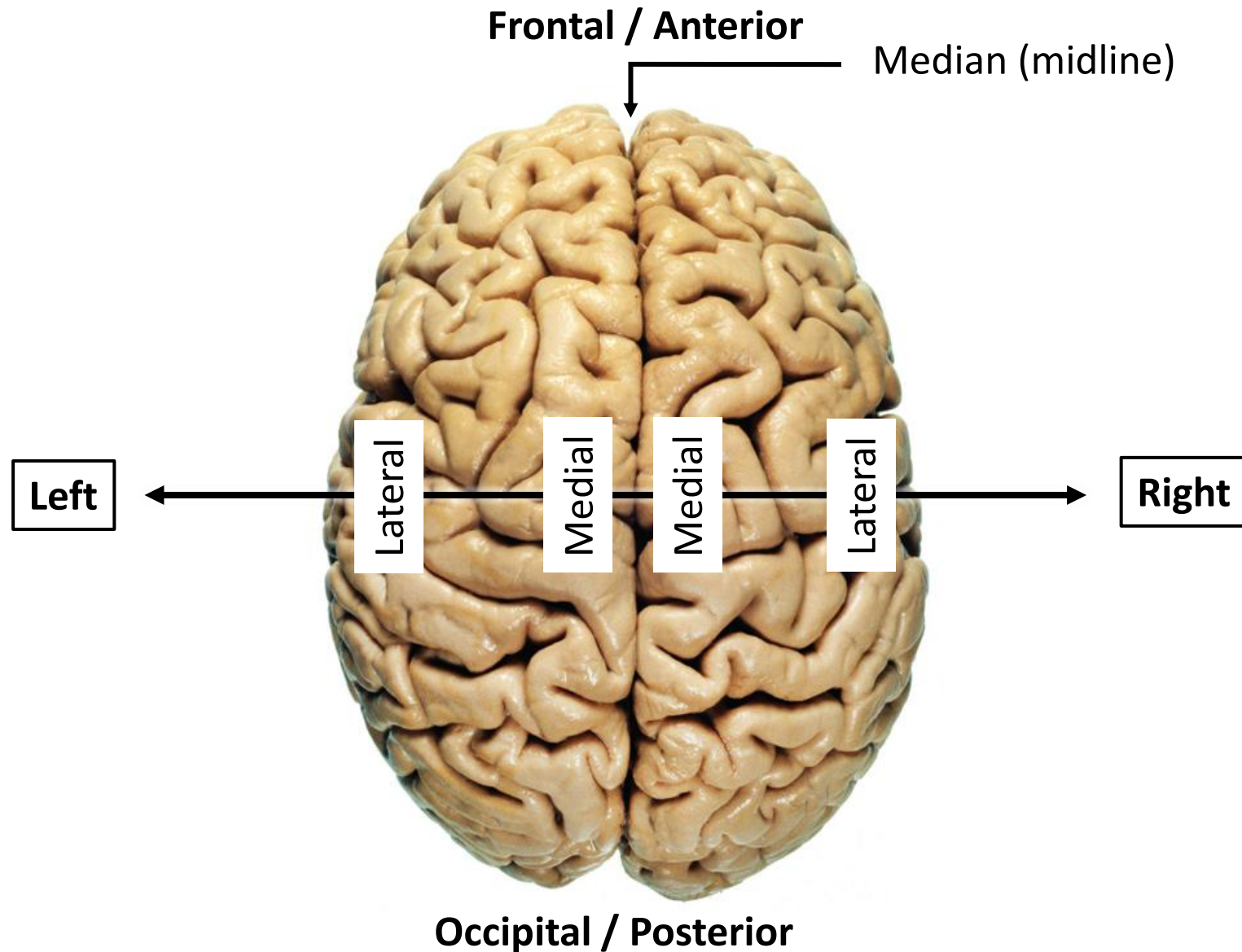


- During evolution, humans developed an upright position, using their lower extremities (the hindlegs of our ancestors) for locomotion, and freeing up upper extremities for other tasks
- Alongside, the rostro-caudal axis developed a bend at the diencephalon/mesencephalon level, which requires some adjustments of our terminology

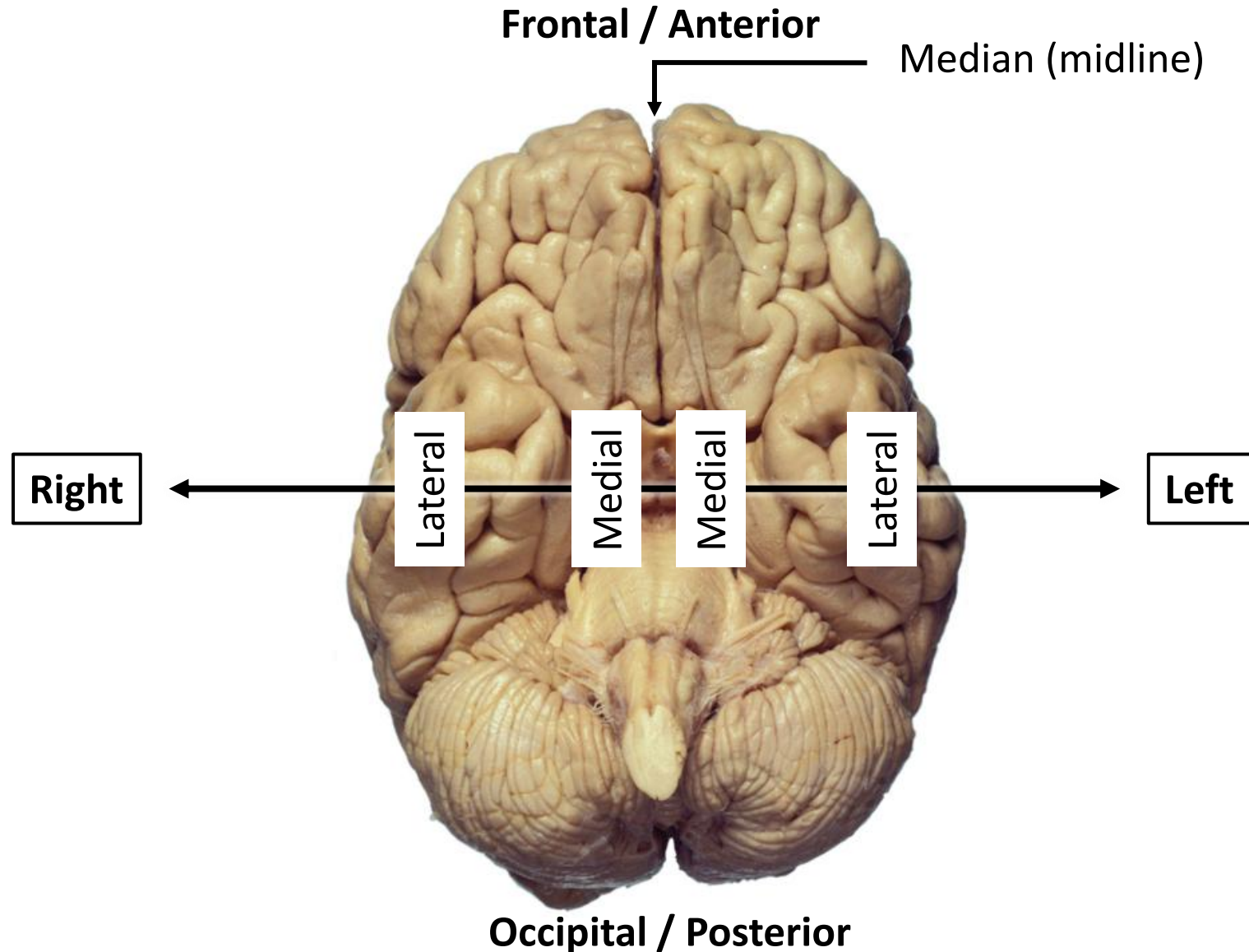
Right Half of the Brain – Orientation Medial (Midsagittal) View



Superior View of the Brain – Orientation



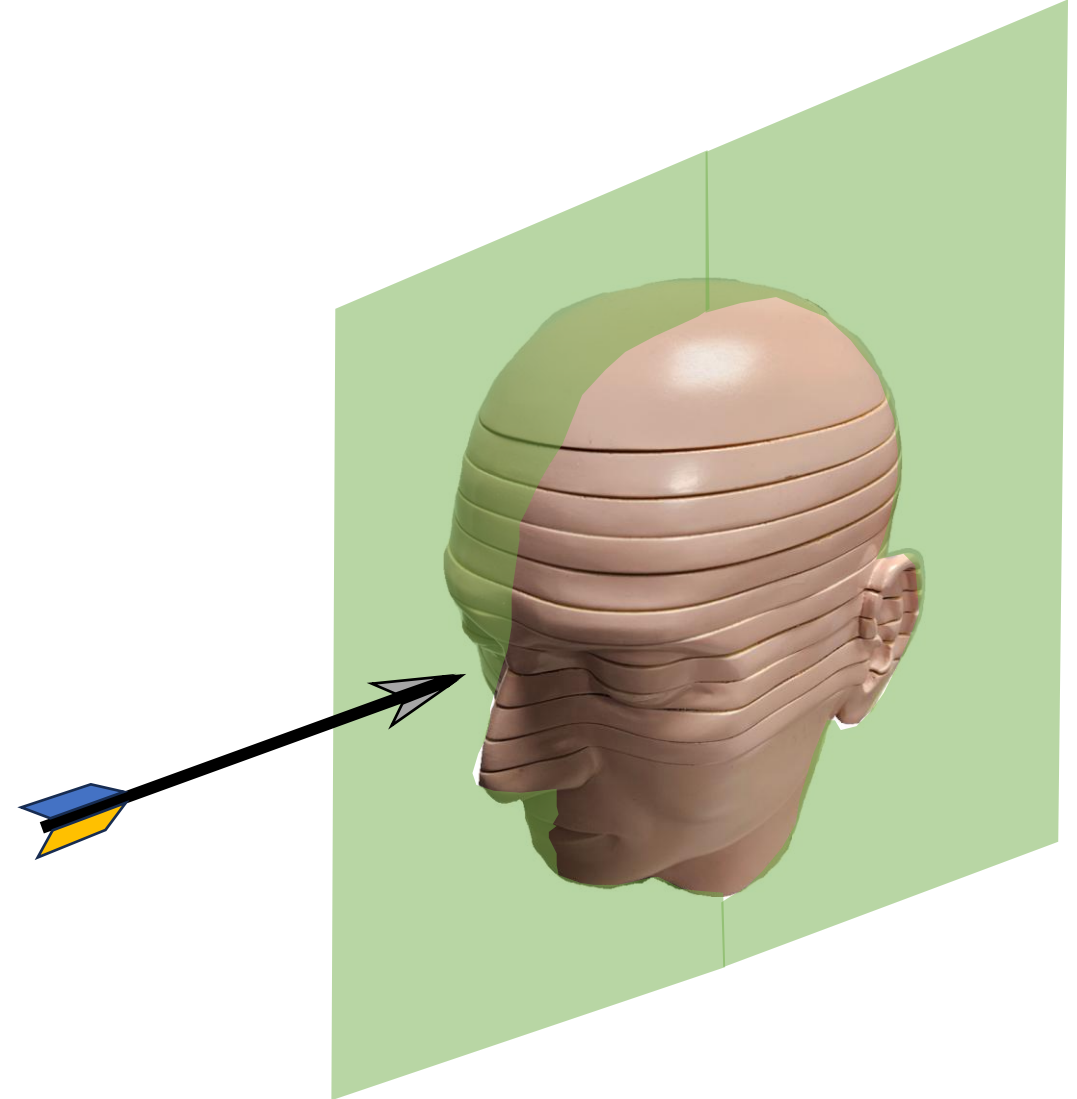
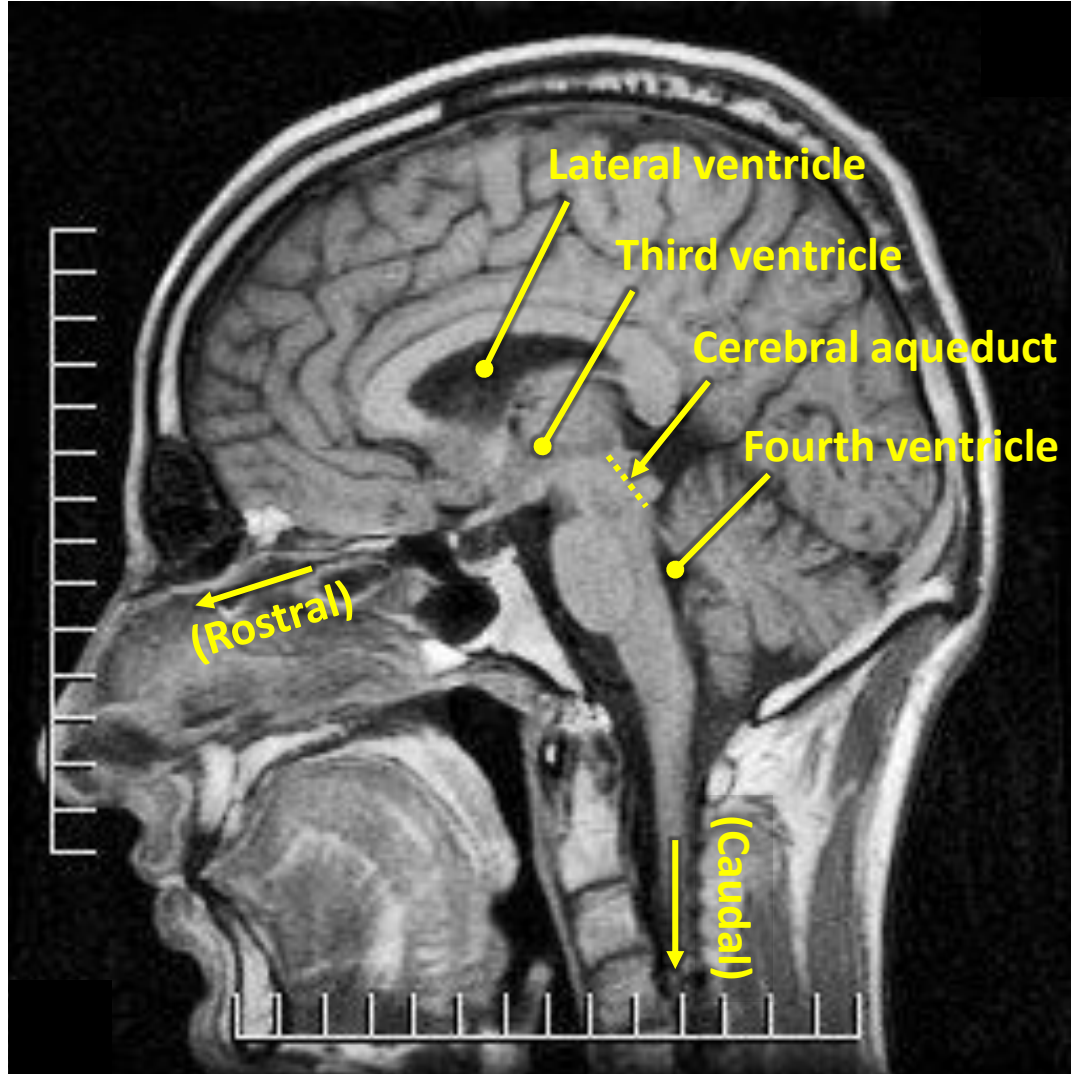
Inferior View of the Brain – Orientation



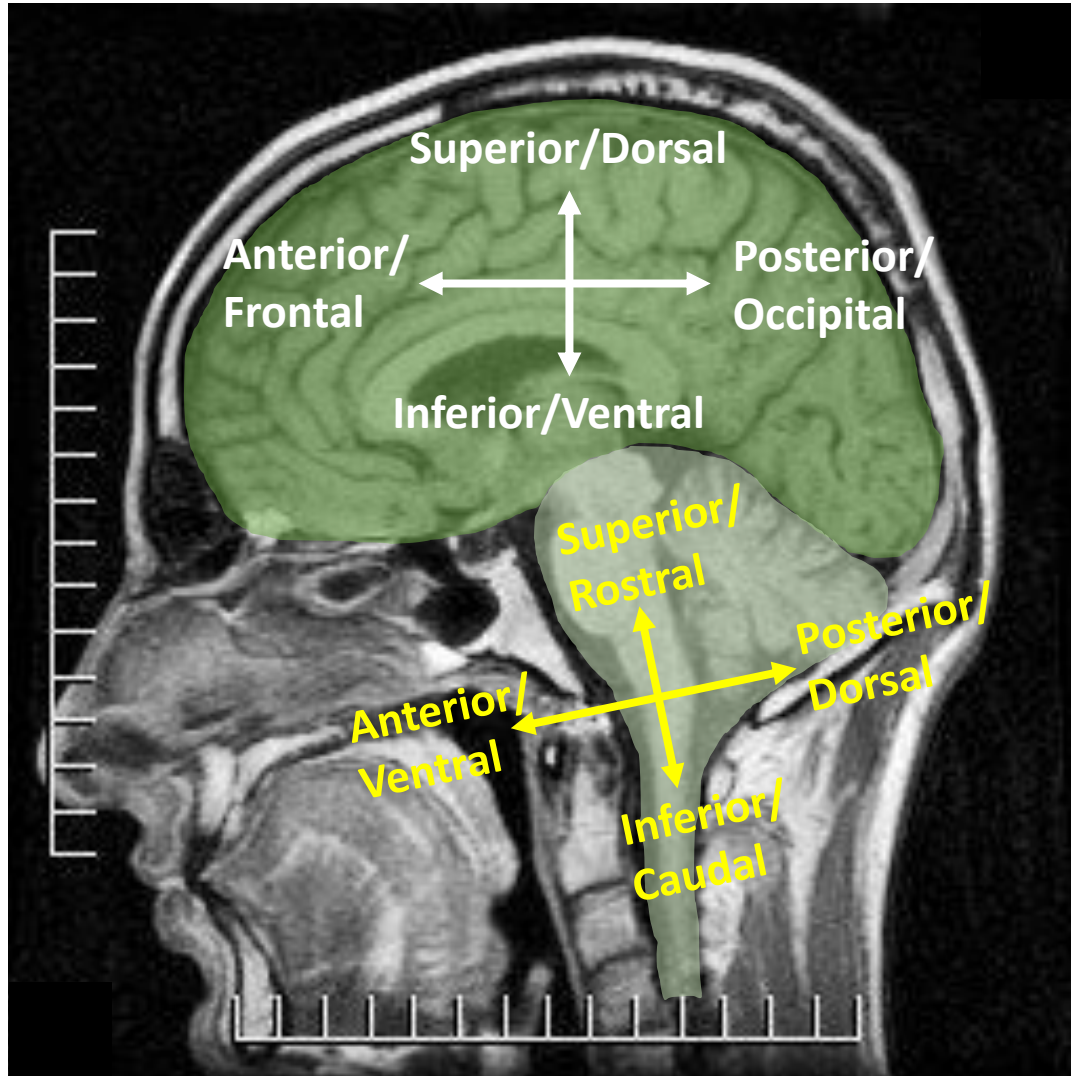
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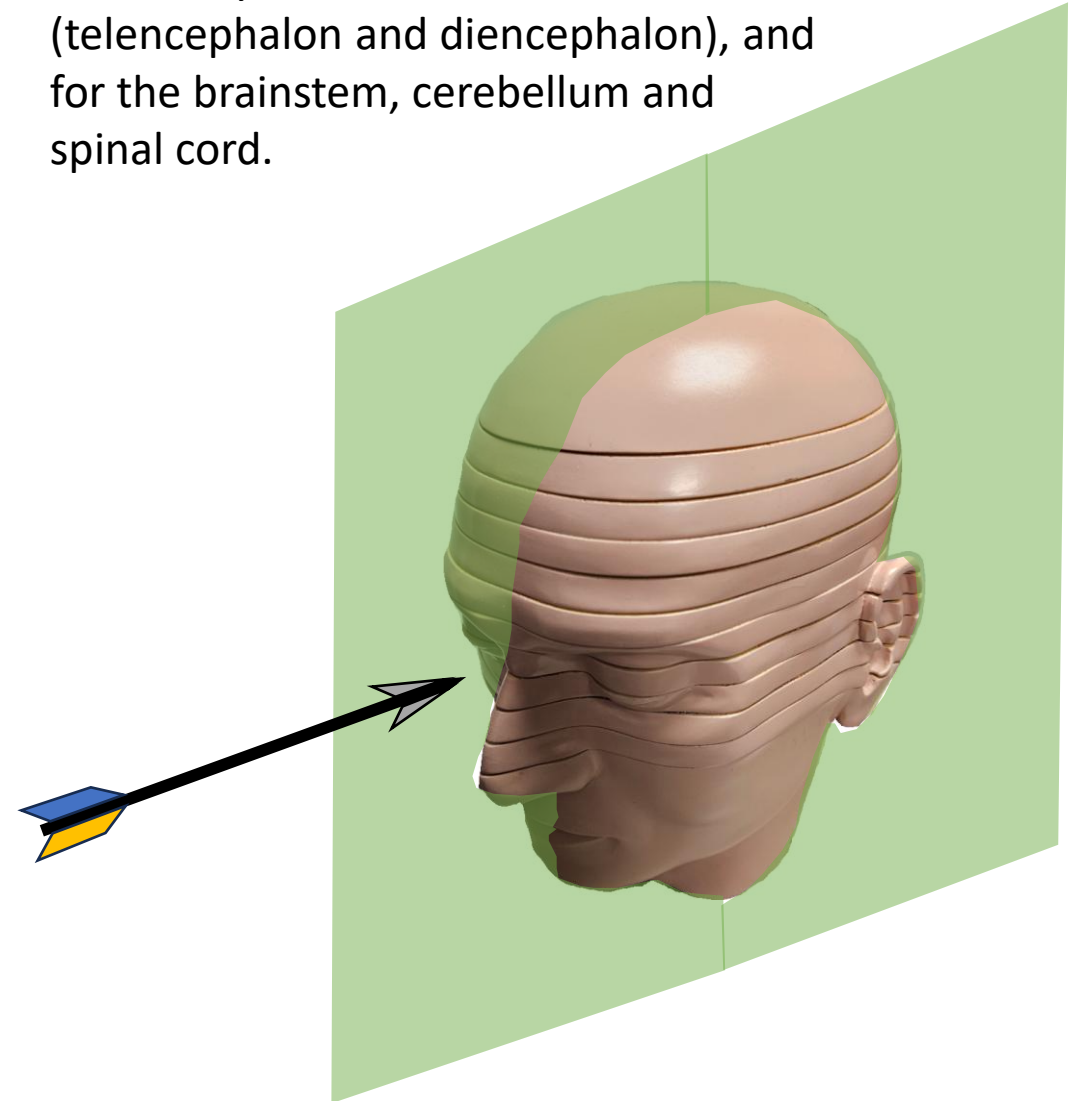
Sagittal Section Plane



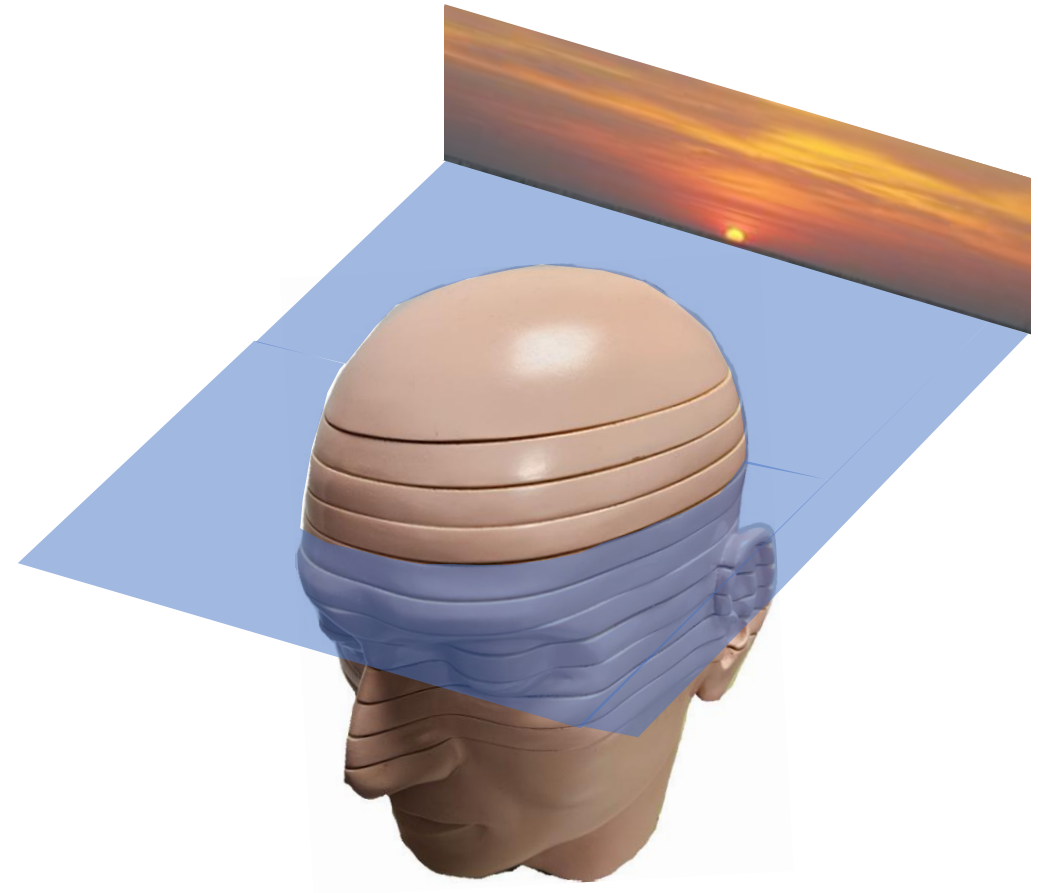
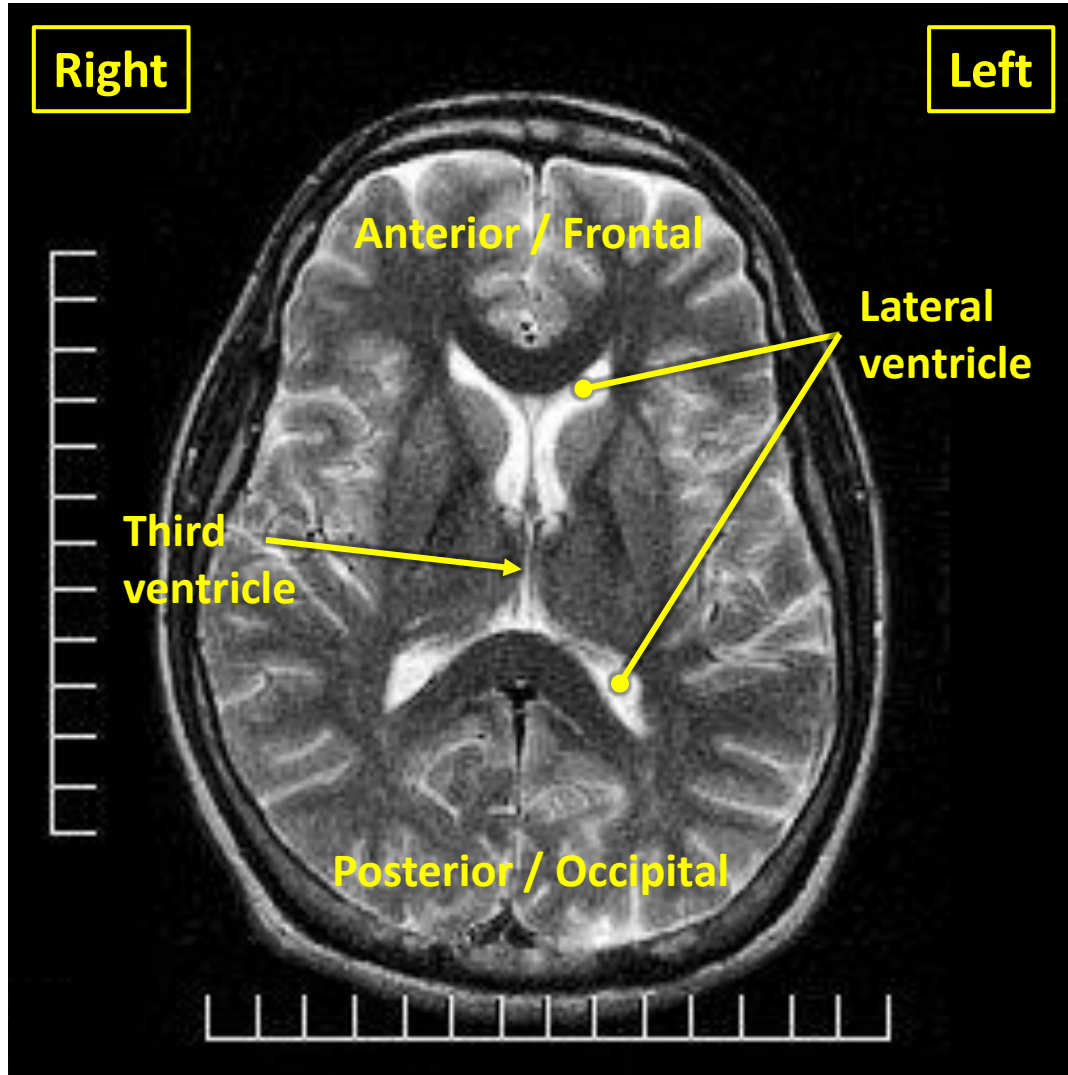
Sagittal Section Plane



Note: Different anatomical vectors are commonly used for the forebrain (telencephalon and diencephalon), and for the brainstem, cerebellum and spinal cord.

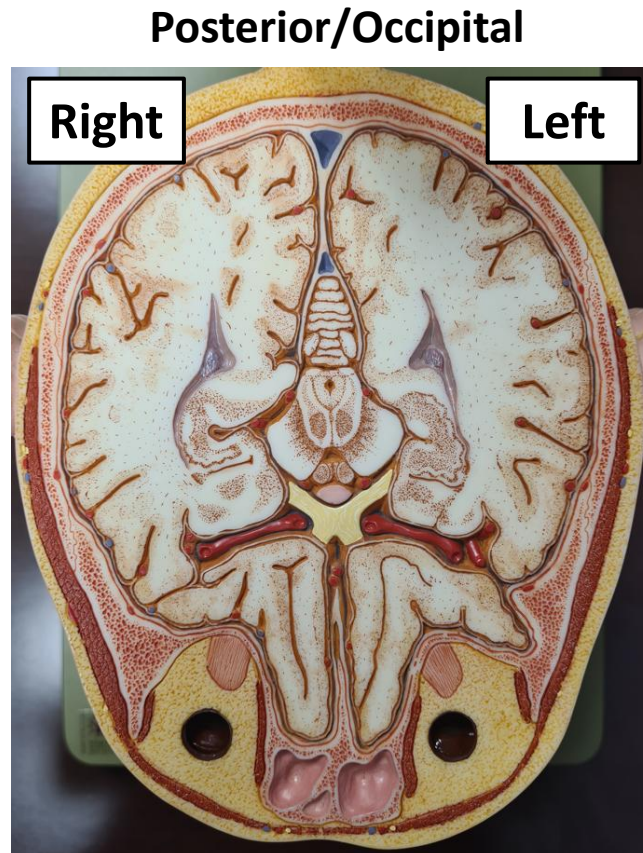


Horizontal (Axial) Section Plane

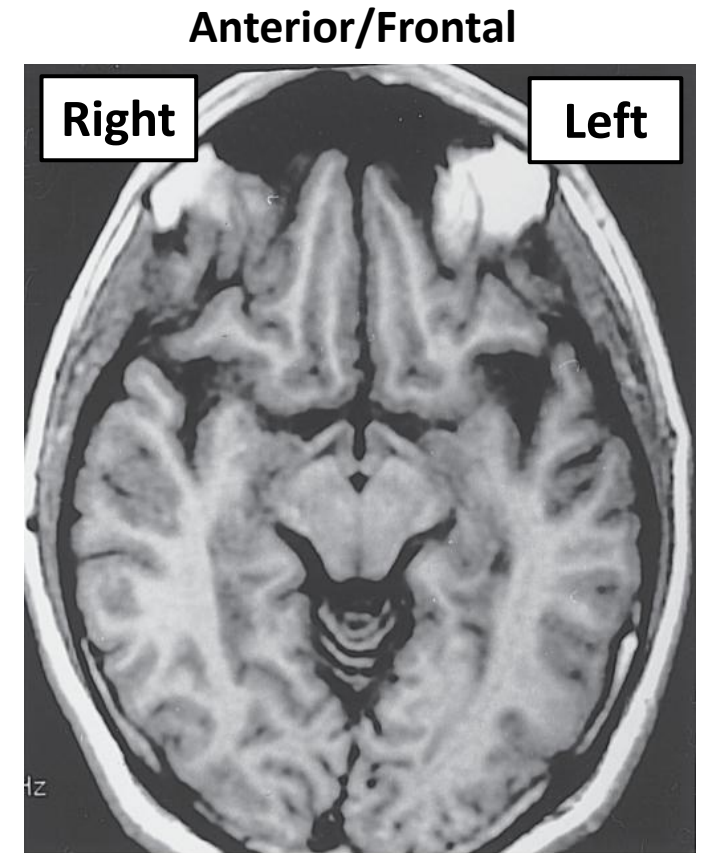


Orientation of Horizontal (Axial) Sections

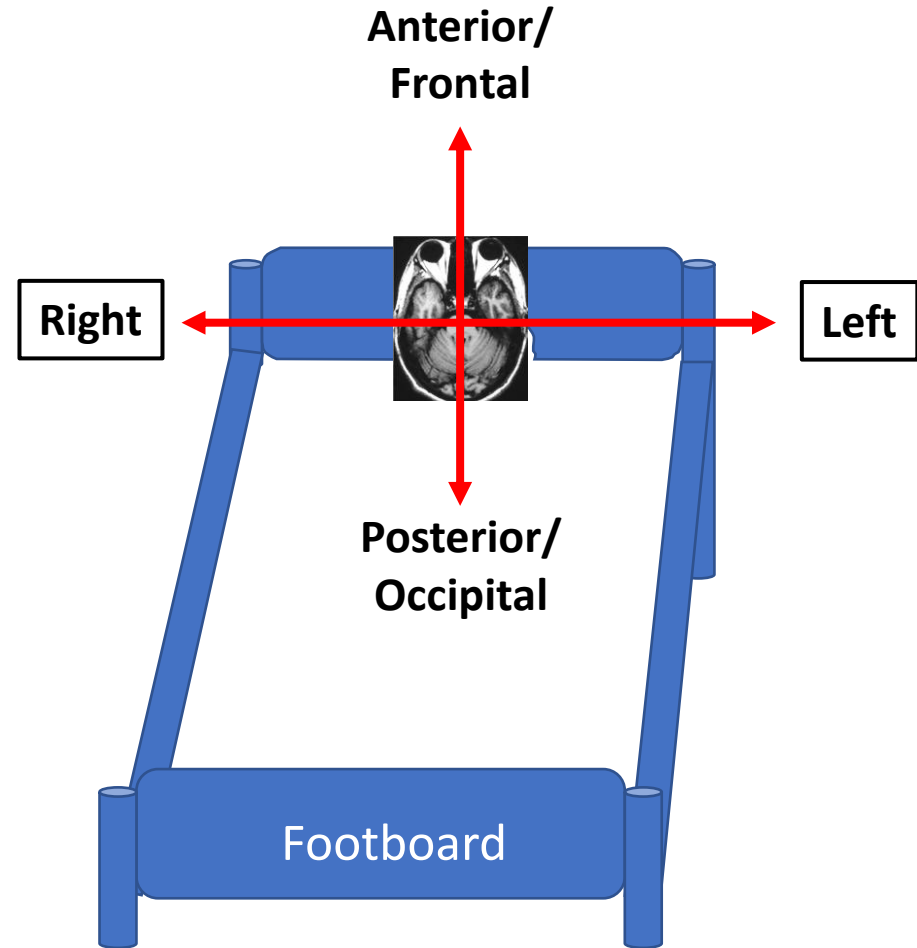
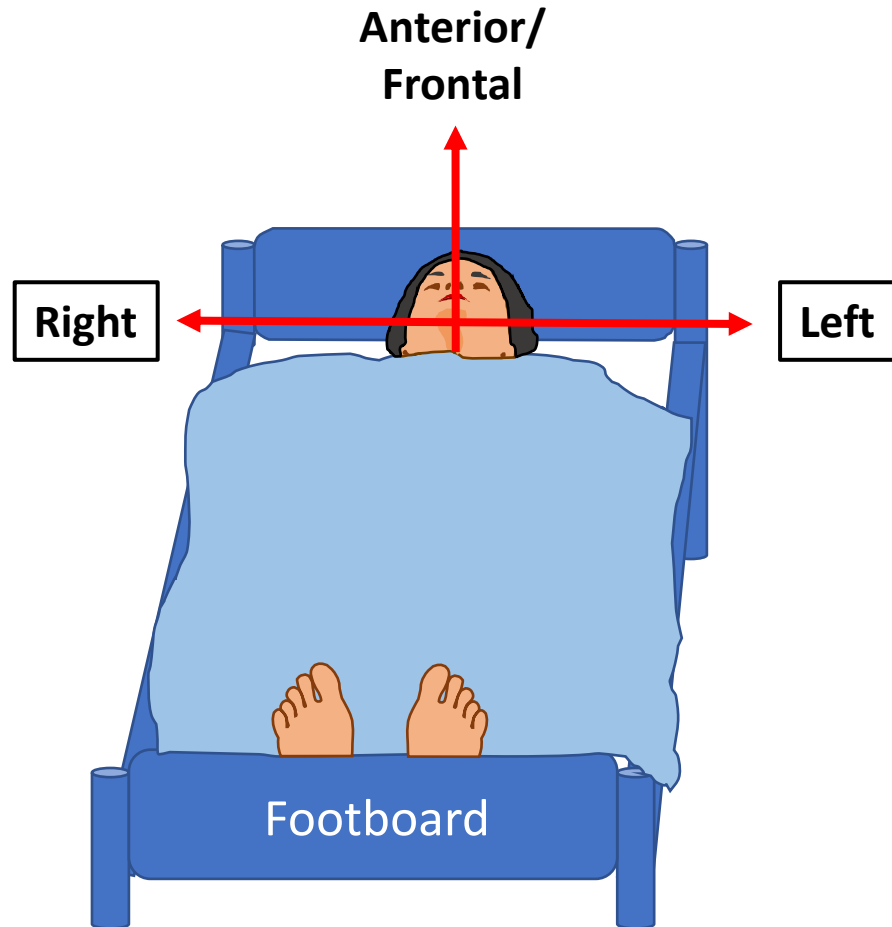
Anatomical Orientation



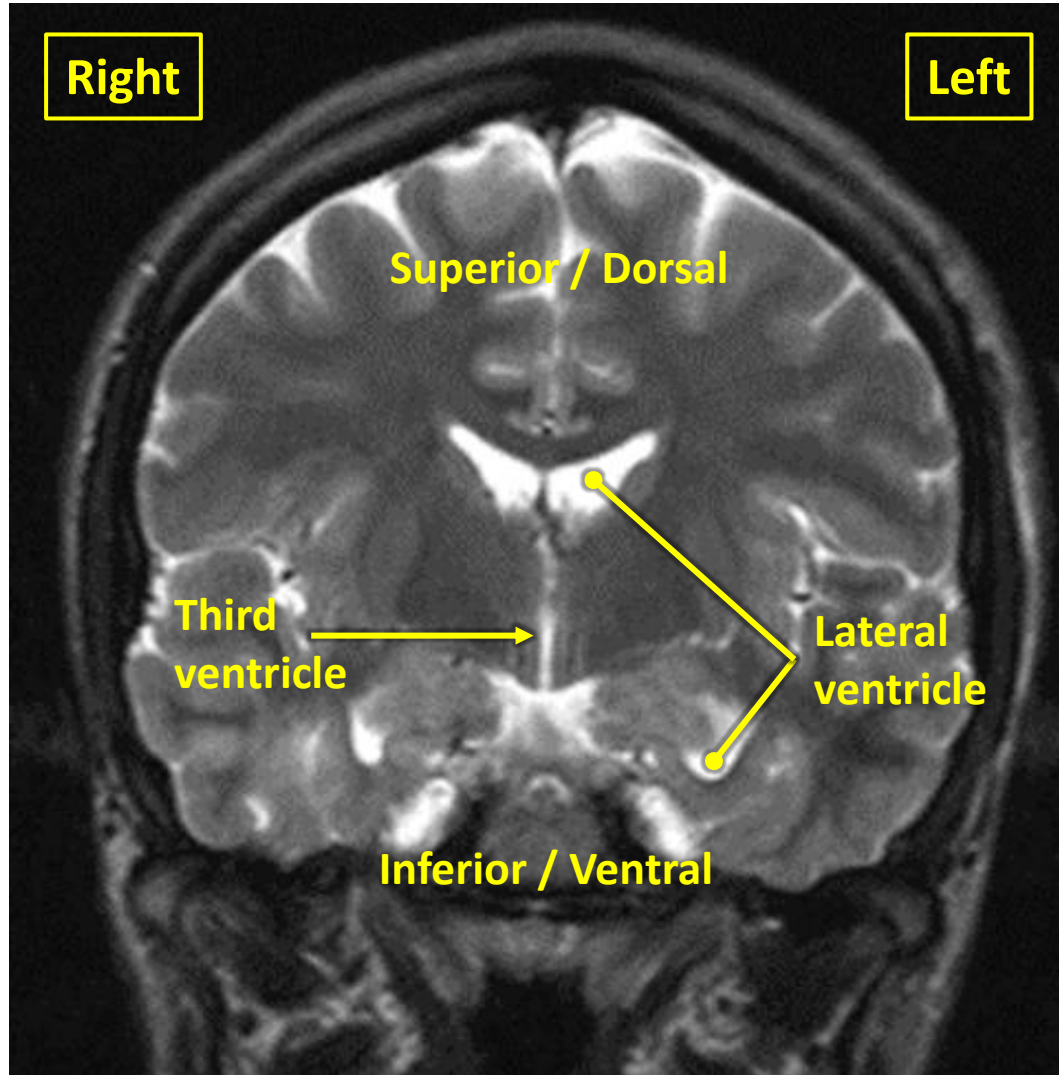
Clinical Orientation



Clinical Orientation of Horizontal (Axial) Sections



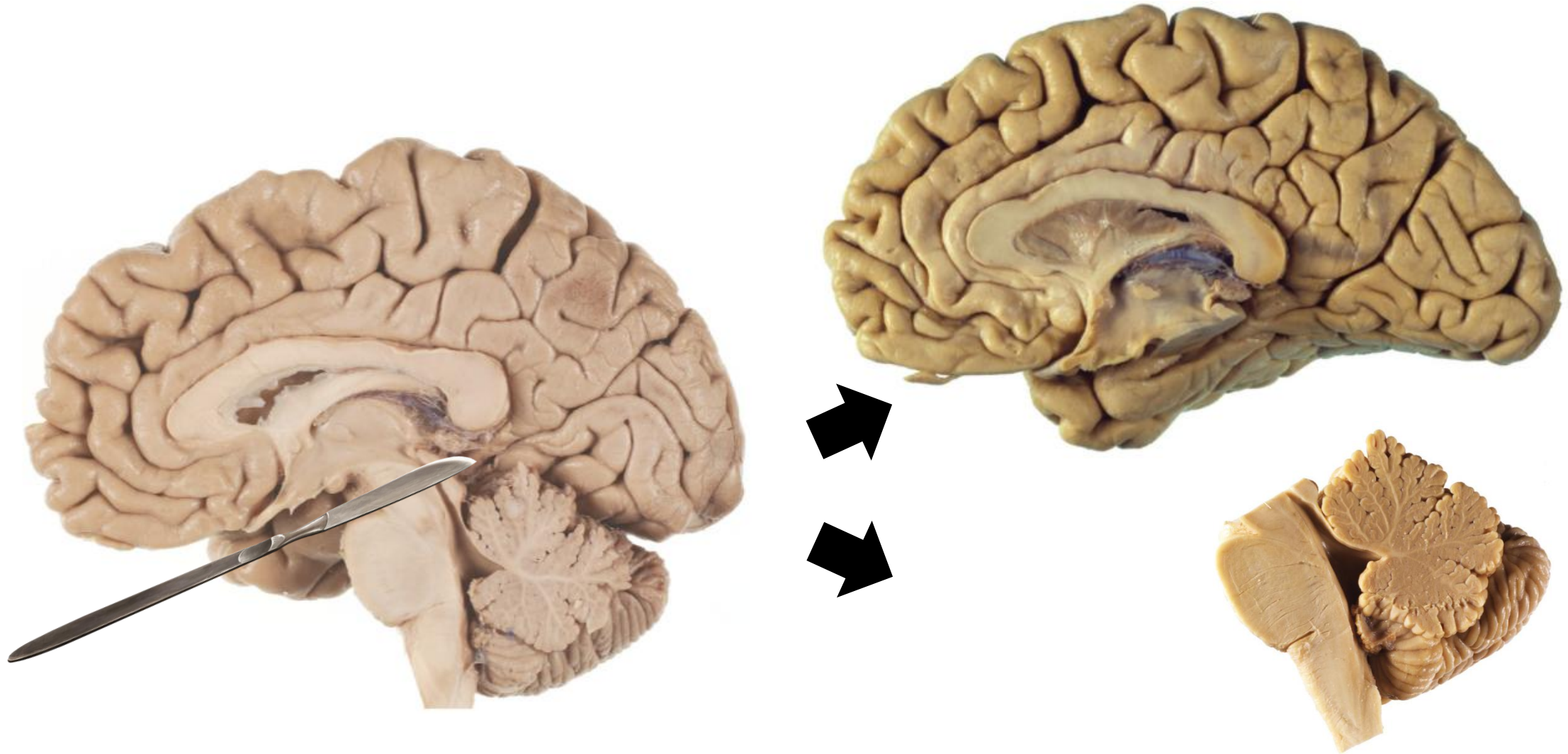
Coronal Section Plane



Learning Objectives

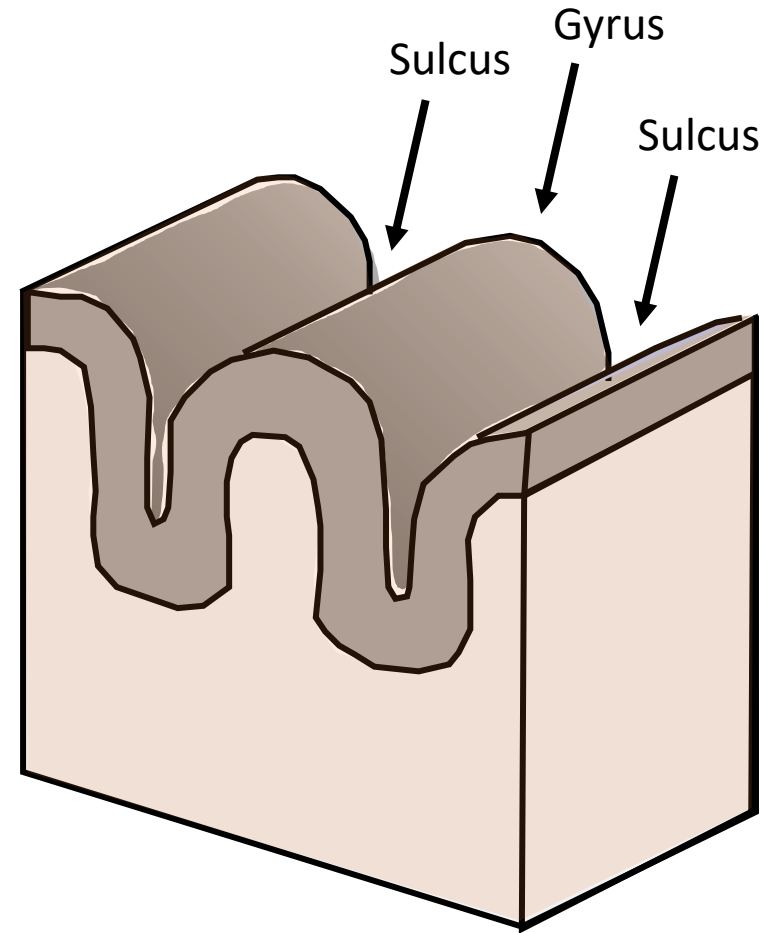
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Upcoming Slides: Brainstem and Cerebellum removed from Cerebral Hemispheres

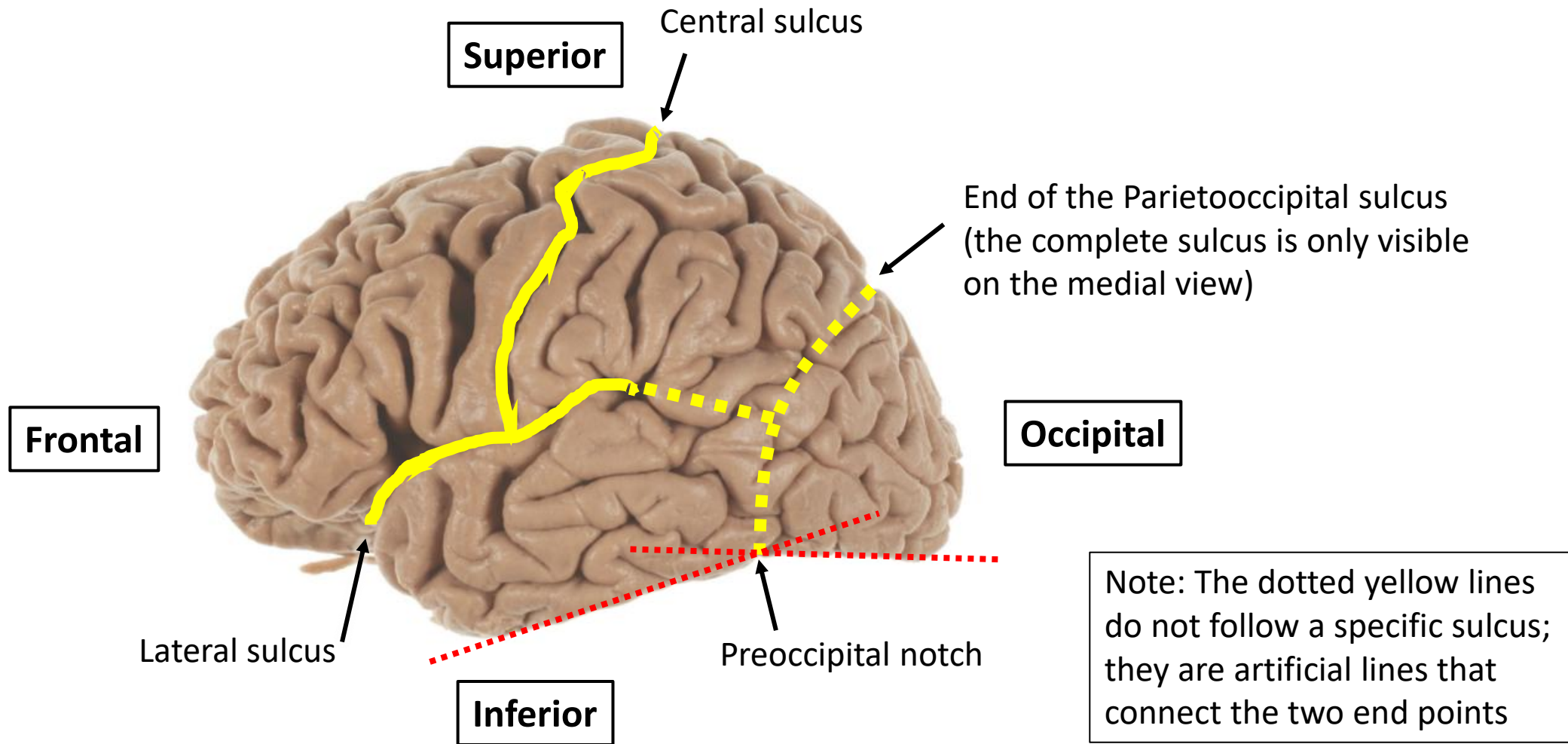


Surface of the Telencephalon – Gyri and Sulci

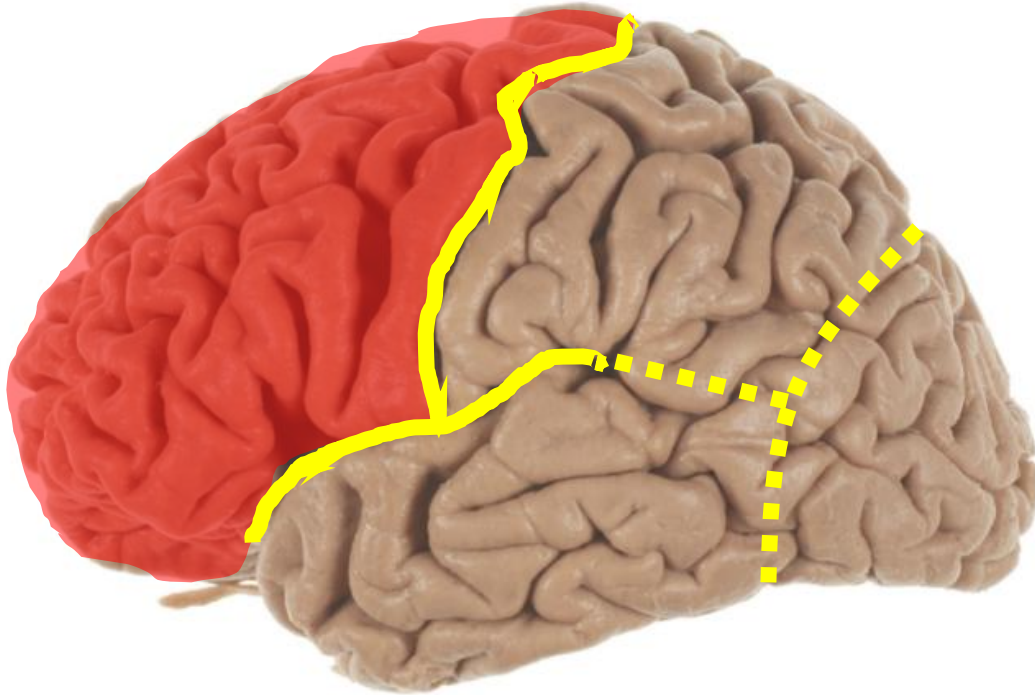
- Gyri and sulci are the two main types of surface folds of the telencephalon
- A gyrus is an elevation of the cerebral cortex
- A sulcus is a groove or indentation that separates the gyri from each other



Lateral View of the Cerebral Hemisphere - Landmarks



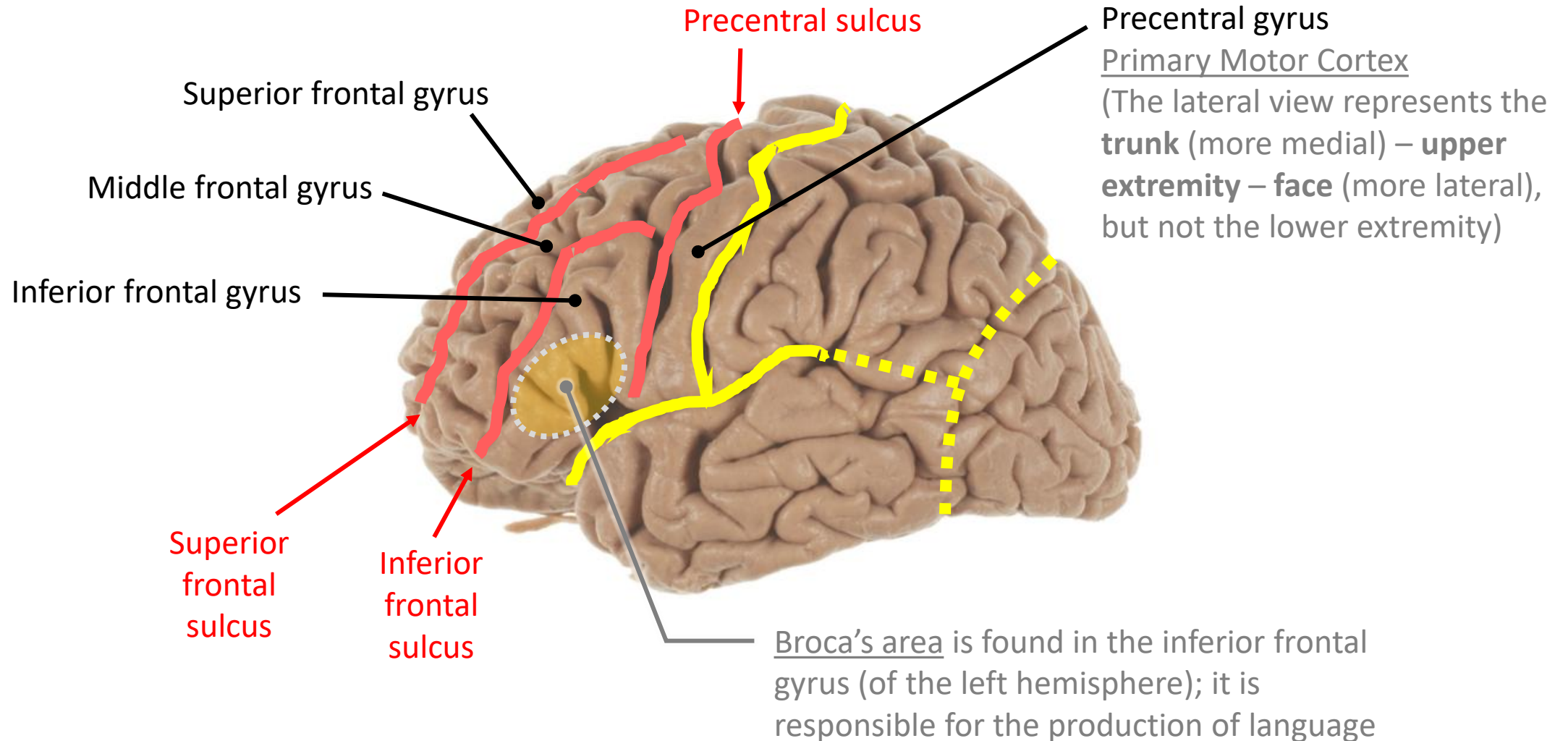
Lobes of the Telencephalon – Frontal Lobe



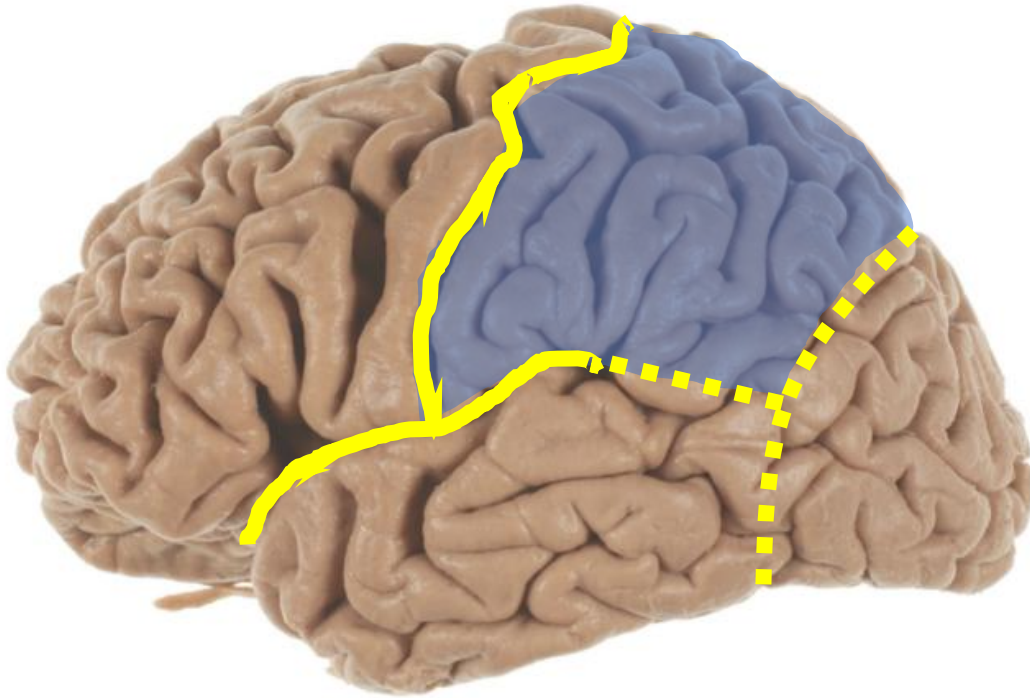
Functions of the frontal lobe:

- Voluntary movement
- Production of language
 - Predominantly left hemisphere
- Higher executive functions including
 - Planning,
 - Initiating and
 - Self-monitoring behavior

Sulci and Gyri of the Frontal Lobe (Lateral View)



Telencephalon – Parietal Lobe



Functions of the parietal lobe:

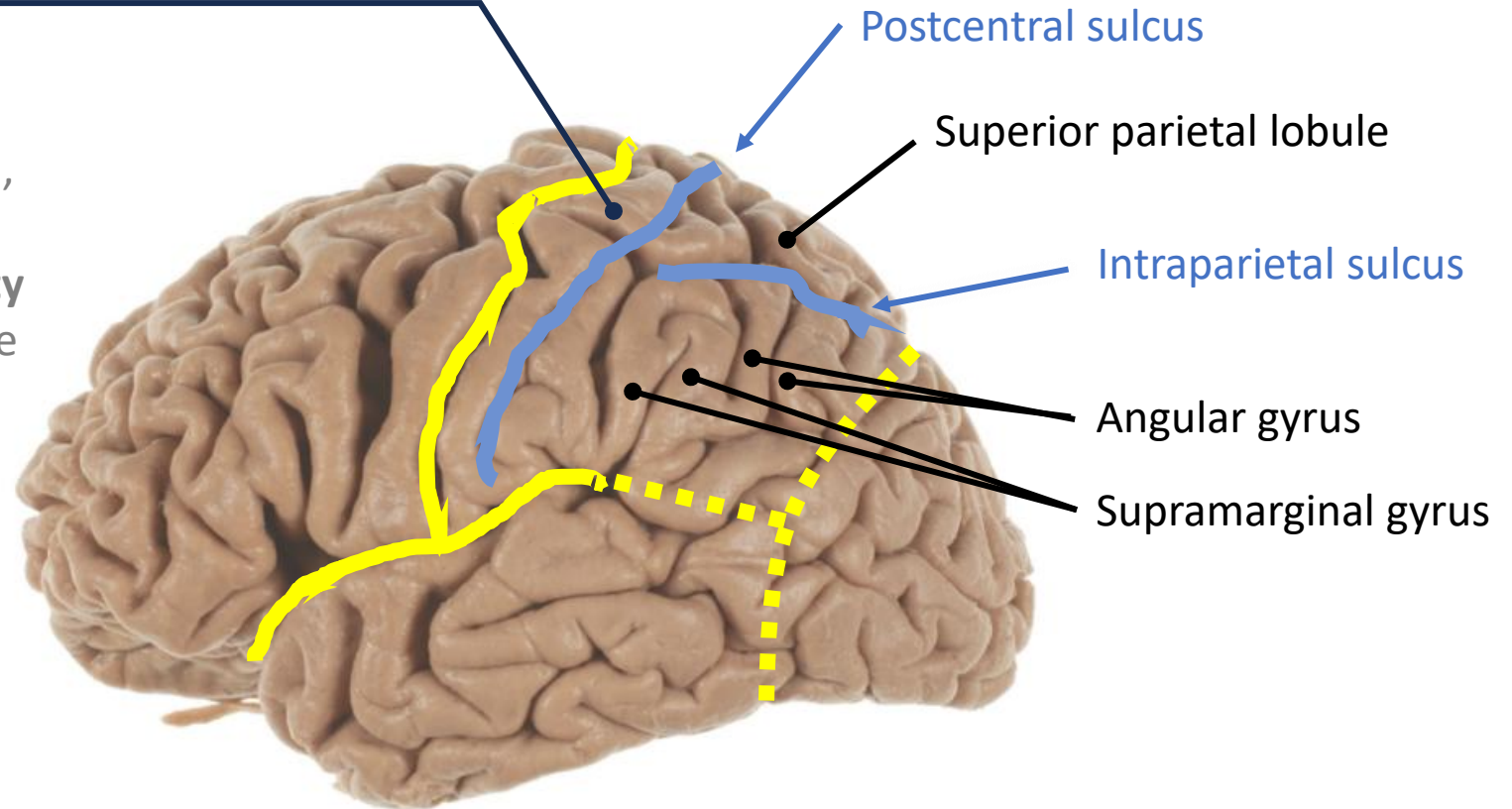
- Processing of somatosensory information
 - Touch, vibration and proprioception
 - Pain and temperature
- Sensory integration

Sulci and Gyri of the Parietal Lobe (Lateral View)

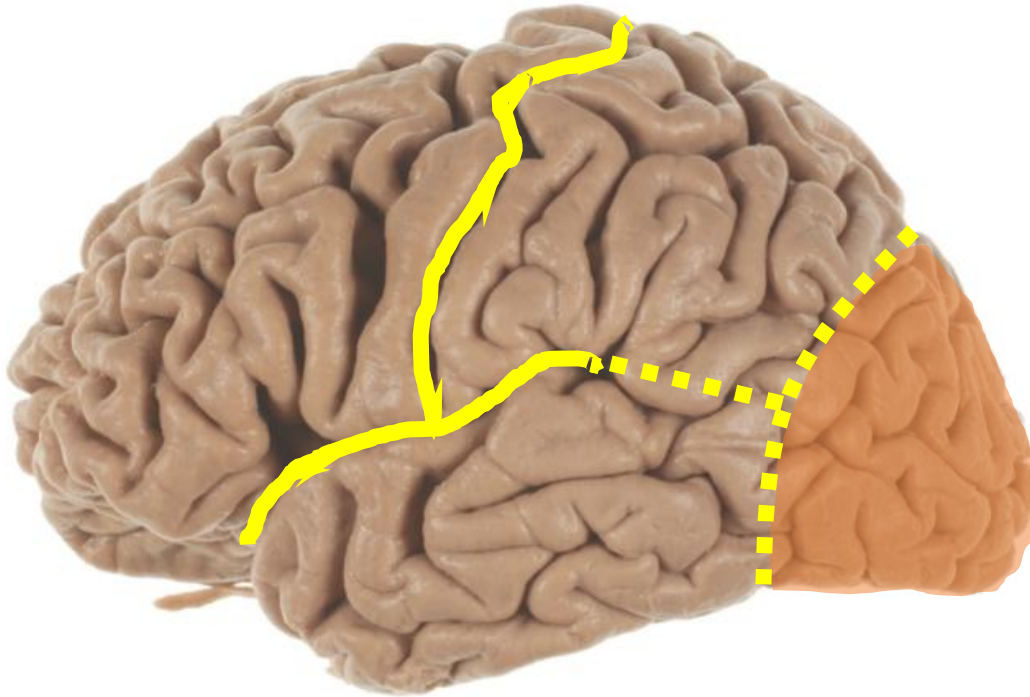
Postcentral gyrus

Primary Somatosensory Cortex

(The lateral view represents touch, vibration, proprioception, pain and temperature of **trunk** (more medial) – **upper extremity** – **face** (more lateral), but not the lower extremity)



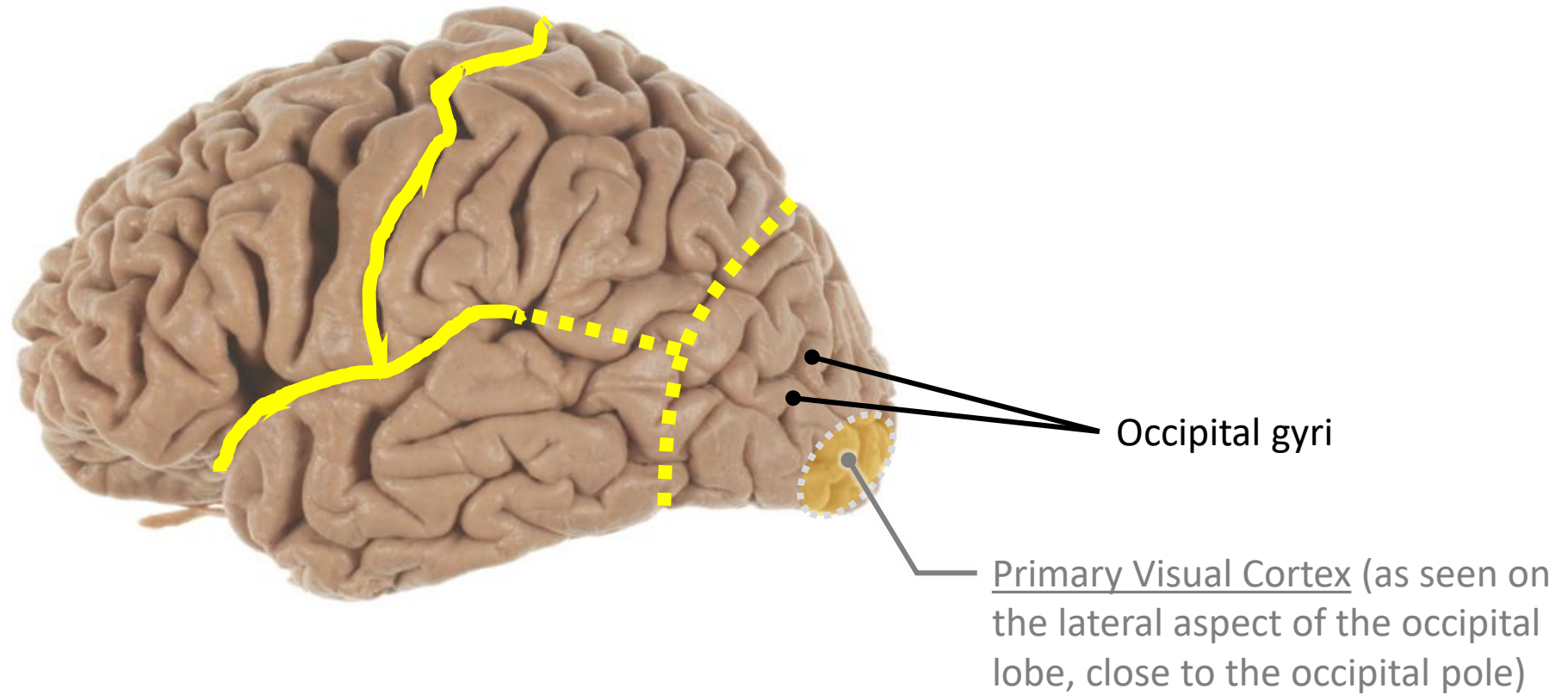
Telencephalon – Occipital Lobe



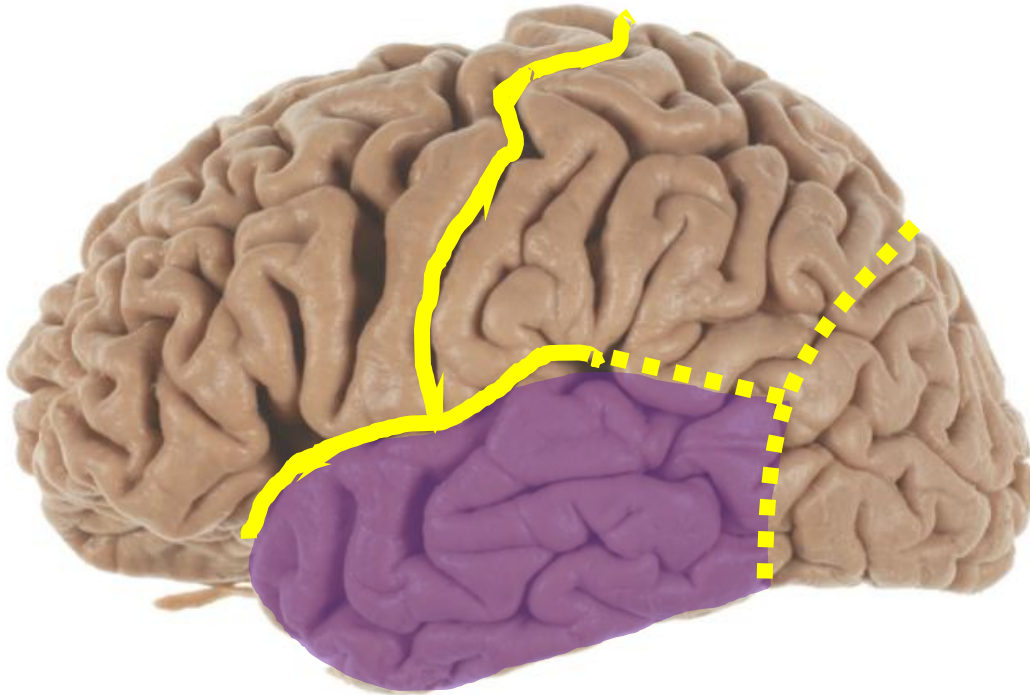
Functions of the occipital lobe:

- Processing of visual information, including
 - Color
 - Shape
 - Object recognition
 - Spatial position
 - Movement

Sulci and Gyri of the Occipital Lobe – Lateral View



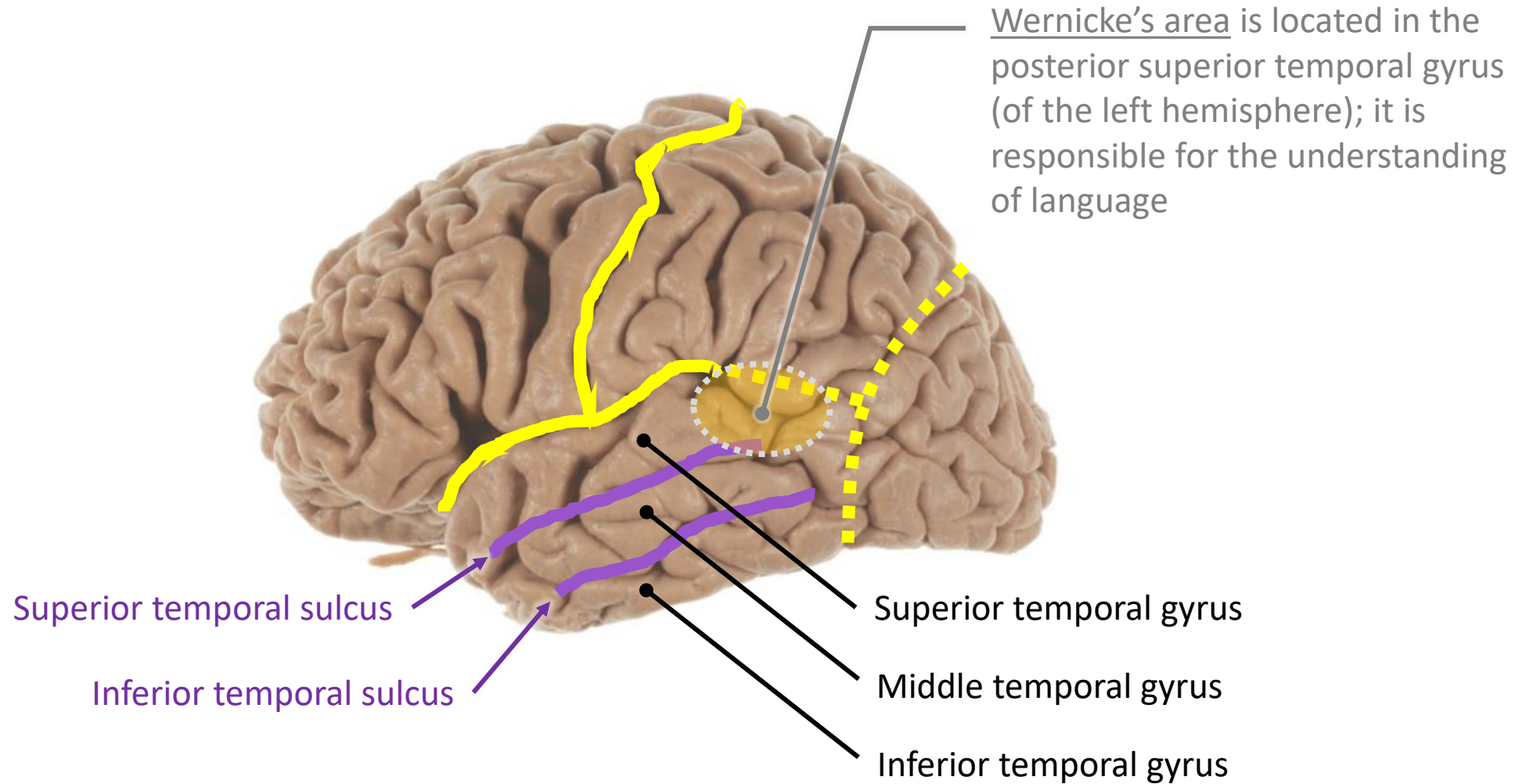
Telencephalon – Temporal Lobe



Functions of the temporal lobe:

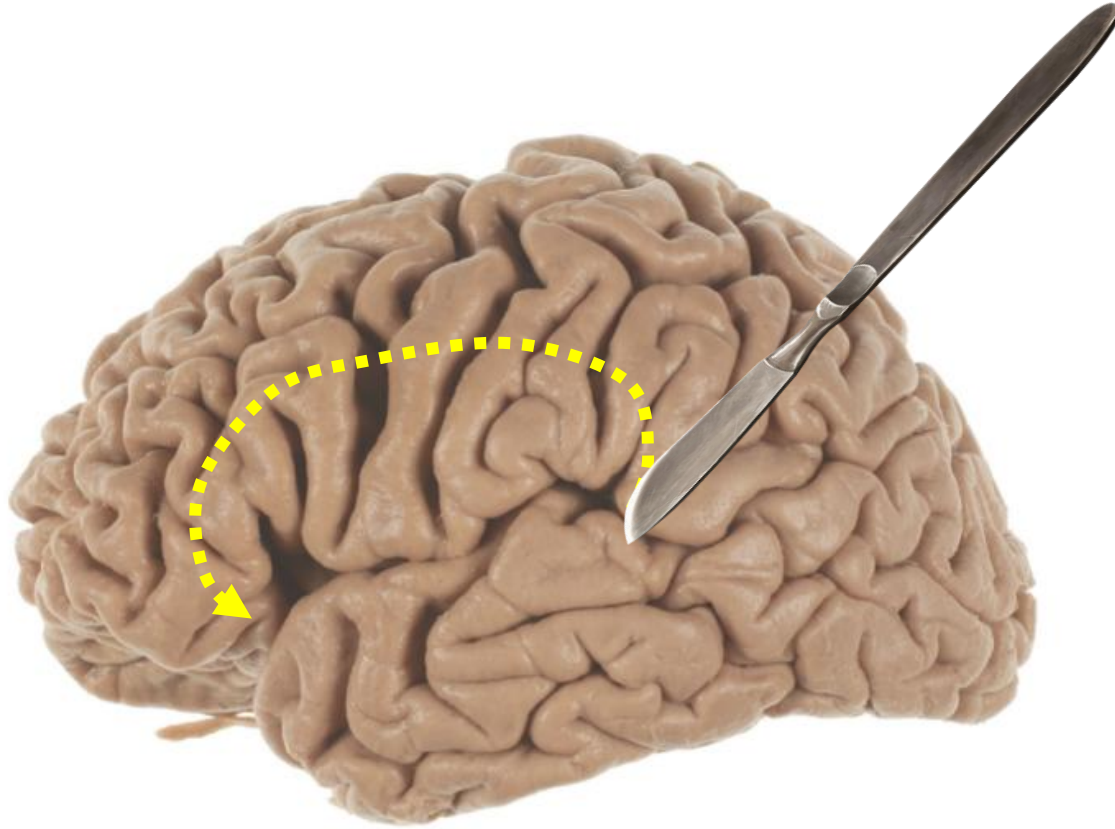
- Processing of auditory information (hearing)
- Understanding of language
 - Predominantly left hemisphere
- Memory

Sulci and Gyri of the Temporal Lobe (Lateral View)

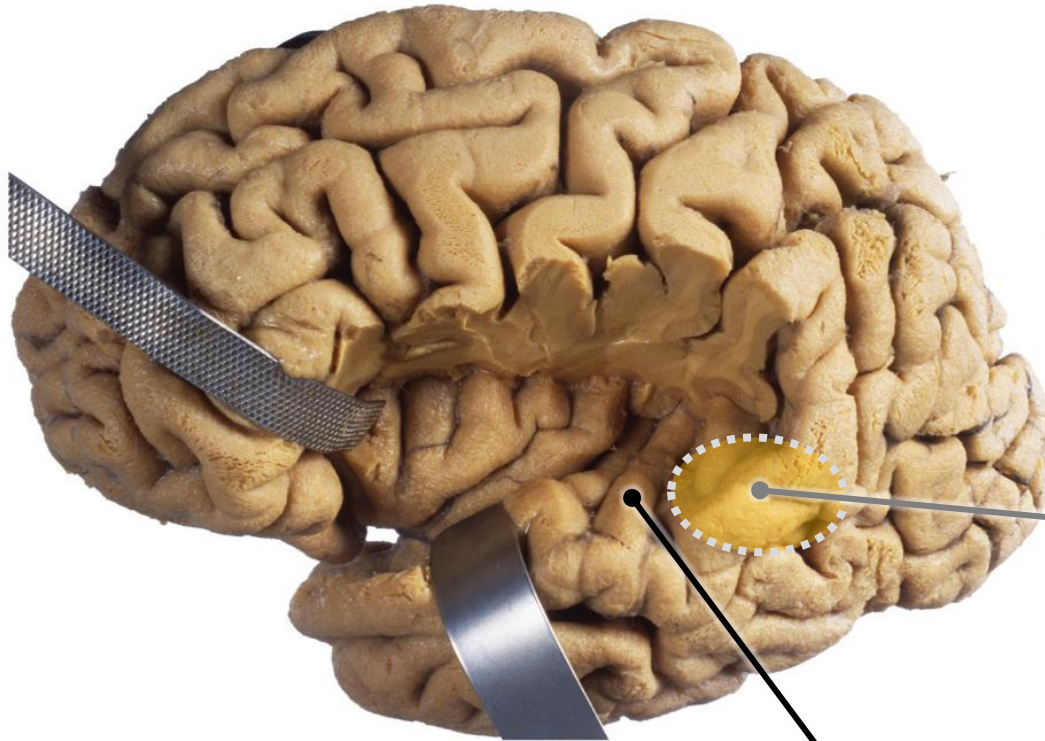


Upcoming Slides:

Parts of the Frontal and Parietal Lobes Removed

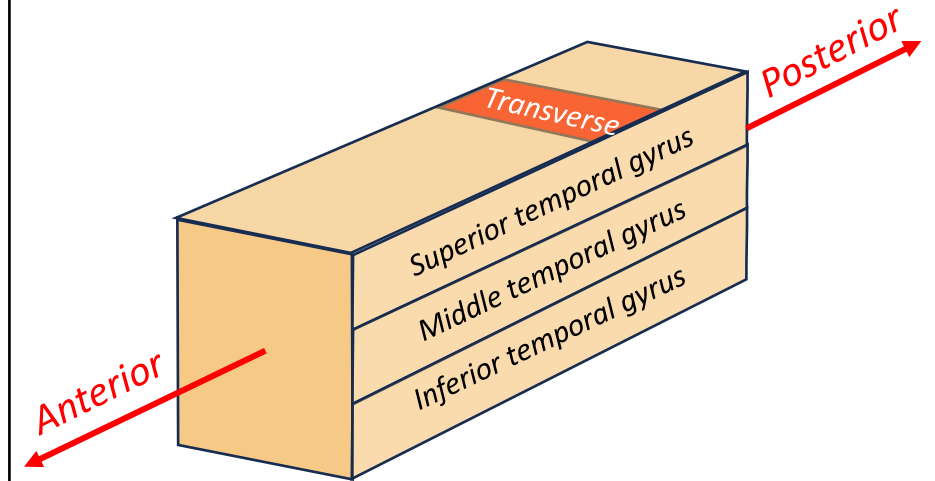


Telencephalon – Temporal Lobe



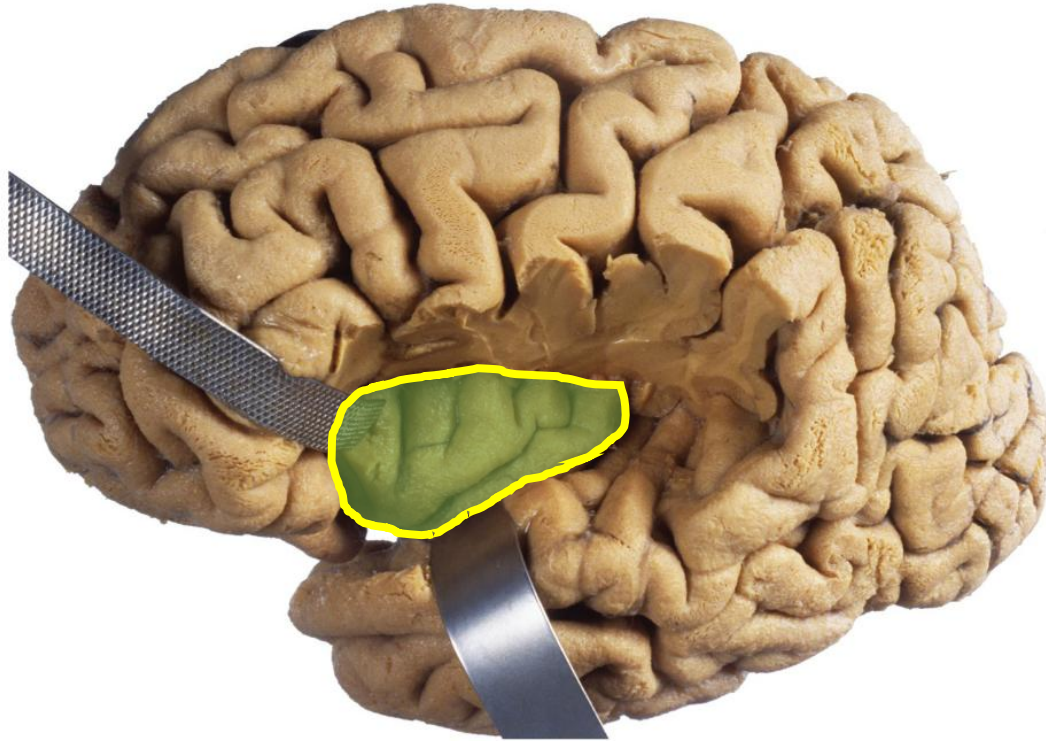
Transverse temporal gyrus of Heschl
(Primary auditory cortex)

Orientation of the transverse temporal gyrus:



Wernicke's area is located in the posterior superior temporal gyrus (of the left hemisphere); it is responsible for the understanding of language

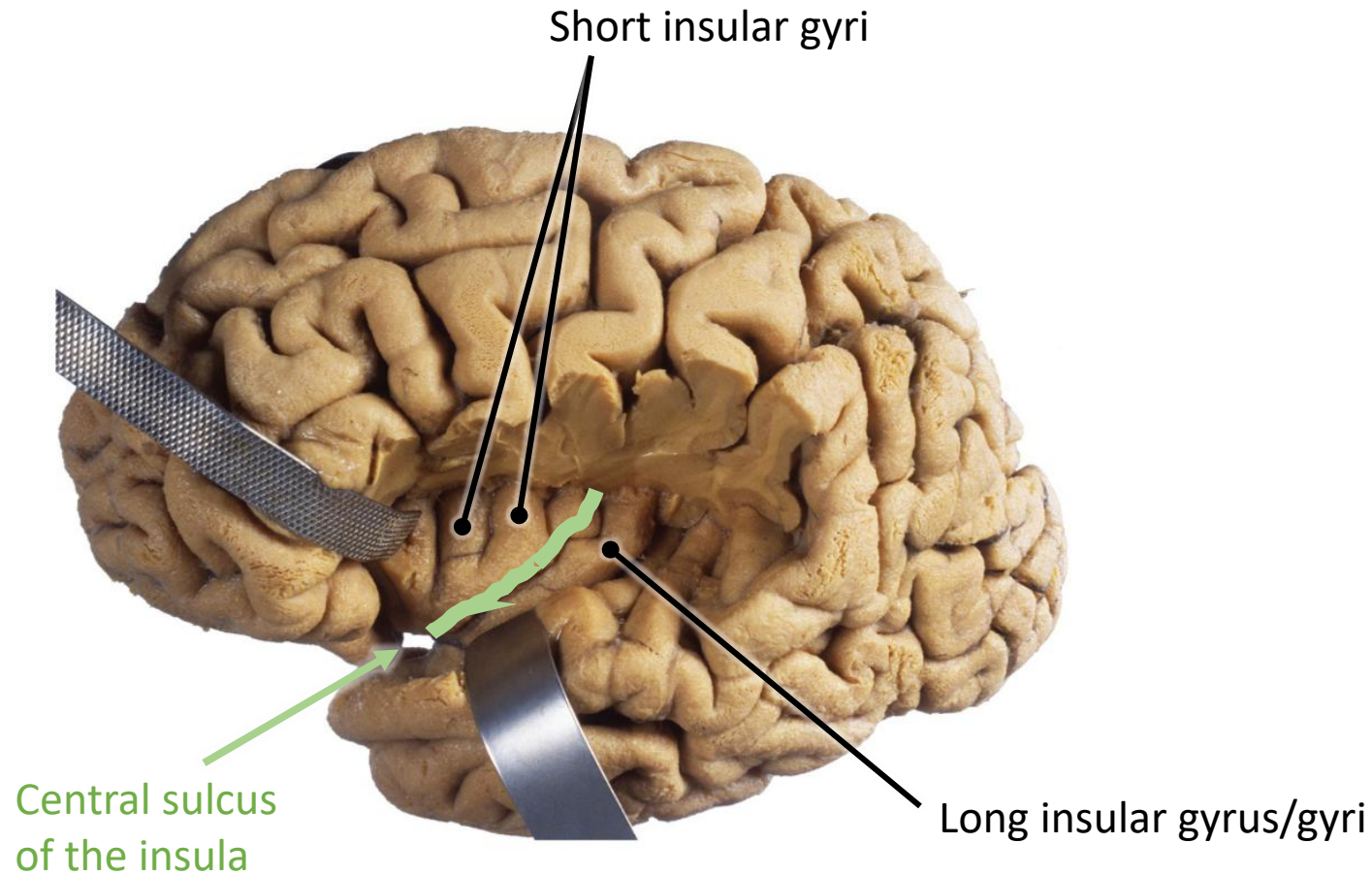
Telencephalon – Insular Lobe



Functions of the insular lobe:

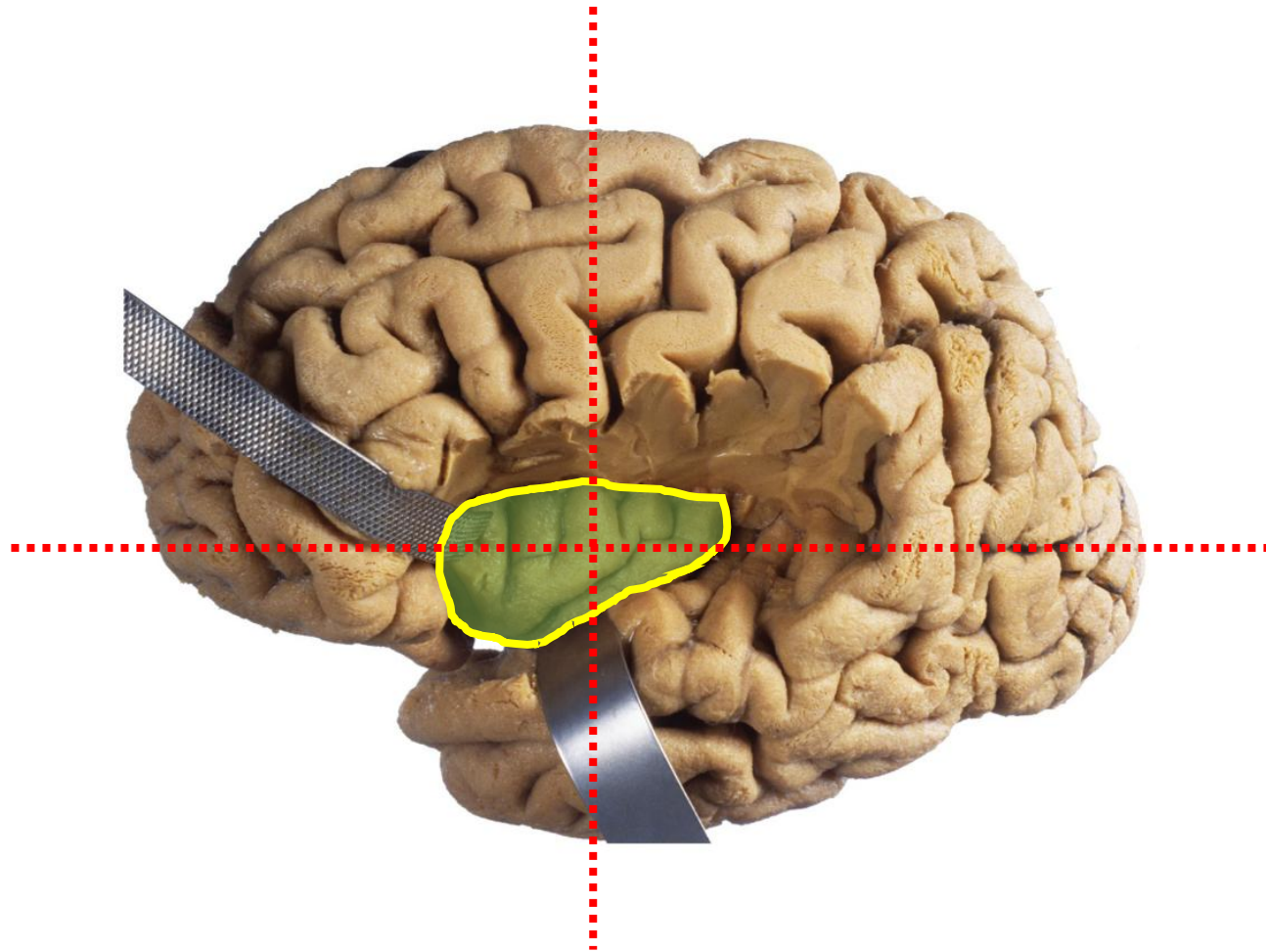
- Processing of gustatory information (taste)
- Interoception (processing of visceral sensations)

Sulci and Gyri of the Insular Lobe



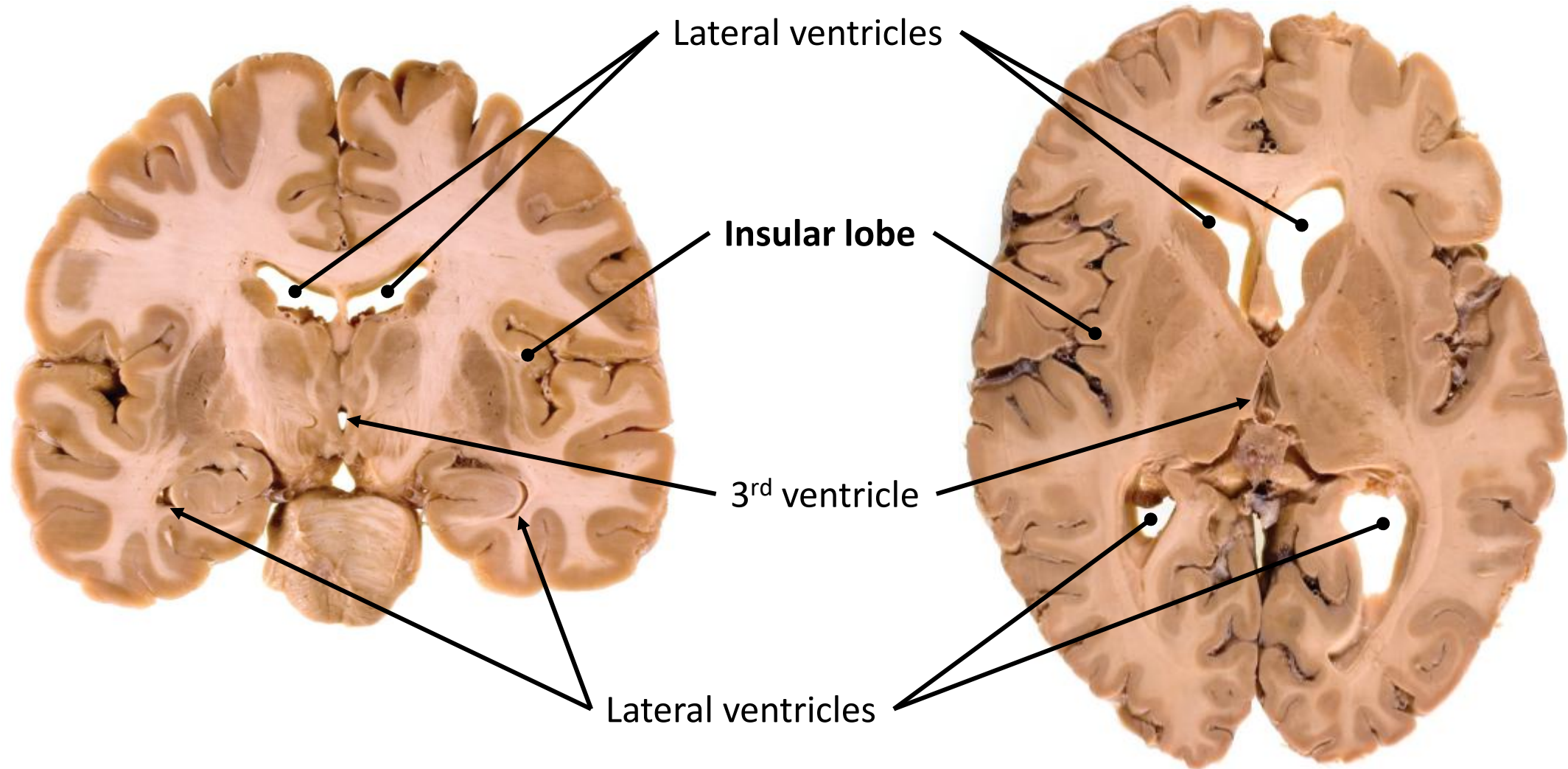
The Following Slide will show Coronal and Horizontal Sections

Coronal Section

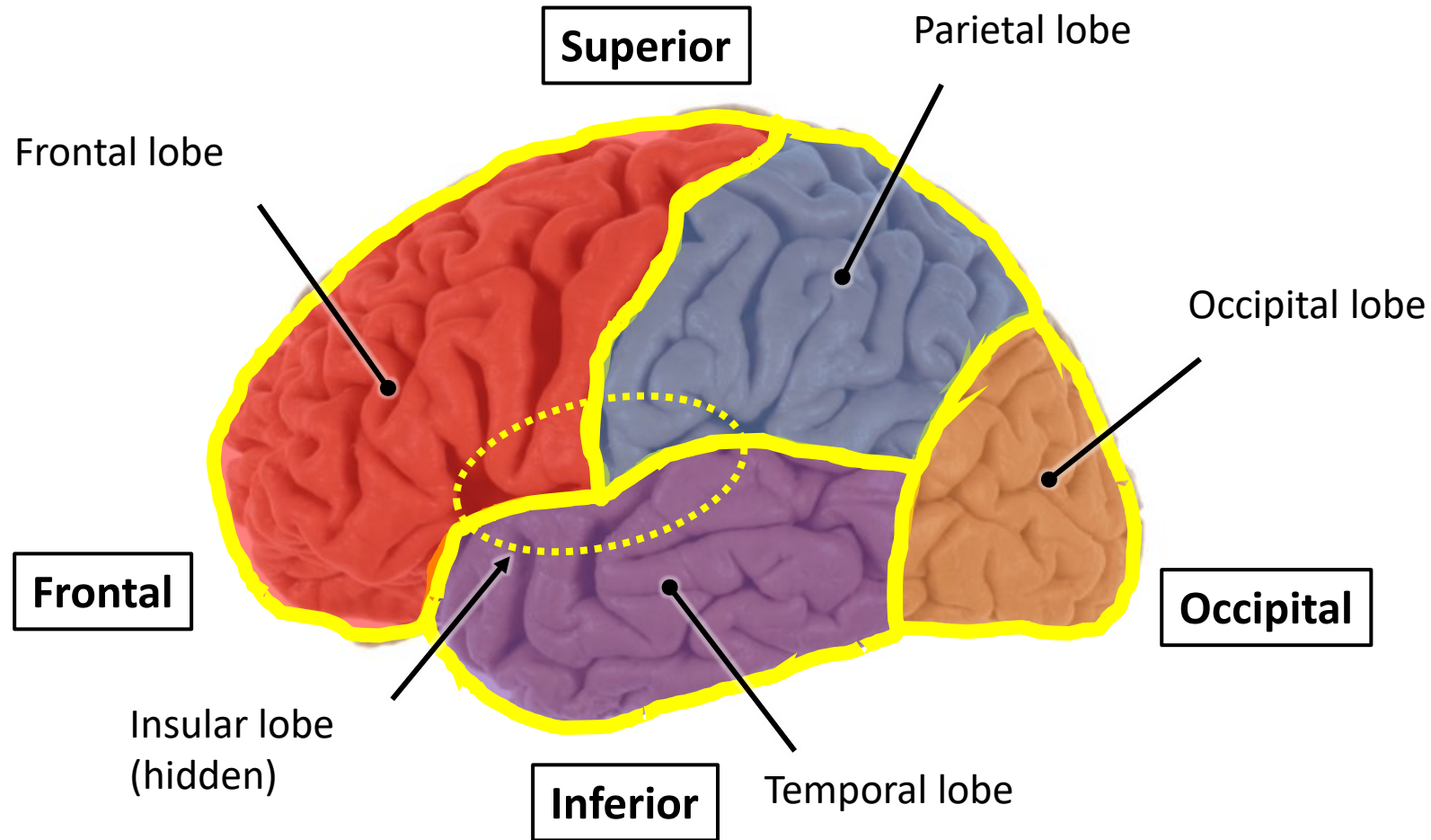


Horizontal (Axial) Section

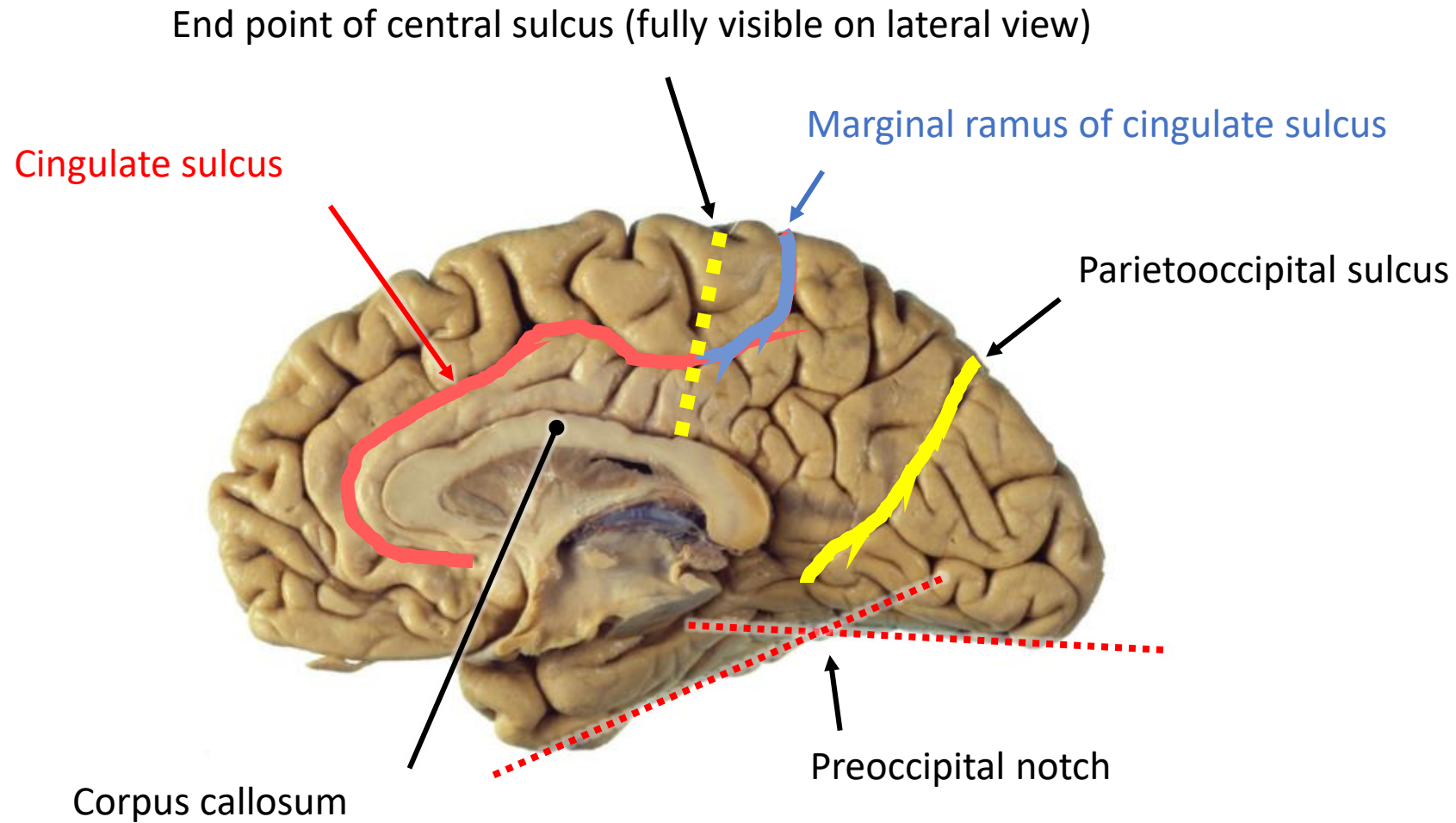
Insular Lobe in Coronal and Horizontal Sections



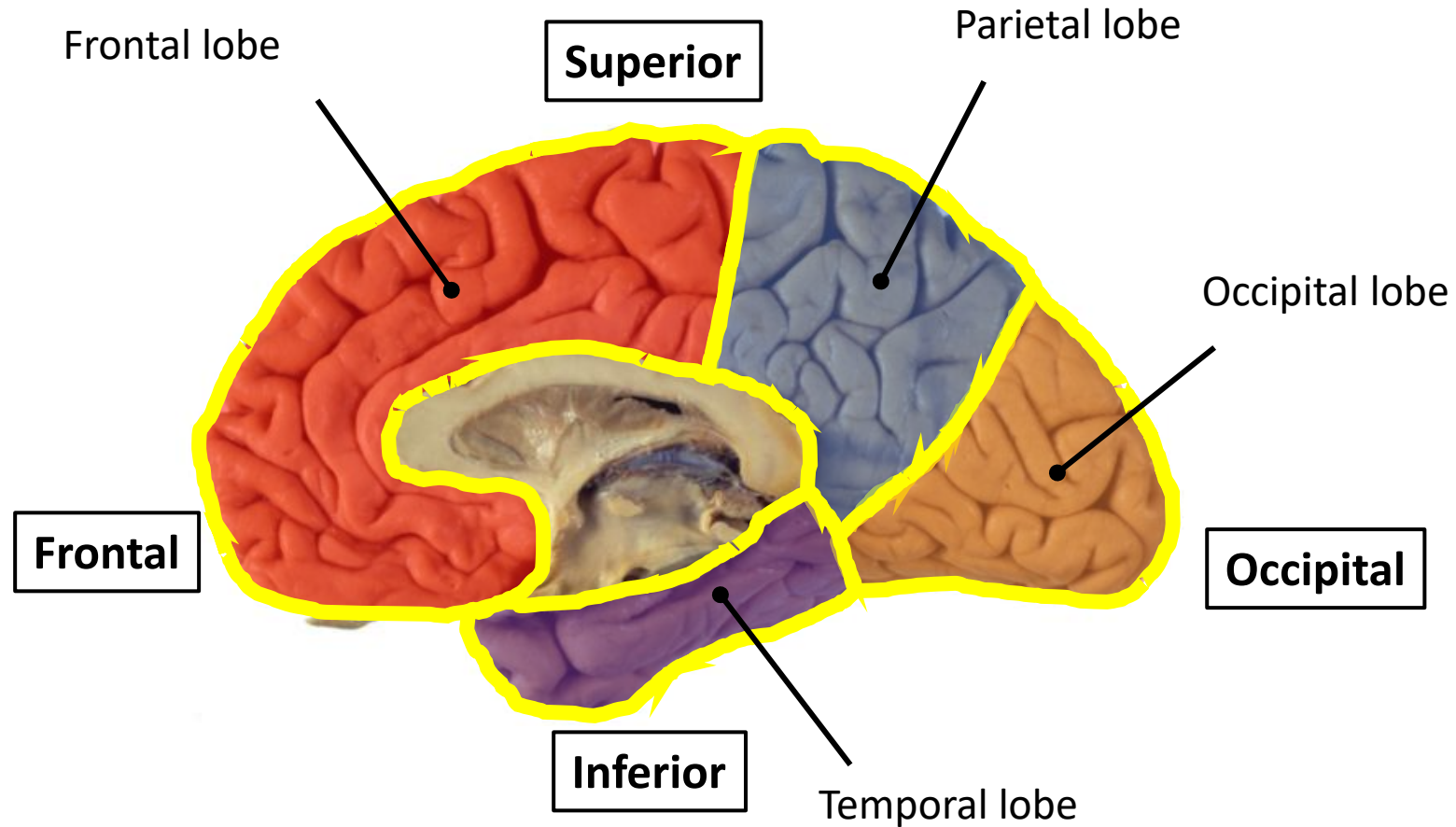
5 Lobes of the Telencephalon – Lateral View Summary



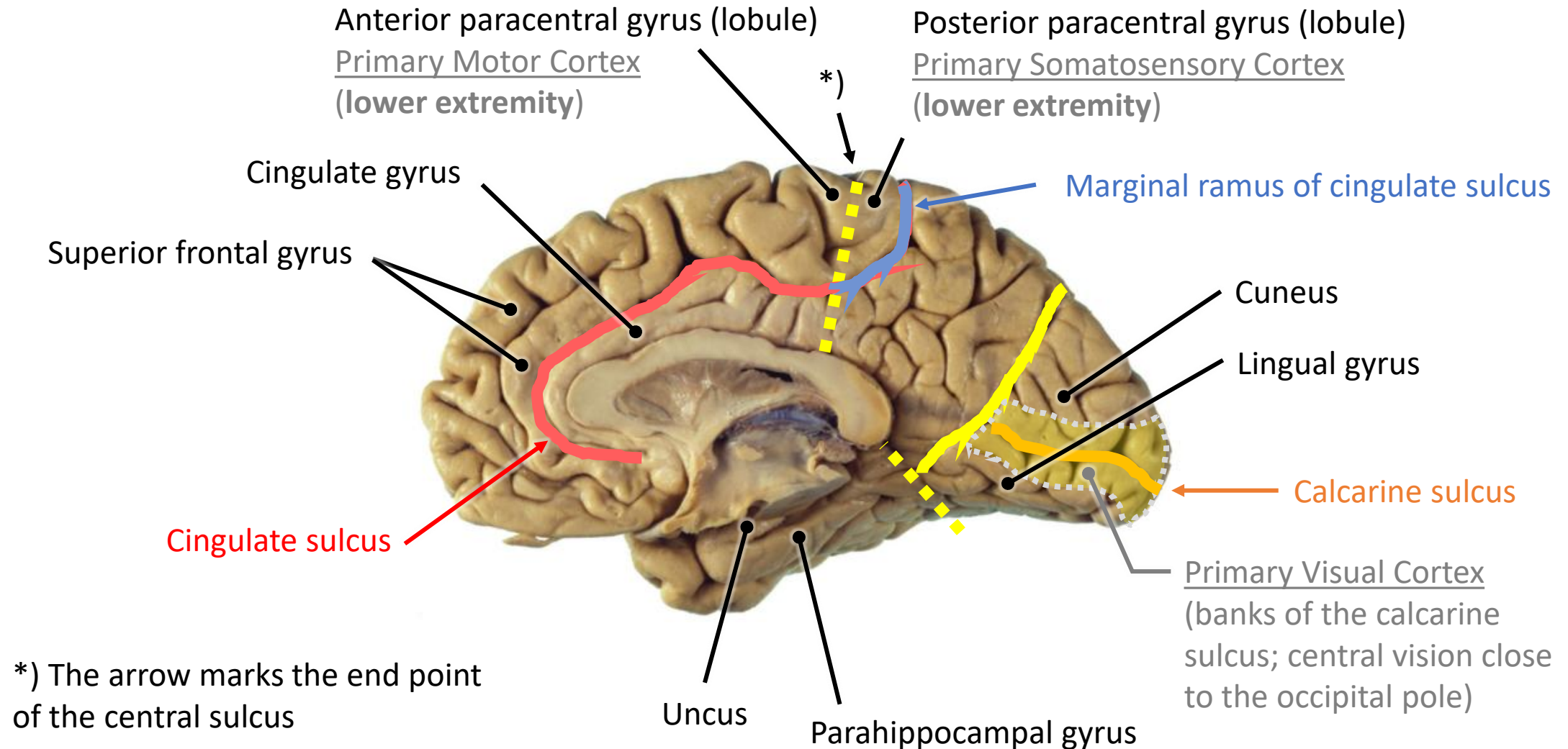
Medial View of the Cerebral Hemisphere – Landmarks



Lobes of the Telencephalon – Medial View Summary



Sulci and Gyri of the Cerebral Cortex (Medial View)



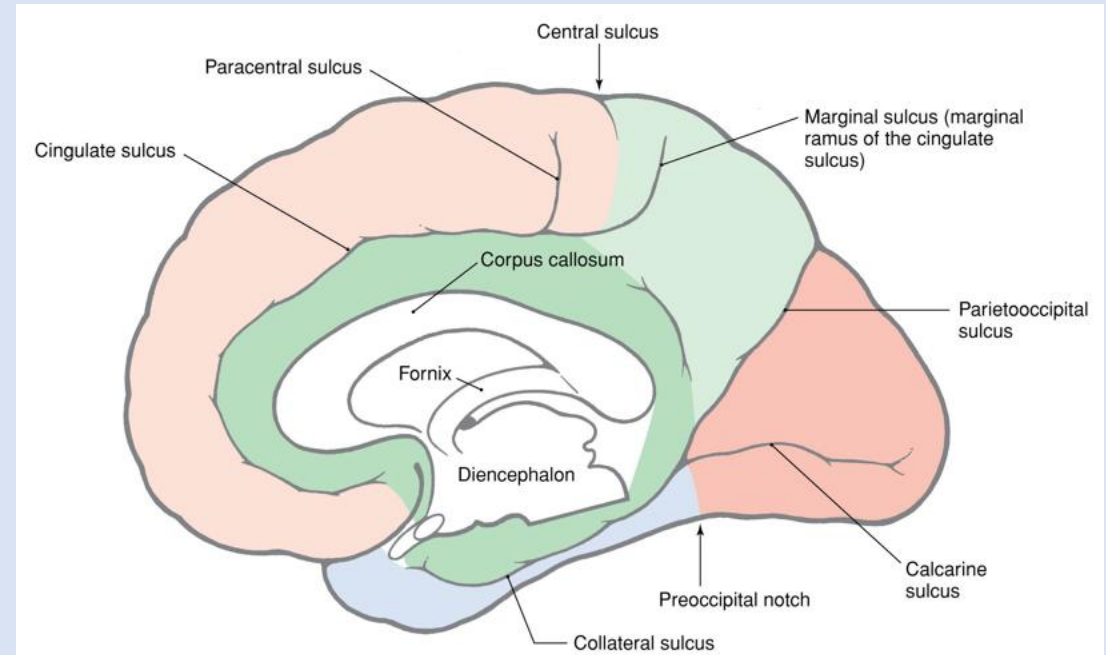
Historical Variation of the Medial View

You may come across a variation of the medial view in some textbooks, showing an additional lobe, the limbic lobe. An example is a figure from your Haines Atlas shown on the right.

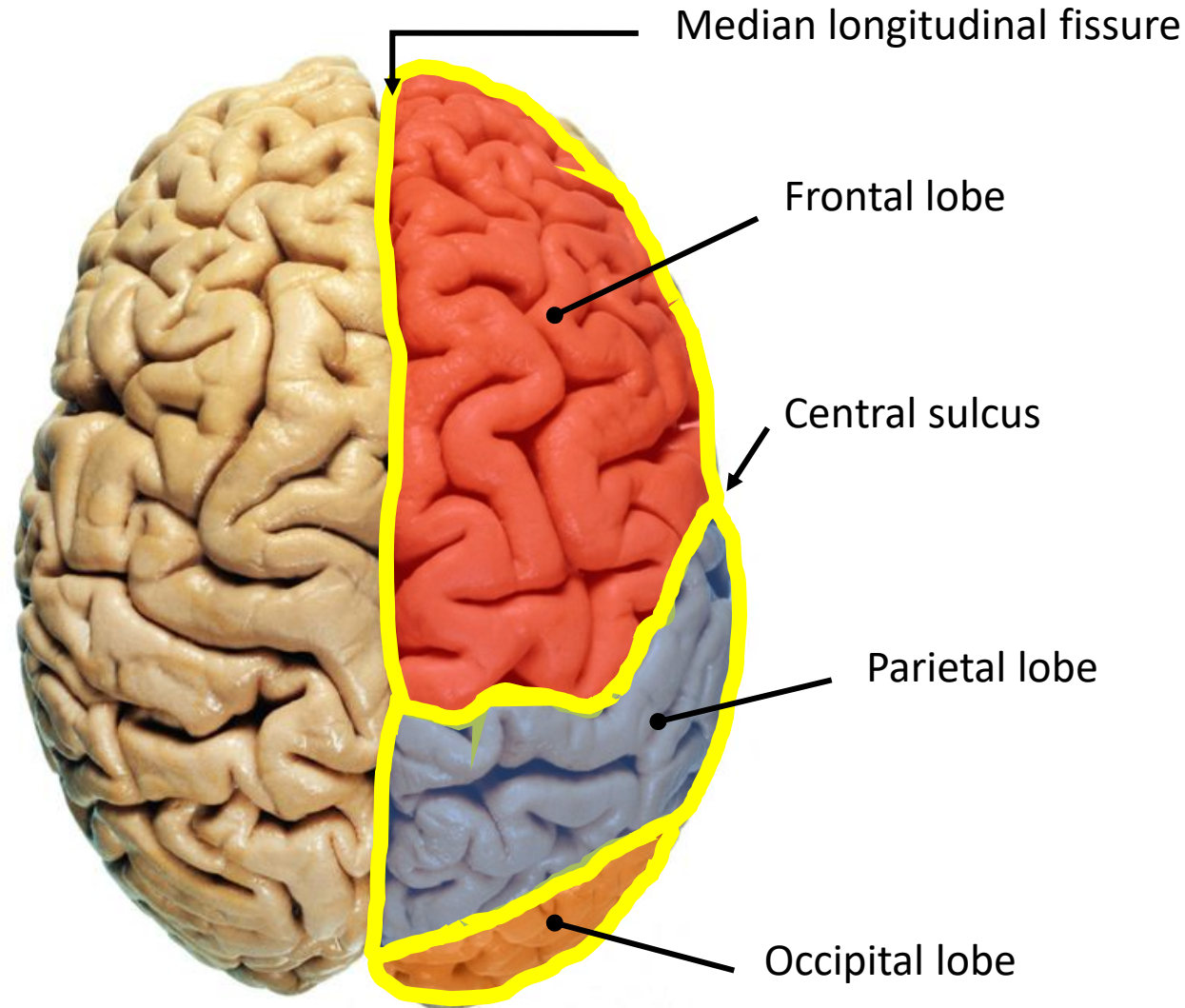
The limbic lobe was named by Paul Broca, a 19th century French physician, as a cortical region that surrounds the corpus callosum and includes elements of the frontal, parietal and temporal lobes.

At present, the term limbic lobe is no longer considered an accurate term. Modern functional imaging has shown that the functions traditionally associated with the limbic lobe, such as emotions, could no longer be limited to this structure alone, but needed to be expanded to a wider variety of cortical and subcortical elements.

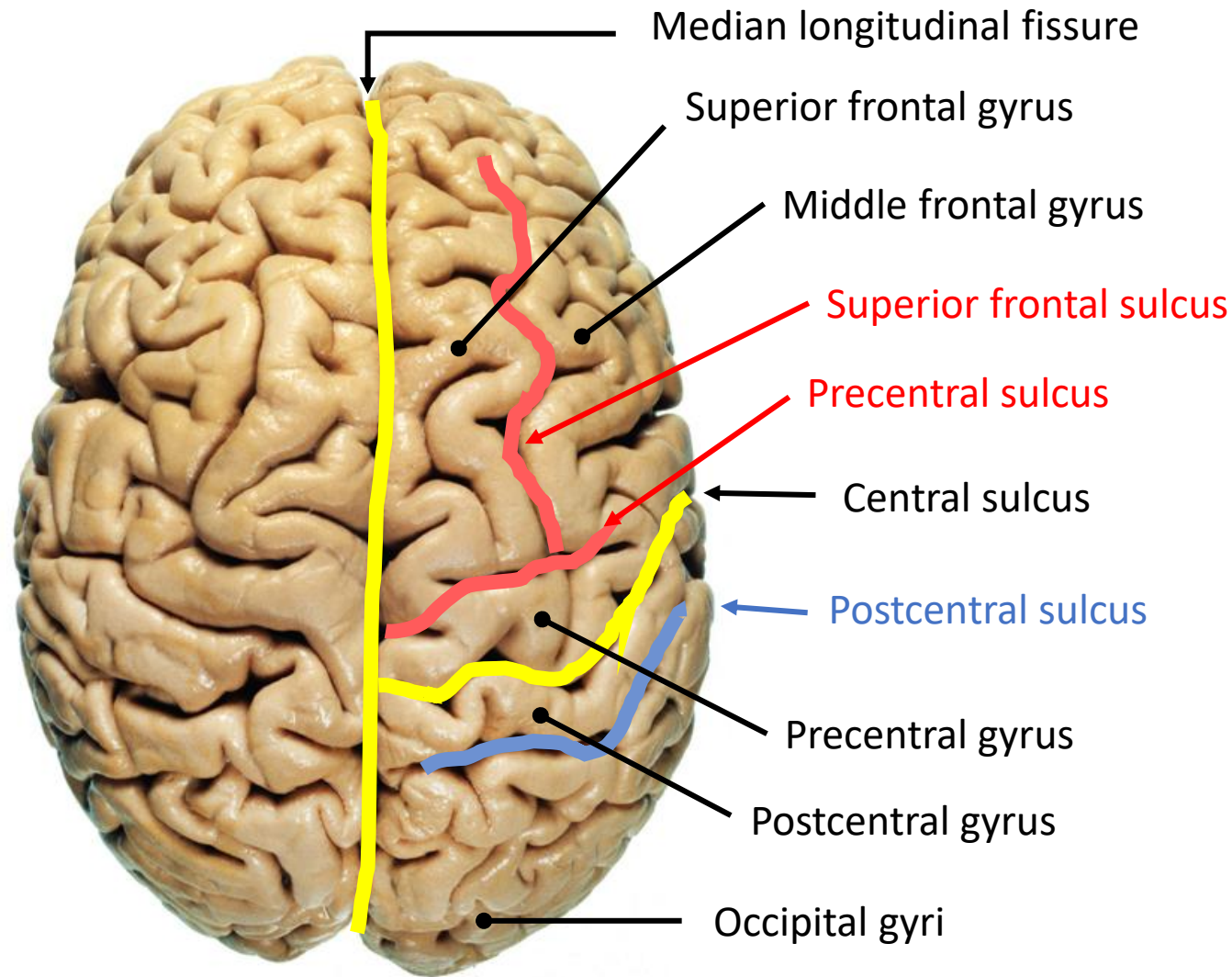
We acknowledge the relevance of the limbic lobe as a historical concept.



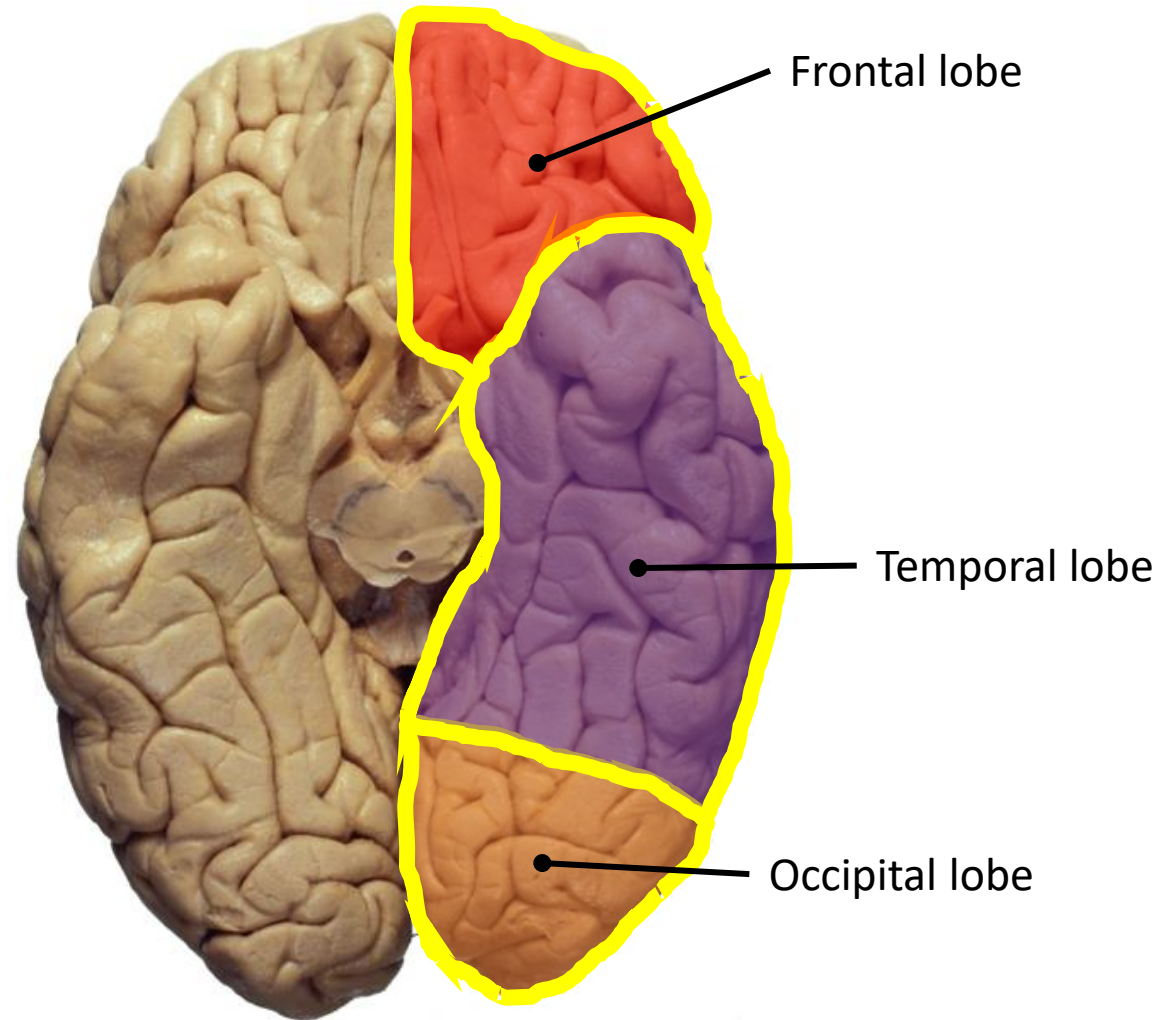
Lobes of the Telencephalon – Superior View Summary



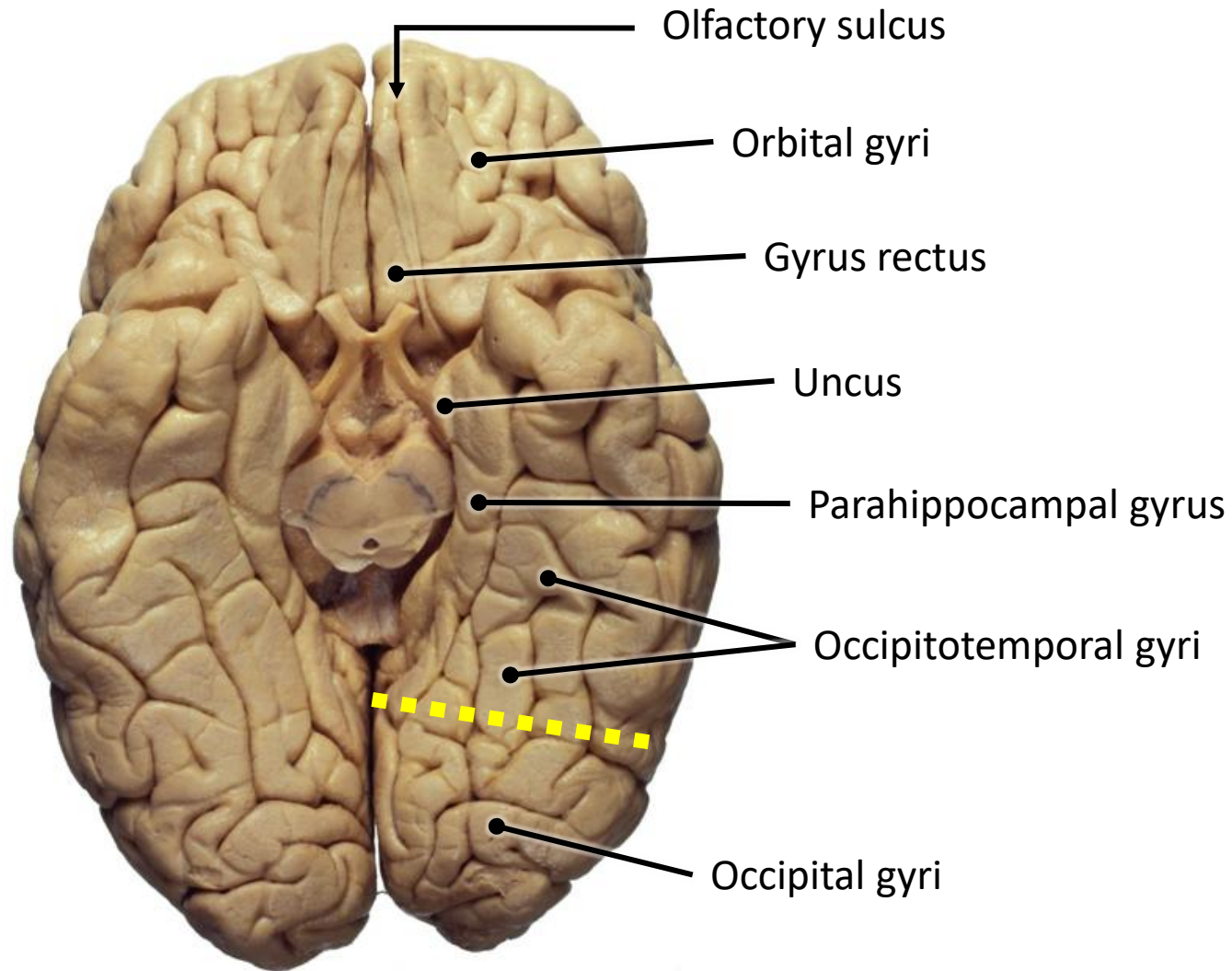
Sulci and Gyri of the Cerebral Hemispheres (Superior View)



Lobes of the Telencephalon – Inferior View (Brainstem and Cerebellum Removed)

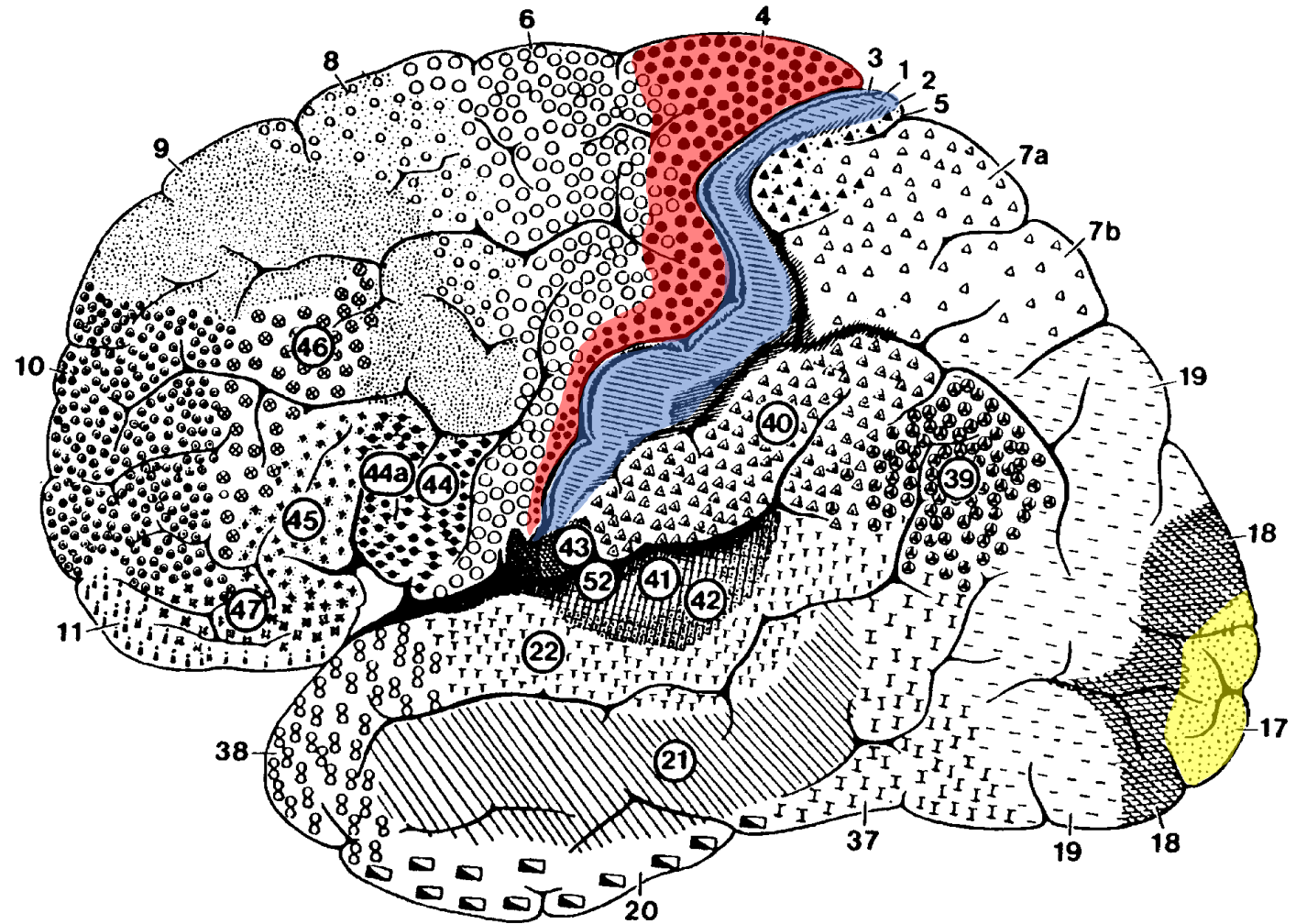


Sulci and Gyri of the Cerebral Hemispheres (Inferior View)



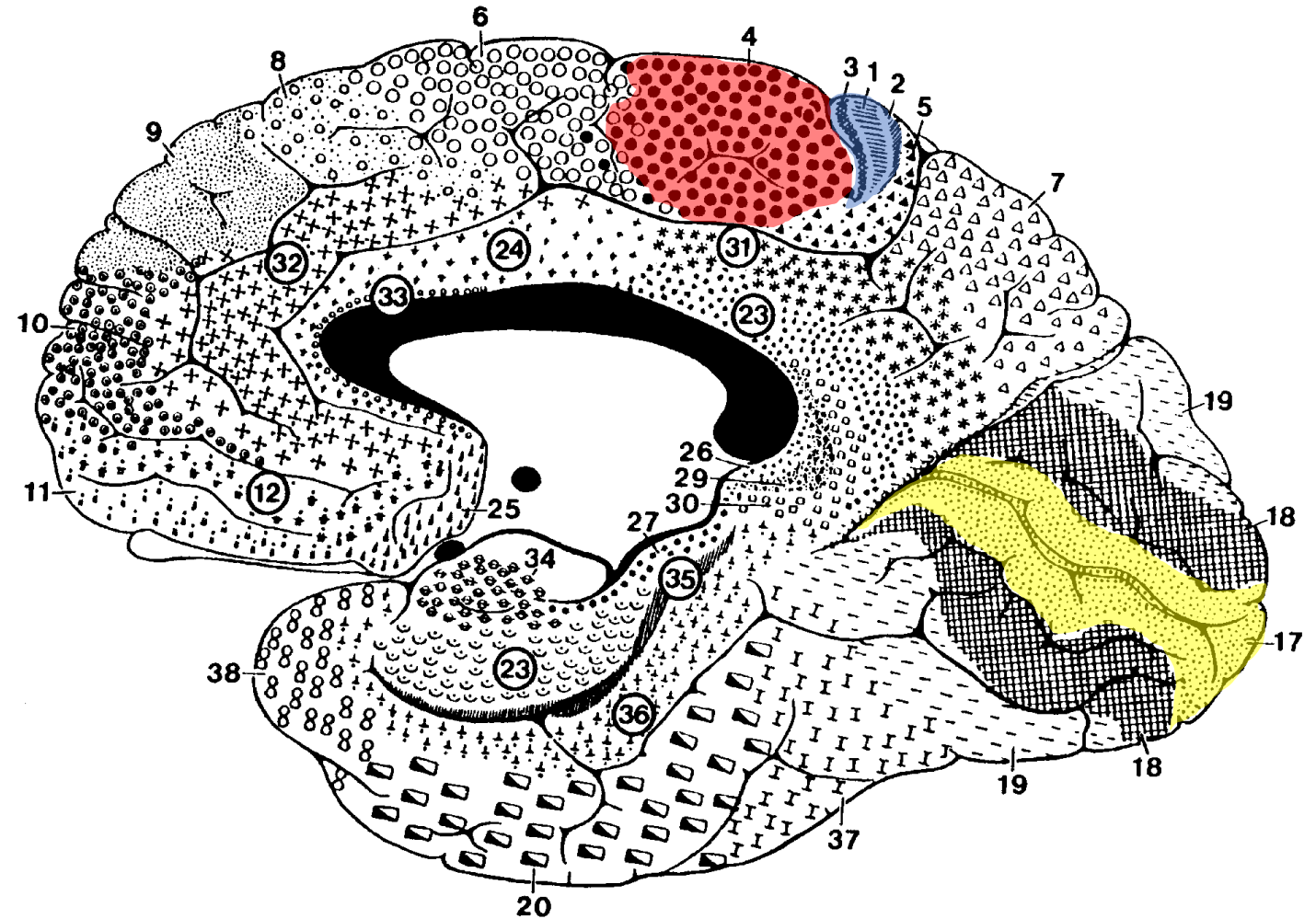
Classification of the Cerebral Cortex based on Histology: Brodmann Areas (Lateral View)

- Korbinian Brodmann (1868-1918), classified the cerebral cortex into more than 50 areas, based on histological similarities
- At the time, he did not know the functional significance of these areas
- The classification is still in use, since many of the areas he described have been linked to specific functions



Classification of the Cerebral Cortex based on Histology: Brodmann Areas (Medial View)

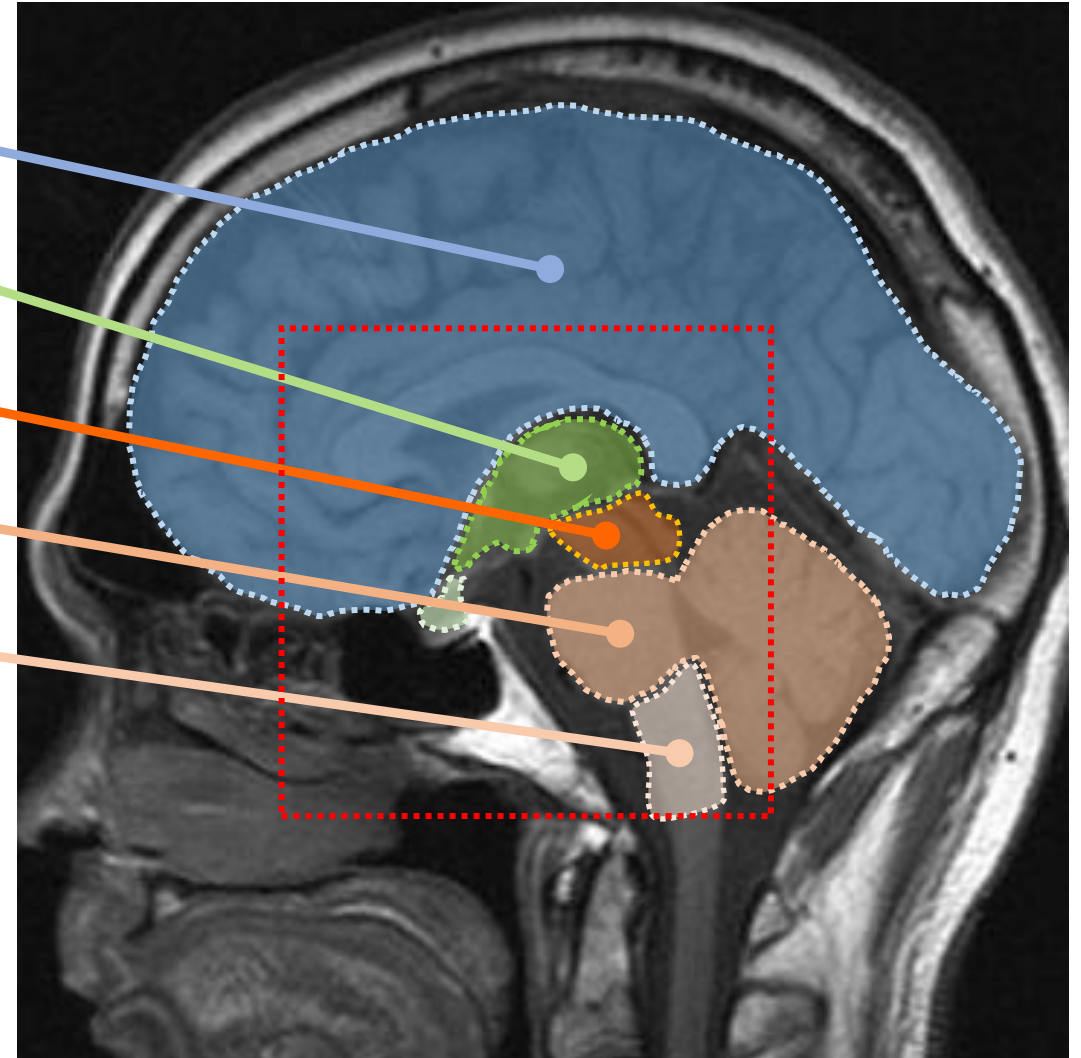
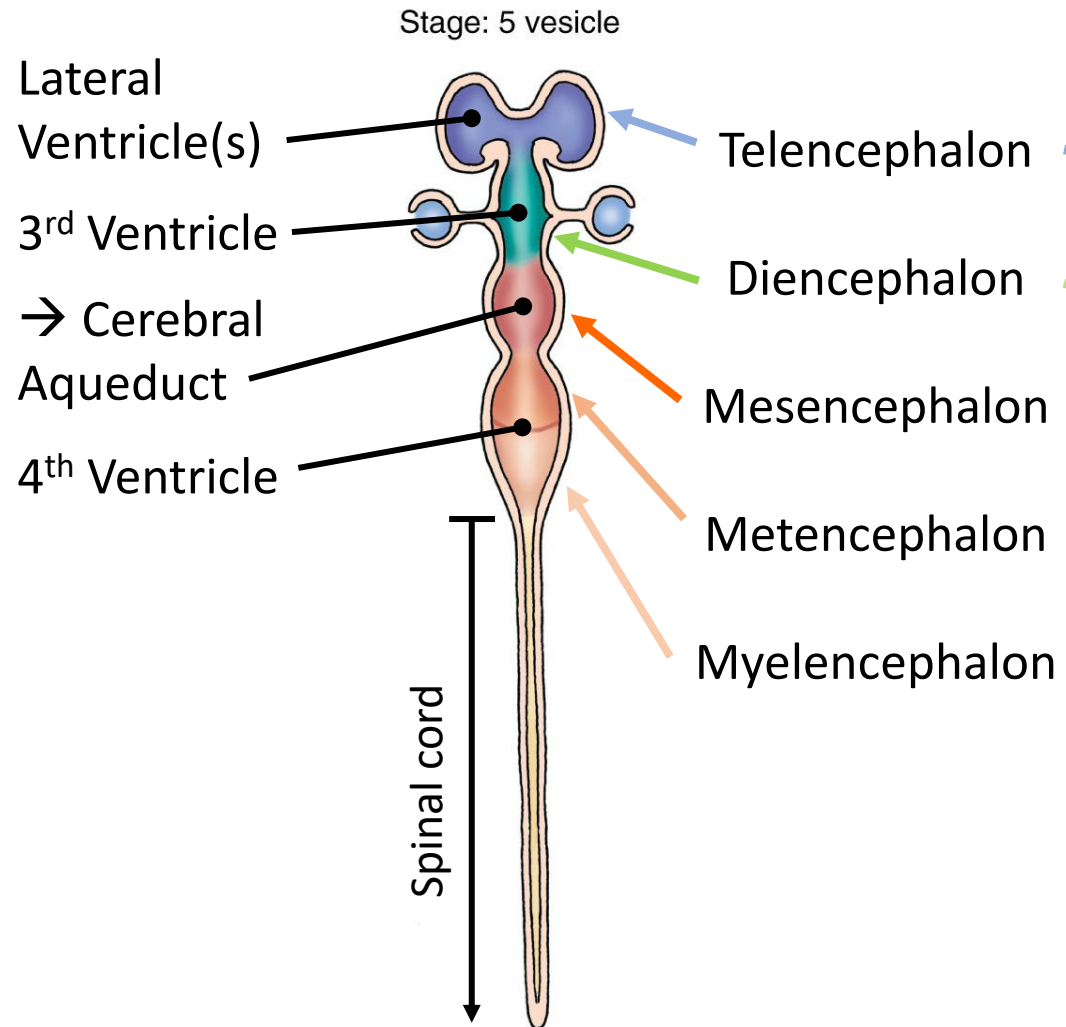
- Examples of the functional significance of Brodmann areas:
 - Area 4: primary motor cortex
 - Areas 1, 2, 3: primary somatosensory cortex
 - Area 17: primary visual cortex



Learning Objectives

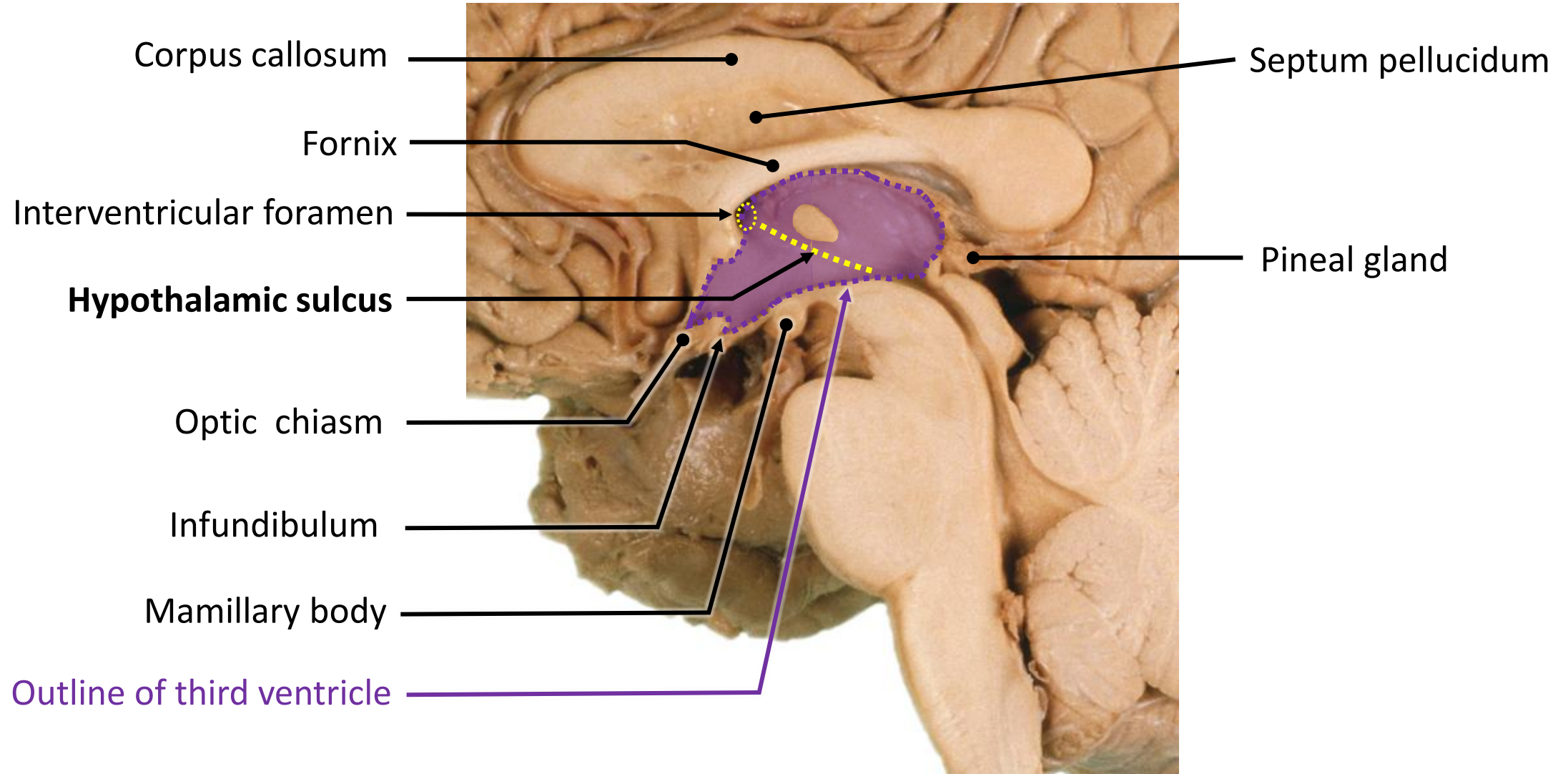
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Five Parts of the Brain



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Medial View Landmarks / Reference to Diencephalon



Diencephalon

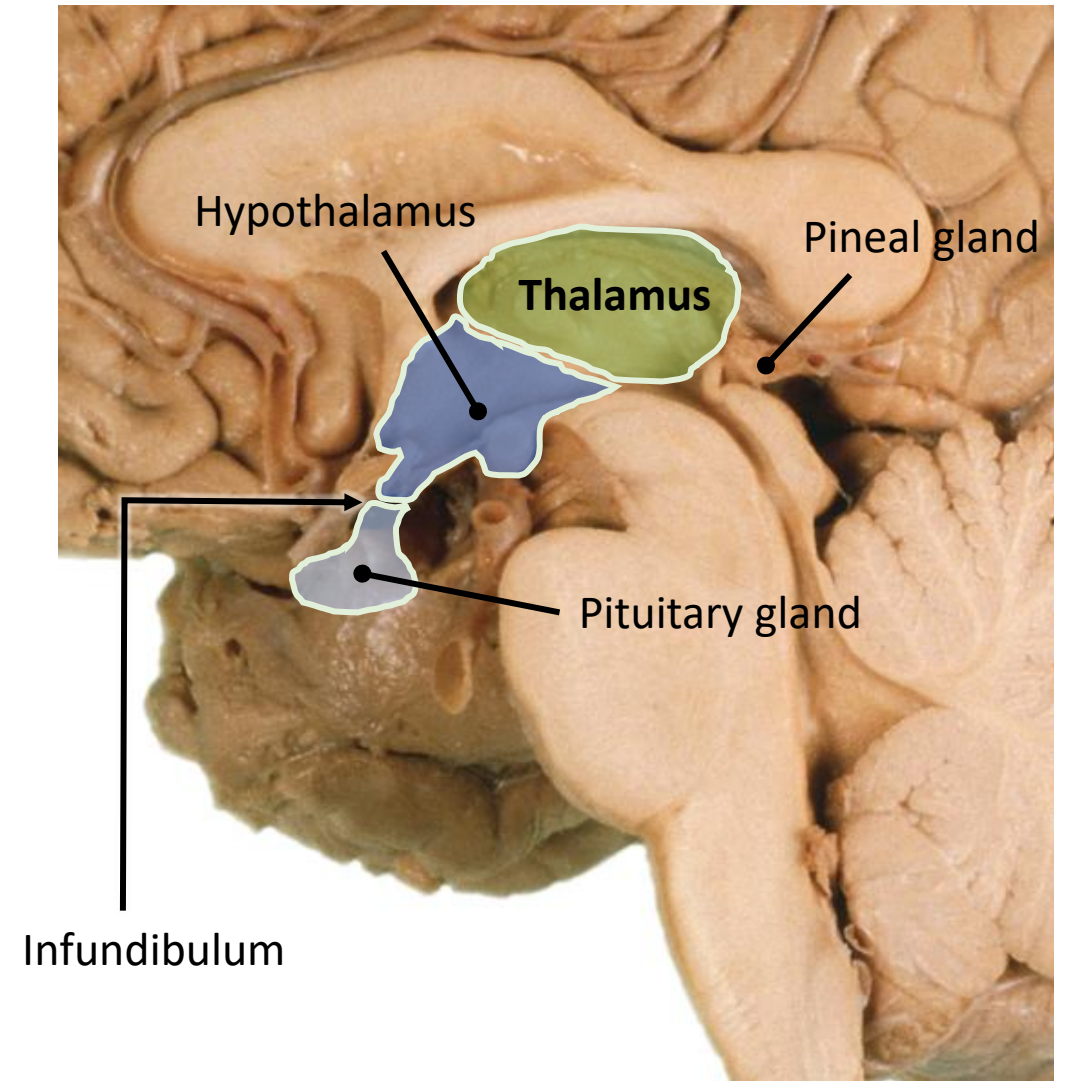
- **Thalamus**

- **Location:**

- above the hypothalamic sulcus in each hemisphere, lateral to the third ventricle

- **Function:**

- Relaying sensory information (except smell) to the sensory cortex areas
 - Modulating movement



Diencephalon

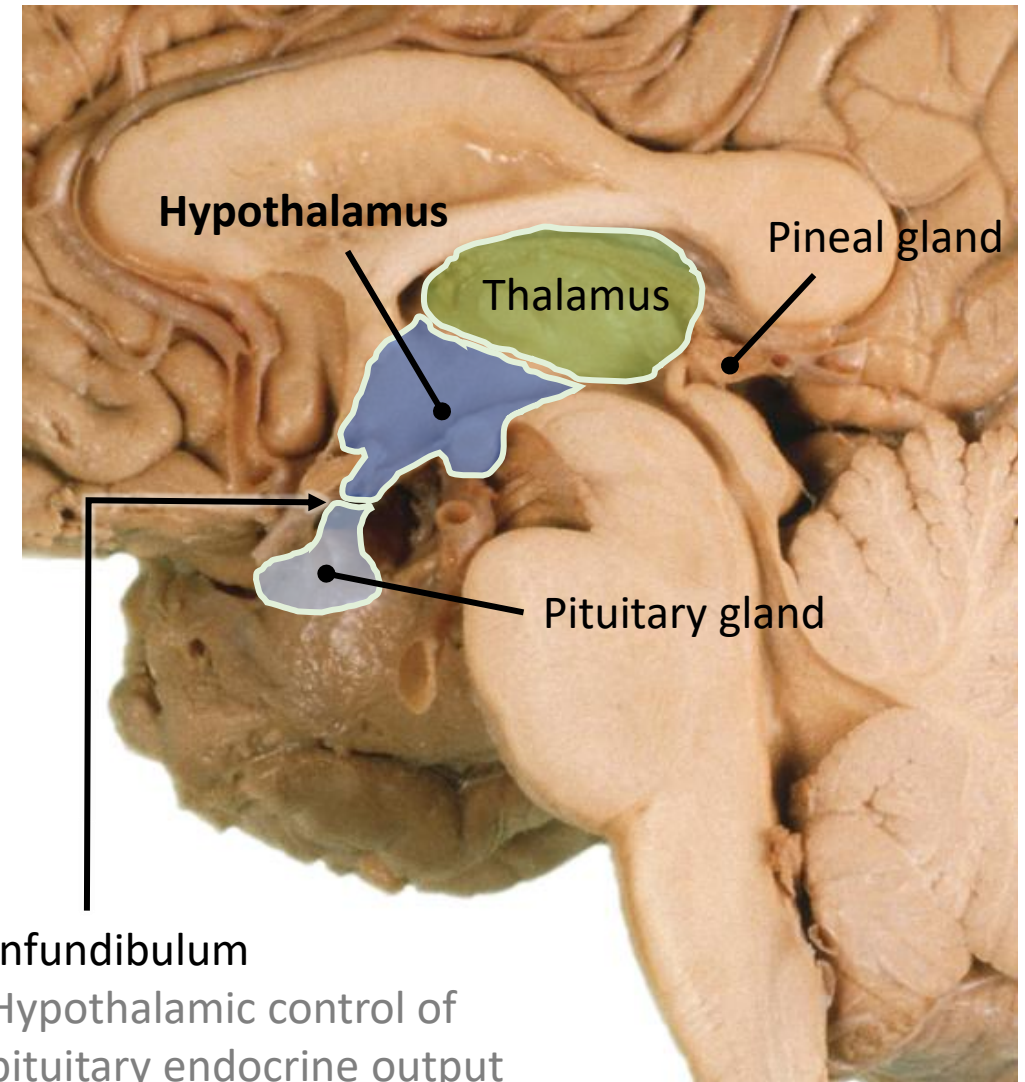
- **Hypothalamus**

- **Location:**

- below the hypothalamic sulcus and above the midbrain, lateral to the third ventricle

- **Functions:**

- **Homeostasis** of vital functions, including body temperature
 - **Neuroendocrine control** of hormones of the pituitary gland
 - Regulation of **eating** and **drinking** behaviors



Infundibulum
Hypothalamic control of
pituitary endocrine output

Diencephalon

- **Subthalamus**

- **Location:**

- Near the junction between midbrain and thalamus

- **Function:**

- Regulation of **movement** through the **subthalamic nucleus**, part of the **basal ganglia circuitry**

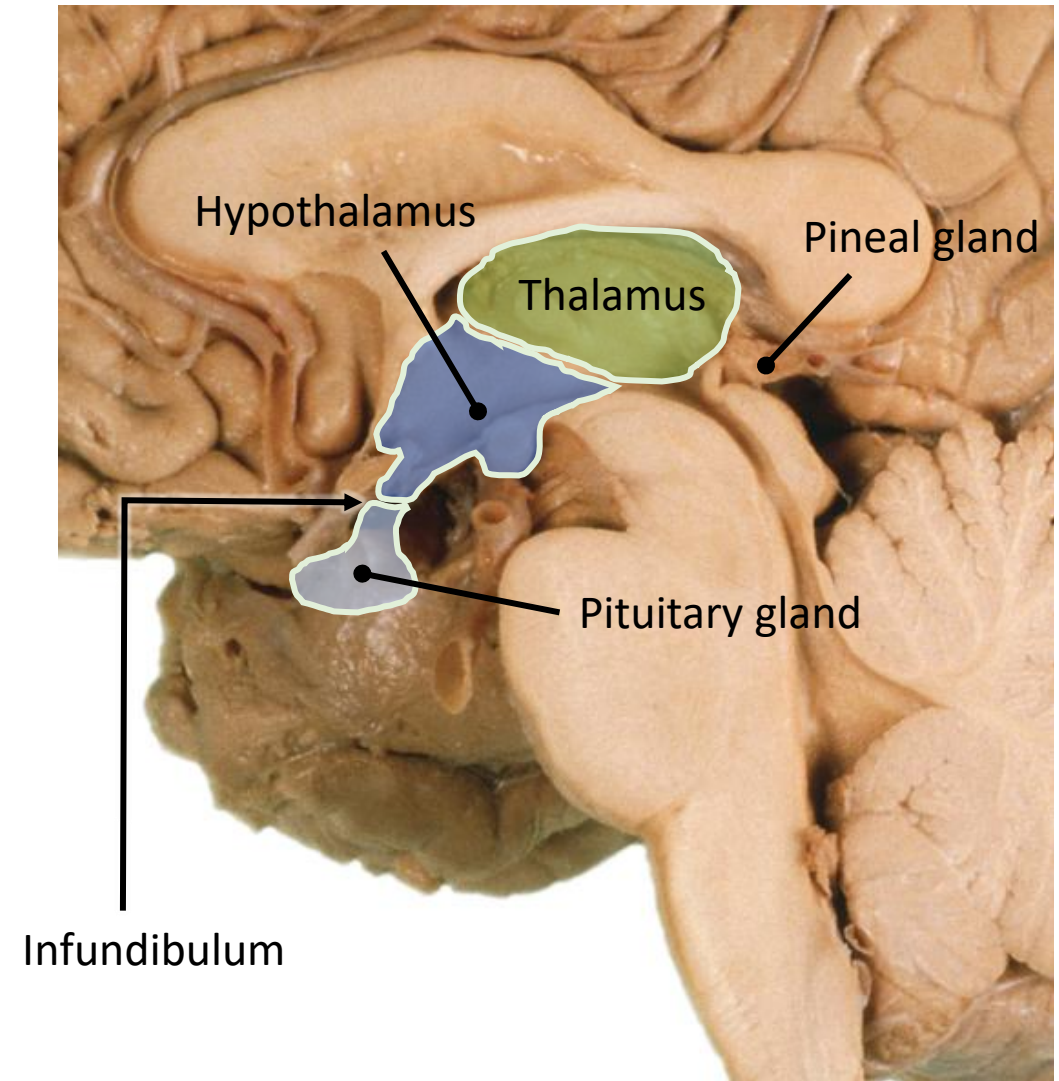
- **Epithalamus**

- **Location:**

- Near the roof of the third ventricle, includes the **pineal gland**

- **Functions:**

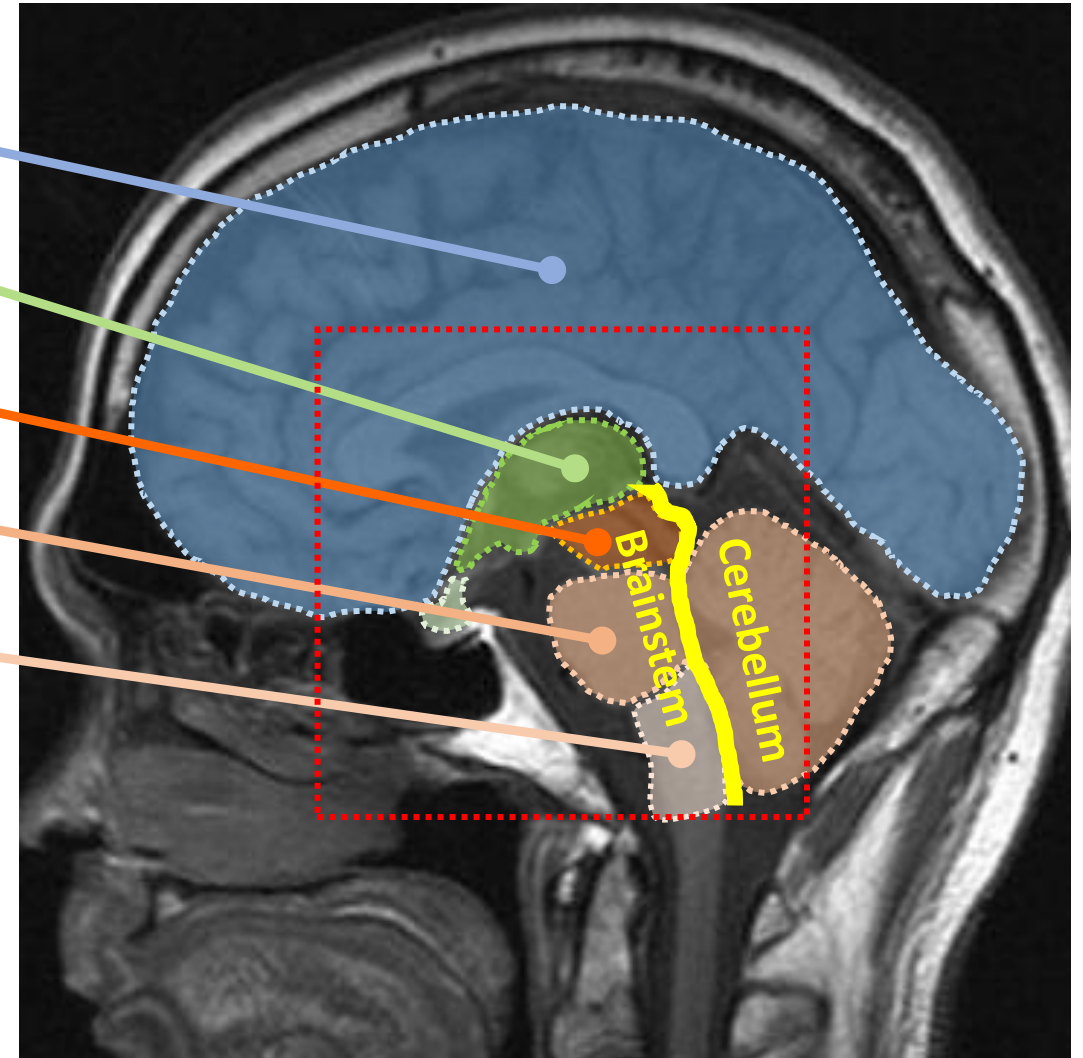
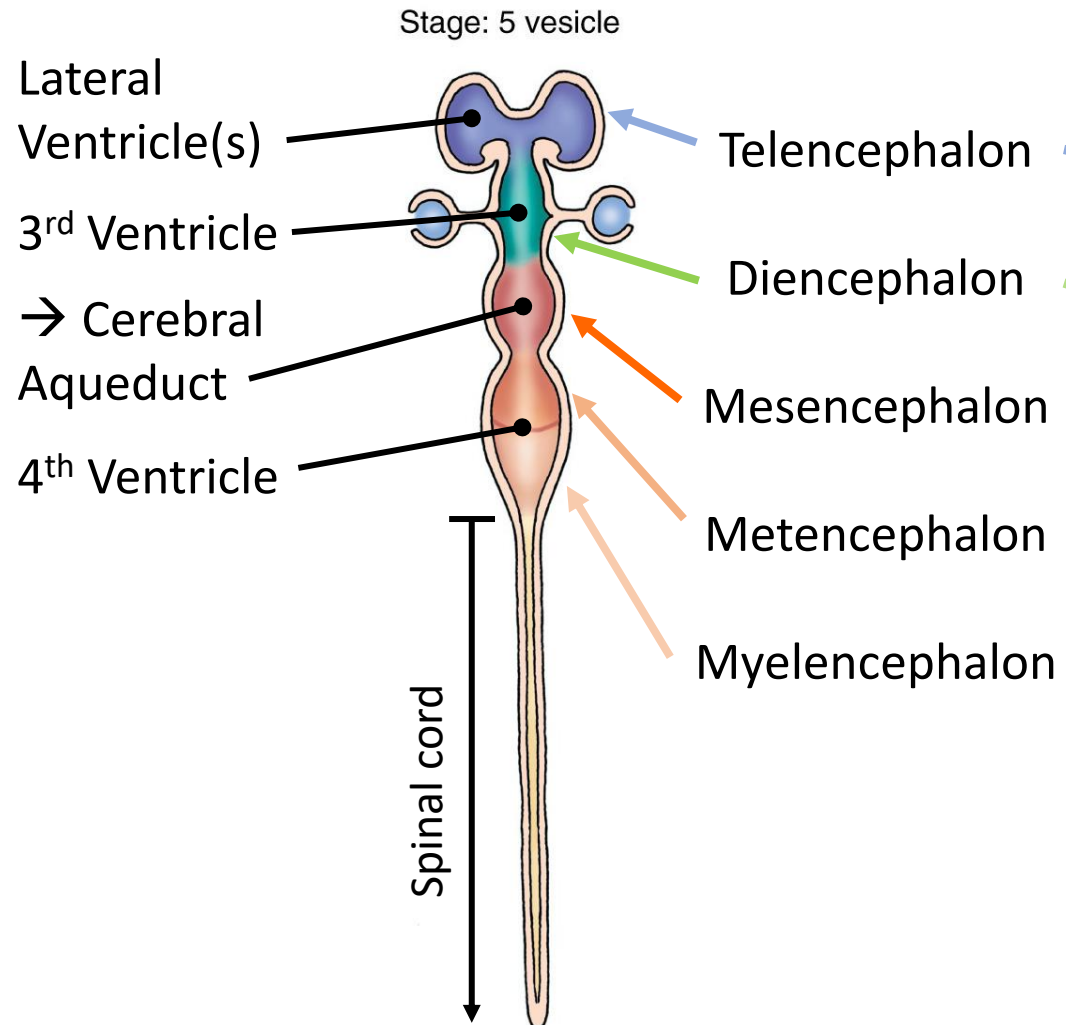
- Production of **melatonin**, which regulates the **circadian rhythm** and the **sleep-wake cycle**



Learning Objectives

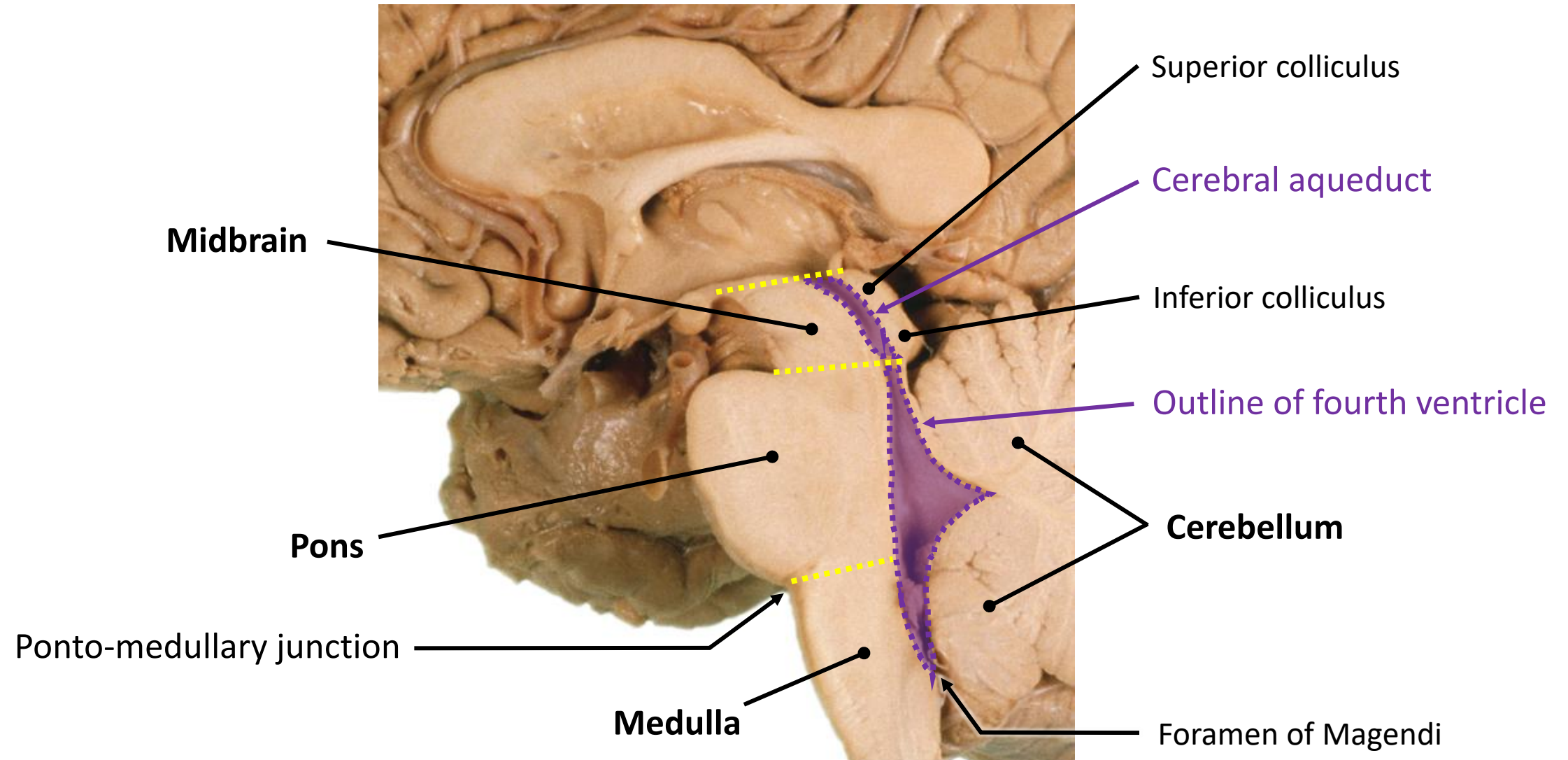
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Five Parts of the Brain



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Midsagittal View Landmarks / Reference to Brainstem and Cerebellum



Underlying image from: Haines, Neuroanatomy Atlas, © Wolters Kluwer
<https://sgu.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/content.aspx?bookid=2473>, Figure 2-30

Midbrain

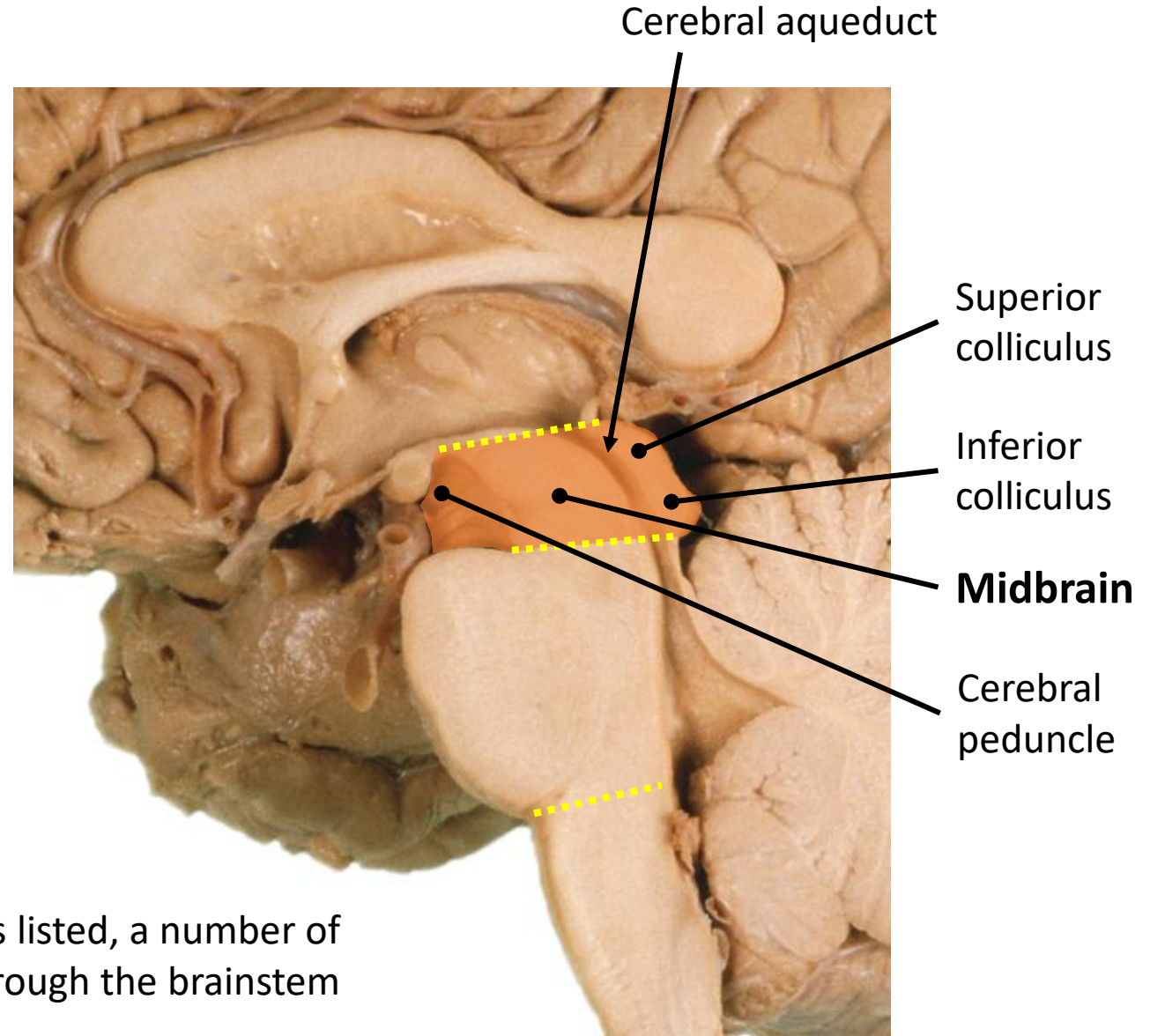
- **Location**

- Between diencephalon and pons
- Anterior and posterior to the cerebral aqueduct

- **Functions**

- Eye movements
- Near accommodation (near vision)
- Pupillary constriction

Please note that in addition to the functions listed, a number of ascending and descending pathways run through the brainstem



Pons

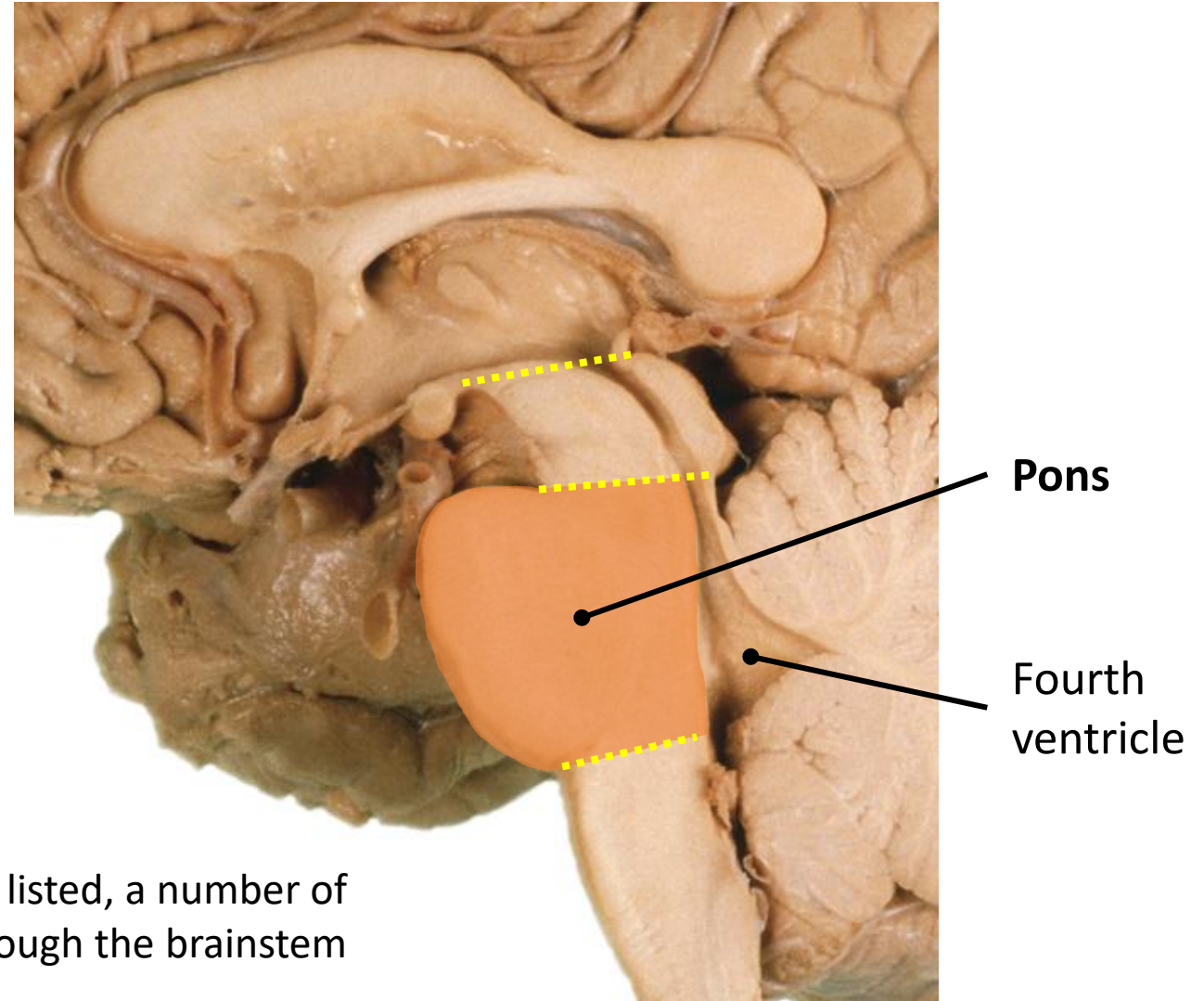
- **Location**

- Between midbrain and medulla
- Anterior to the fourth ventricle

- **Functions**

- Lateral eye movements
- Facial expression
- Chewing (innervation of muscles of mastication)
- Processing of facial touch, vibration and proprioception

Please note that in addition to the functions listed, a number of ascending and descending pathways run through the brainstem



Medulla – Midsagittal View

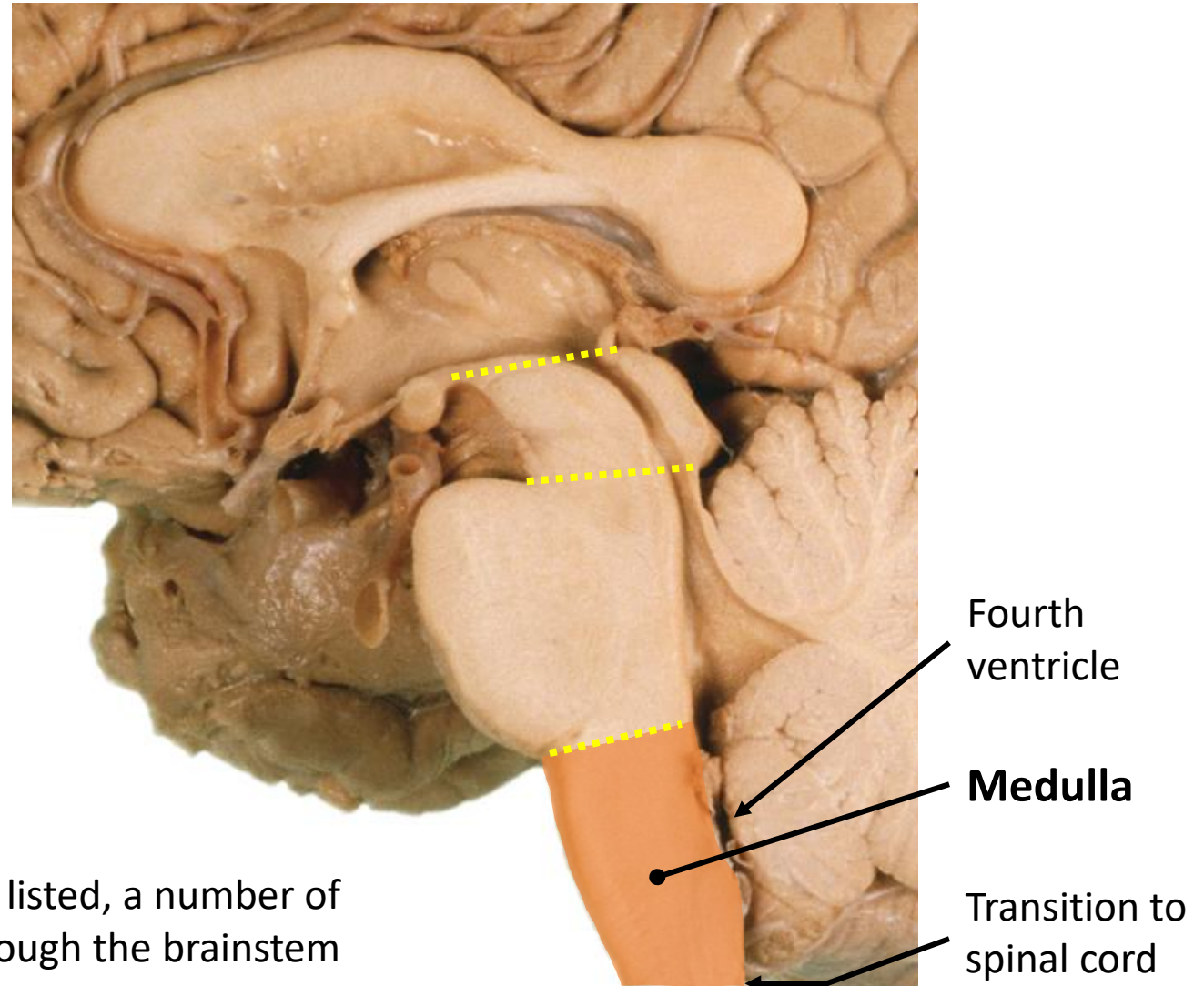
- **Location**

- Between pons and spinal cord
- Anterior to the fourth ventricle

- **Functions**

- Regulating heart and respiratory function
- Processing of facial pain and temperature sensations
- Processing of touch, vibration and proprioception from the body

Please note that in addition to the functions listed, a number of ascending and descending pathways run through the brainstem



Terminology applied throughout the Brainstem

- **Midbrain**

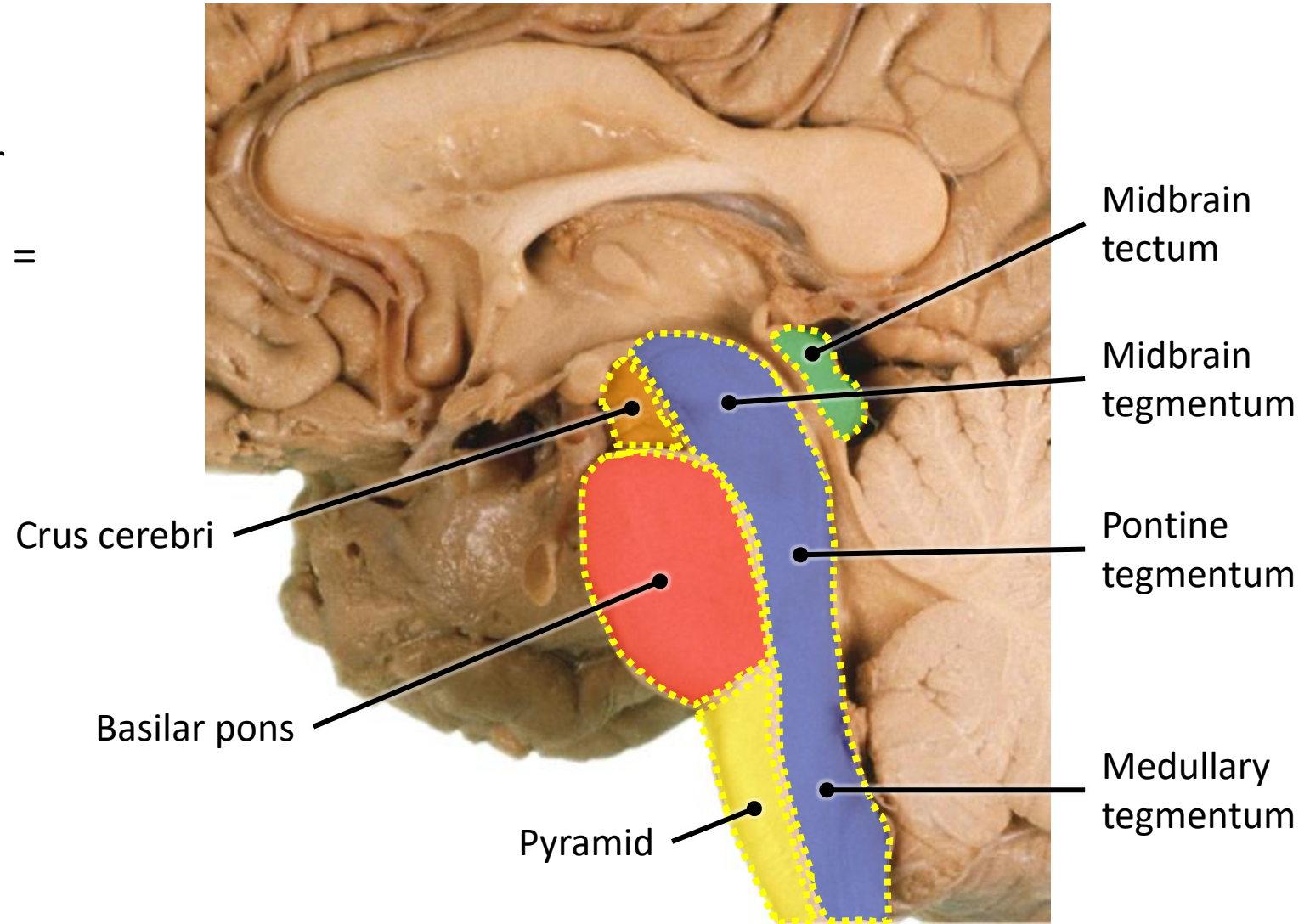
- Tectum (roof)
- Tegmentum (under the roof)
- Crura (legs) cerebri = cerebral peduncles

- **Pons**

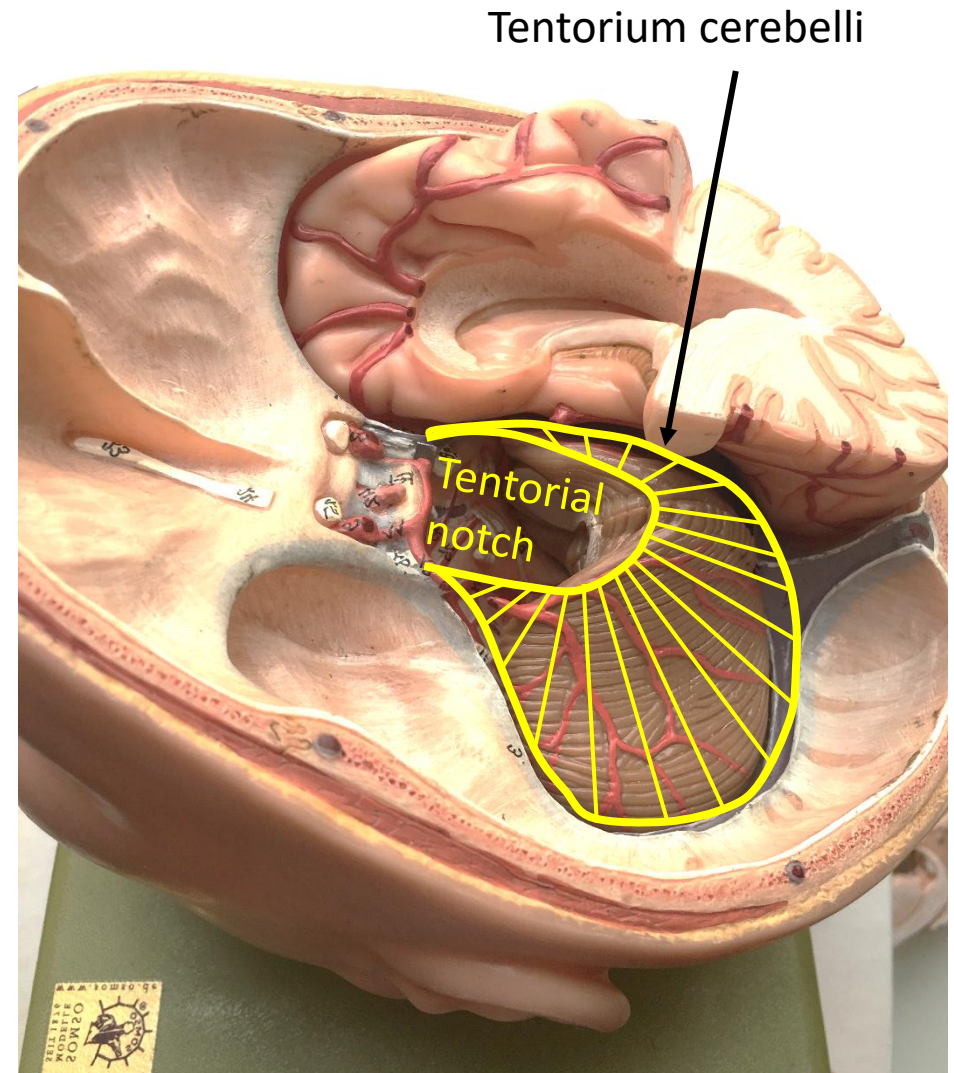
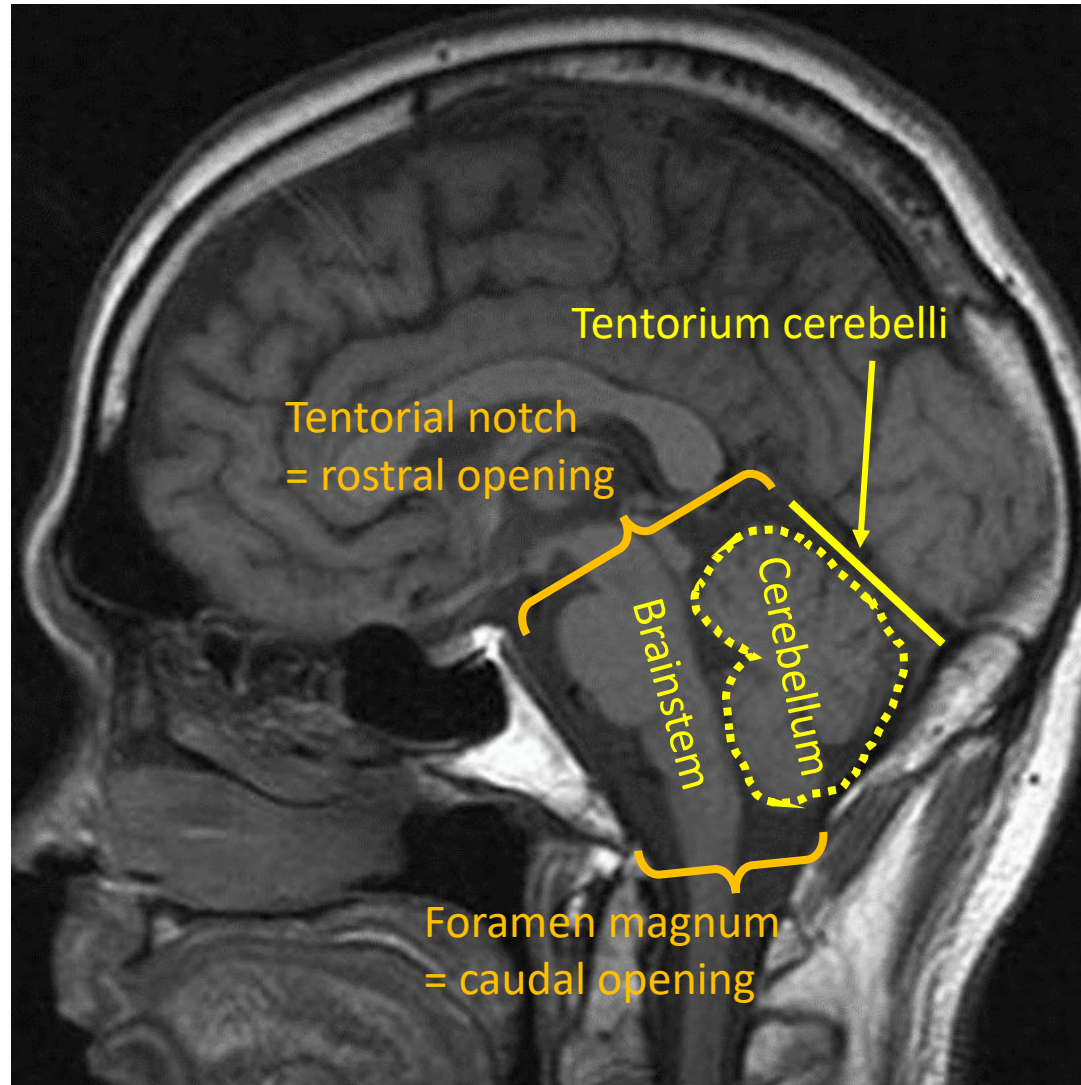
- Tegmentum
- Basilar pons

- **Medulla**

- Tegmentum
- Pyramids



Landmarks – Reference to the Cerebellum



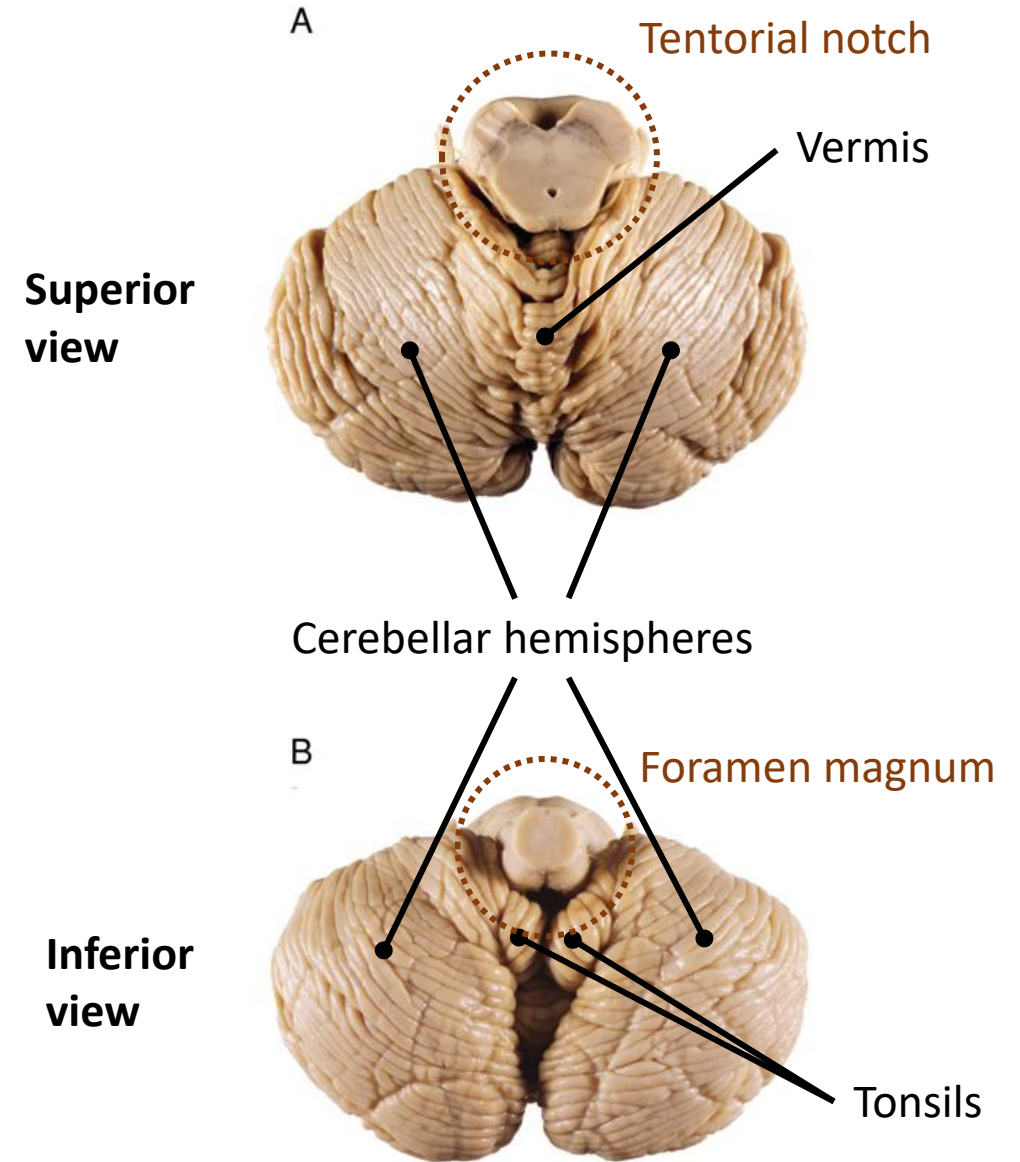
Cerebellum

- **Location**

- Posterior cranial fossa, covered by tentorium cerebelli
- Posterior to fourth ventricle, which separates it from pons and medulla
- Connected with the brainstem through superior, middle and inferior cerebellar peduncles

- **Functions**

- Regulating the precision of movements, including the precision of eye movements



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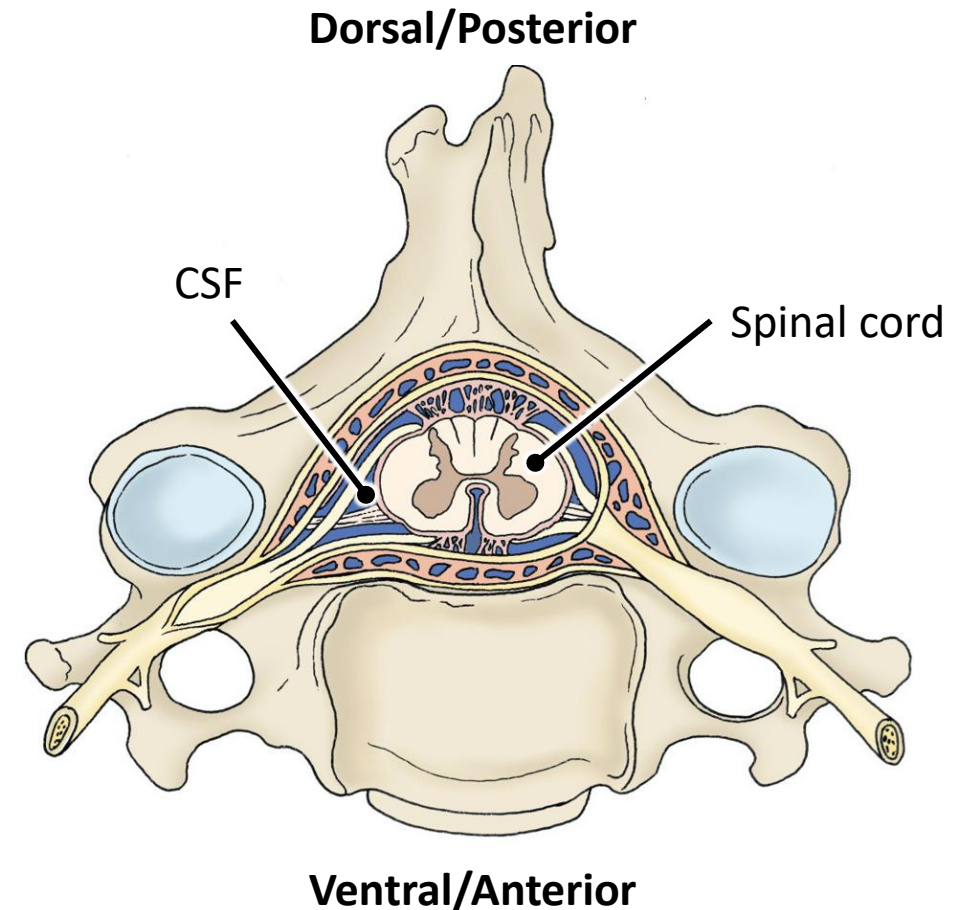
Spinal Cord

- **Location**

- The spinal cord continues where the medulla ends (at the foramen magnum)
- It runs inside the vertebral canal and is surrounded by cerebrospinal fluid (CSF)

- **Functions**

- Receives sensory information from the body and carries the information to the brain
- Receives motor commands from the brain and transmits them to the muscles of the body
- Forms the circuits of spinal reflexes, integrating sensory and motor functions



The End of General Morphology of the Nervous System

If you have any questions, please send an email to:

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